notebook

August 29, 2025

1 To do

- Find the alignment problem of 150 ms (motor?) in group 4 (experiment=2AFC_4)
- ~~Repeat all analyses per mouse (figure with a subplots)~~
- Check ILI distributions using only the first interval (between 1st and 2nd licks) instead of the mean
- Use raw distributions instead of KDE when not comparing distributions. We have **a ton** of trials, so we can afford it
- Log scale in y-axis in RT distributions to stretch around the peak
- Check if the 2 peaks in bimodal RT distributions correspond to the first interval (between 1st and 2nd licks) of the ILI. It could be due to preparatory licks (mouse start licking before the lickport is available and when the lickport does come sometimes it register the 1st lick of the train and sometimes the 2nd). If not:
  - Check split in engaged/disengaged to explain bimodality
- ~~Plot lick rate as discrete distributions (it's not possible to do 0.5 licks). As mean lick rate per ILD too for cleaner plot~~
- Check if bimodality is there in groups with
  - Delay (#4-6)
  - Fixed XYZ positions (#6)
  - No evidence levels (# 5-6)

```
The autoreload extension is already loaded. To reload it, use:
  %reload_ext autoreload
```

2 Load behavioral data

3 Filter data

```
211k trials
```

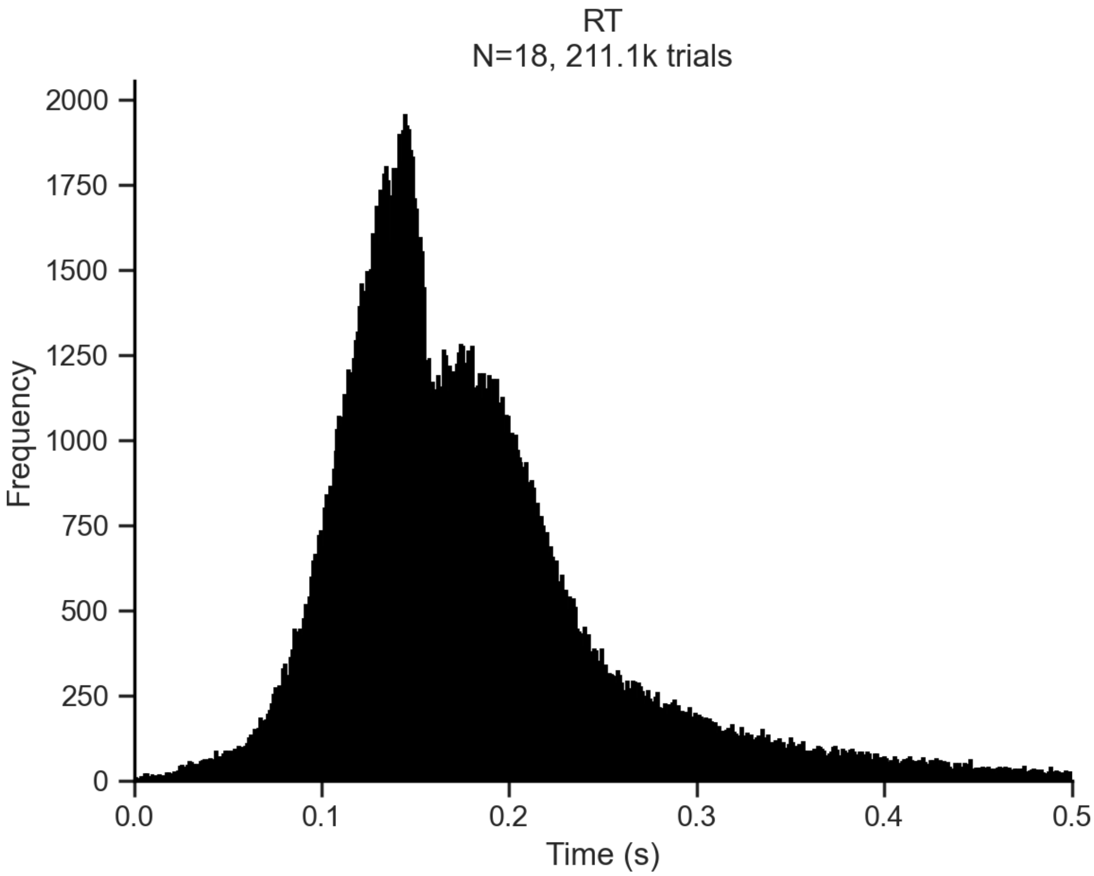
4 Add lick data

```
0.16% of premature lick trials (before response window)
Port1In or Port2In are already lists
['absILD', 'LicksLeft', 'LicksRight', 'nLicksLeft', 'nLicksRight', 'nLicks',
'leftILI', 'rightILI', 'ILI', 'RT', 'RT2', 'SessionHalf']
```

5 Reaction Times (RTs)

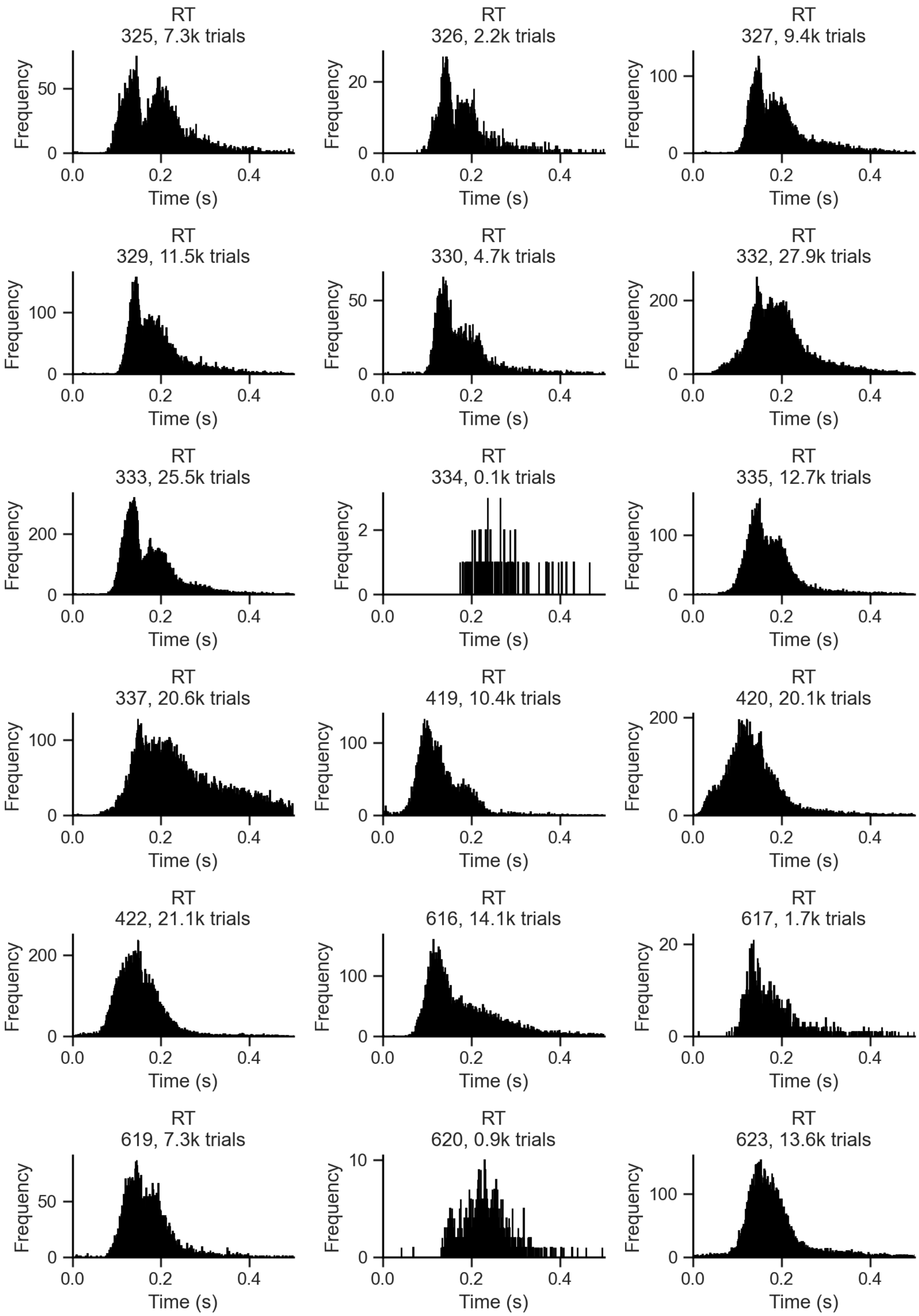
5.1 Distributions

5.1.1 All trials



This is the rawest representation of RT data. It includes all trials from all mice. There is a bimodal distribution of RTs across all mice. The overall distribution has a shoulder-like second peak (lower) after the first one. It is maybe due to the presence of two different strategies in the task: one that involves a fast response and another that involves a slower response. Let's rule out other parsimonious hypotheses first (more likely, easier and faster to check).

First, let's check if this bimodality exist in single subjects, or if in turn it is an averaging artifact.

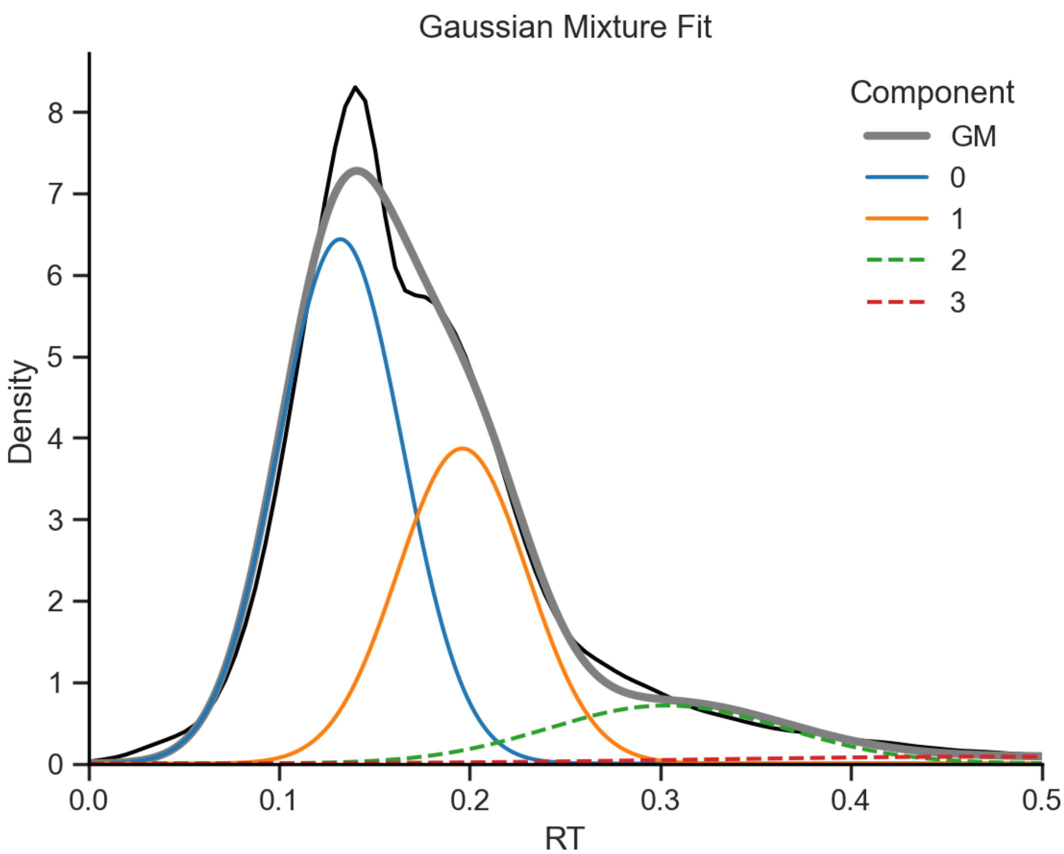

RT
325, 7.3k trials
RT
326, 2.2k trials
RT
327, 9.4k trials
RT
329, 11.5k trials
RT
330, 4.7k trials
RT
332, 27.9k trials
RT
333, 25.5k trials
RT
334, 0.1k trials
RT
335, 12.7k trials
RT
337, 20.6k trials
RT
419, 10.4k trials
RT
420, 20.1k trials
RT
422, 21.1k trials
RT
616, 14.1k trials
RT
617, 1.7k trials
RT
619, 7.3k trials
RT
620, 0.9k trials
RT
623, 13.6k trials
Frequency
Time (s)

The bimodality is indeed present in quite some single subjects.

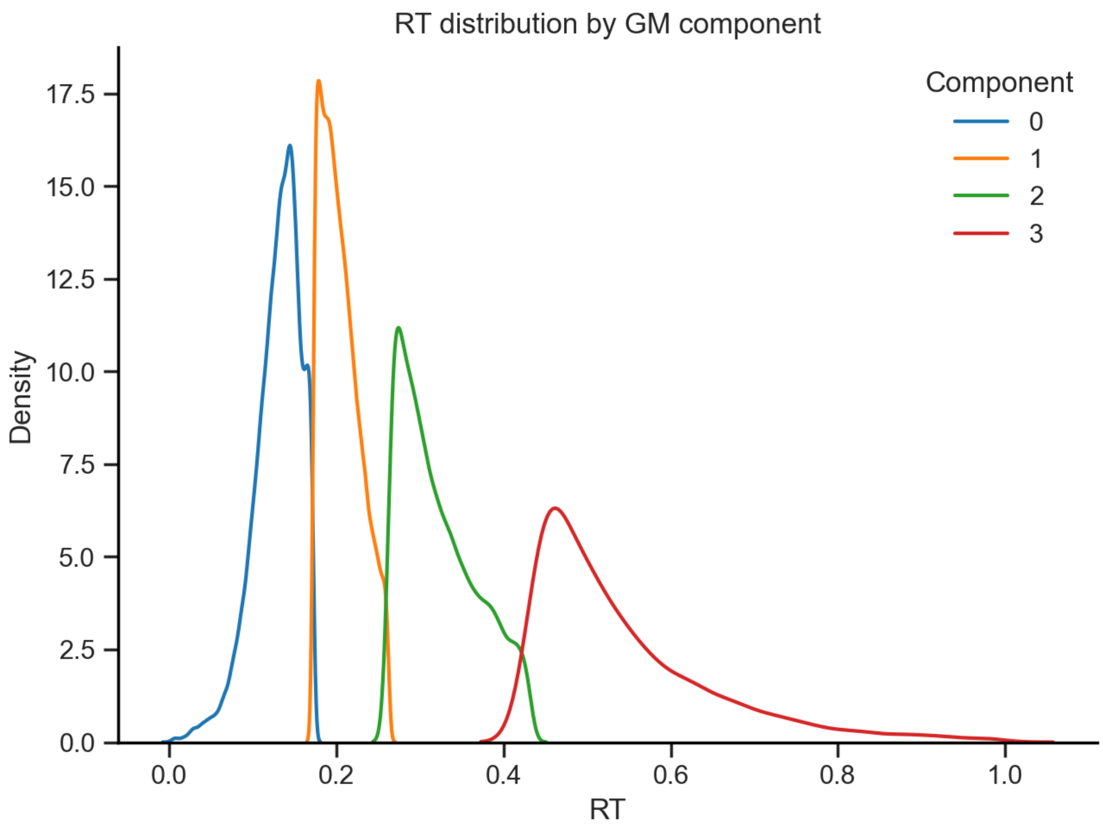
One hypothesis is that the difference evidence levels (ILDs) drives different RTs. To examine this, we can plot a RT distribution per ILD level. To visually compare the distributions, a Kernel Desnity Estimation (KDE) plot is preferred.

Gaussian Mixture fit

```
Component 0: mean=0.13 s, std=0.03 s, weight=0.53
Component 1: mean=0.20 s, std=0.03 s, weight=0.33
Component 2: mean=0.30 s, std=0.06 s, weight=0.11
Component 3: mean=0.47 s, std=0.15 s, weight=0.03
```



Plot RT distributions grouped by component



Screening of variables to explain the difference between the first 2 components (0 and 1)

```
Index(['Trial', 'Side', 'RepTrial', 'Reward', 'Punish', 'Miss', 'WrongLick',
       'Hit', 'AfterHit', 'Choice', 'RepChoice', 'Response', 'TrialStart',
       'TrialEnd', 'TrialLen', 'StimStart', 'StimEnd', 'StimLen',
       'RespWinStart', 'RespWinEnd', 'RespWinLen', 'Filename', 'Filename2',
       'FilesMatch', 'Message', 'MessageFound', 'SoundLeft', 'SoundRight',
       'Sound', 'ILD', 'absILD', 'ILDRep', 'Port1In', 'Port1Out', 'Port2In',
       'Port2Out', 'ValveLeft', 'ValveRight', 'AW', 'AWTrials', 'Switch',
       'Timeout', 'Fixation', 'Stage', 'Motor', 'REC', 'Progression', 'CB',
       'RespWin', 'ITI', 'WarmUp', 'RecoveryMode', 'P', 'PRight', 'Blocks',
       'BlockLen', 'Task', 'Ephys', 'StimDur', 'Delay', 'VarDelay',
       'SerialPort', 'Protocol', 'Creator', 'Project', 'Experiment', 'Board',
       'Setup', 'NetPort', 'Subject', 'BpodApiVersion', 'Session', 'Date',
       'SessionStart', 'SessionEnd', 'LicksLeft', 'LicksRight', 'nLicksLeft',
       'nLicksRight', 'nLicks', 'leftILI', 'rightILI', 'ILI', 'RT', 'RT2',
       'SessionHalf', 'RTlabelsGM'],
      dtype='object')
```

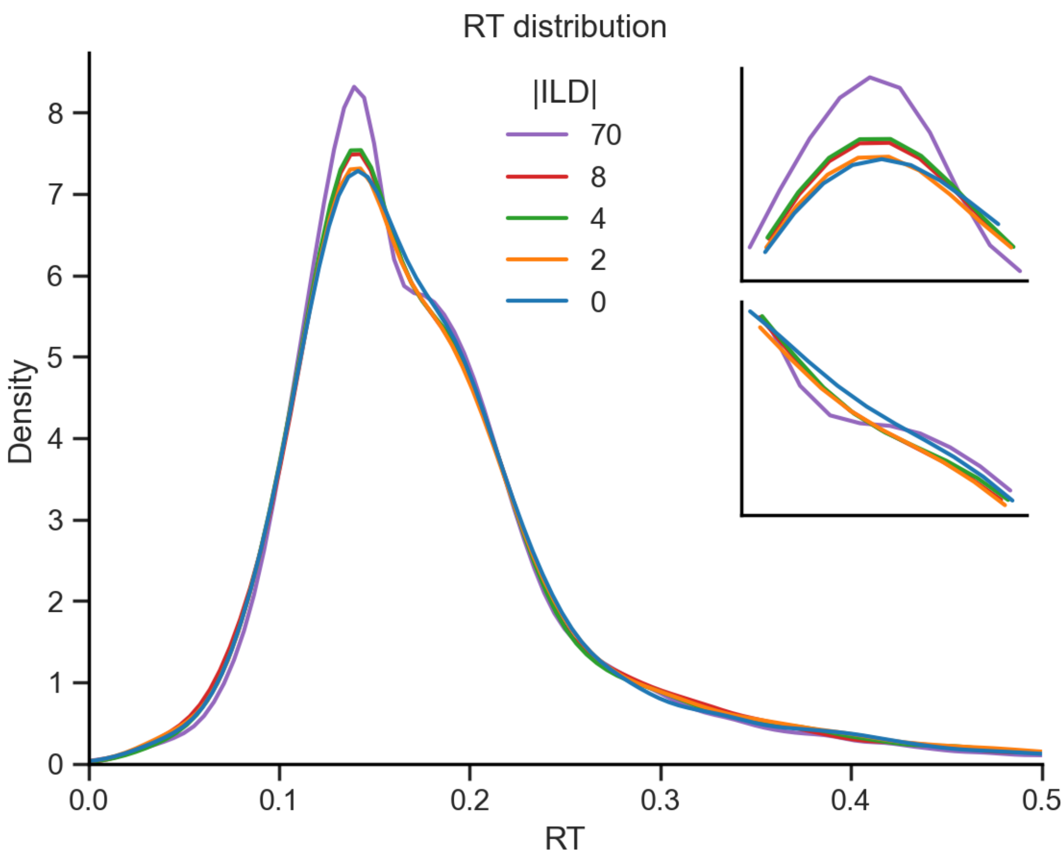
```
 Variable      Type   MeanDiff            pval        pval_adj
```

```
13  SessionHalf  numeric -0.062377  1.596523e-148  2.235132e-147
11  nLicksRight  numeric  0.399763  6.909850e-111  9.673789e-110
10   nLicksLeft  numeric  0.293982   7.558240e-70   1.058154e-68
4      AfterHit  numeric  0.024354   1.191199e-38   1.667679e-37
3           Hit  numeric  0.023235   7.205621e-35   1.008787e-33
0         Trial  numeric -8.370656   4.135594e-31   5.789831e-30
9             P  numeric  0.008412   1.096451e-25   1.535031e-24
7           ILD  numeric  2.415958   1.769892e-16   2.477849e-15
5        Choice  numeric  0.018885   3.883211e-15   5.436495e-14
1          Side  numeric  0.016663   4.239486e-12   5.935281e-11
8        ILDRep  numeric  1.879681   1.865501e-10   2.611701e-09
2      RepTrial  numeric  0.014183   4.572769e-09   6.401877e-08
6     RepChoice  numeric  0.008529   3.395956e-04   4.754339e-03
12       absILD  numeric  0.100285   4.613474e-01   6.458864e+00
```

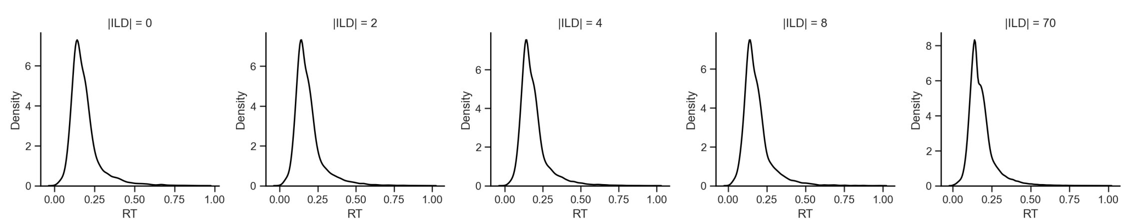
5.1.2 Splits

ILD

```
ILD 70: peak RT = 0.139 s
ILD 8: peak RT = 0.143 s
ILD 4: peak RT = 0.143 s
ILD 2: peak RT = 0.143 s
ILD 0: peak RT = 0.141 s
Mean RT = 0.142 s
```



Too many lines can make on top of each other make it hard to disentangle the ILD impact on RT, so we can plot the RT distribution for each ILD level separately.



Although the bimodality is present in all ILD levels, the second peak is more pronounced for lower ILD levels. This suggests that the bimodality is modulated by evidence.

```
ILD 70: peak RT = 0.136 s
ILD 8: peak RT = 0.144 s
ILD 4: peak RT = 0.126 s
ILD 2: peak RT = 0.134 s
ILD 0: peak RT = 0.140 s
Mean RT = 0.136 s
```

```
ILD 70: peak RT = 0.142 s
ILD 8: peak RT = 0.157 s
ILD 4: peak RT = 0.174 s
ILD 2: peak RT = 0.146 s
ILD 0: peak RT = 0.136 s
Mean RT = 0.151 s
ILD 70: peak RT = 0.148 s
ILD 8: peak RT = 0.153 s
ILD 4: peak RT = 0.148 s
ILD 2: peak RT = 0.151 s
ILD 0: peak RT = 0.149 s
Mean RT = 0.150 s
ILD 70: peak RT = 0.140 s
ILD 8: peak RT = 0.153 s
ILD 4: peak RT = 0.159 s
ILD 2: peak RT = 0.184 s
ILD 0: peak RT = 0.176 s
Mean RT = 0.162 s
ILD 70: peak RT = 0.138 s
ILD 8: peak RT = 0.141 s
ILD 4: peak RT = 0.143 s
ILD 2: peak RT = 0.141 s
ILD 0: peak RT = 0.146 s
Mean RT = 0.142 s
ILD 70: peak RT = 0.153 s
ILD 8: peak RT = 0.173 s
ILD 4: peak RT = 0.170 s
ILD 2: peak RT = 0.185 s
ILD 0: peak RT = 0.188 s
Mean RT = 0.174 s
ILD 70: peak RT = 0.134 s
ILD 8: peak RT = 0.134 s
ILD 4: peak RT = 0.137 s
ILD 2: peak RT = 0.135 s
ILD 0: peak RT = 0.134 s
Mean RT = 0.135 s
ILD 70: peak RT = 0.240 s
ILD 8: peak RT = 0.240 s
ILD 2: peak RT = 0.251 s
ILD 0: peak RT = 0.304 s
Mean RT = 0.259 s
ILD 70: peak RT = 0.143 s
ILD 8: peak RT = 0.149 s
ILD 4: peak RT = 0.148 s
ILD 2: peak RT = 0.148 s
ILD 0: peak RT = 0.151 s
Mean RT = 0.148 s
ILD 70: peak RT = 0.167 s
```

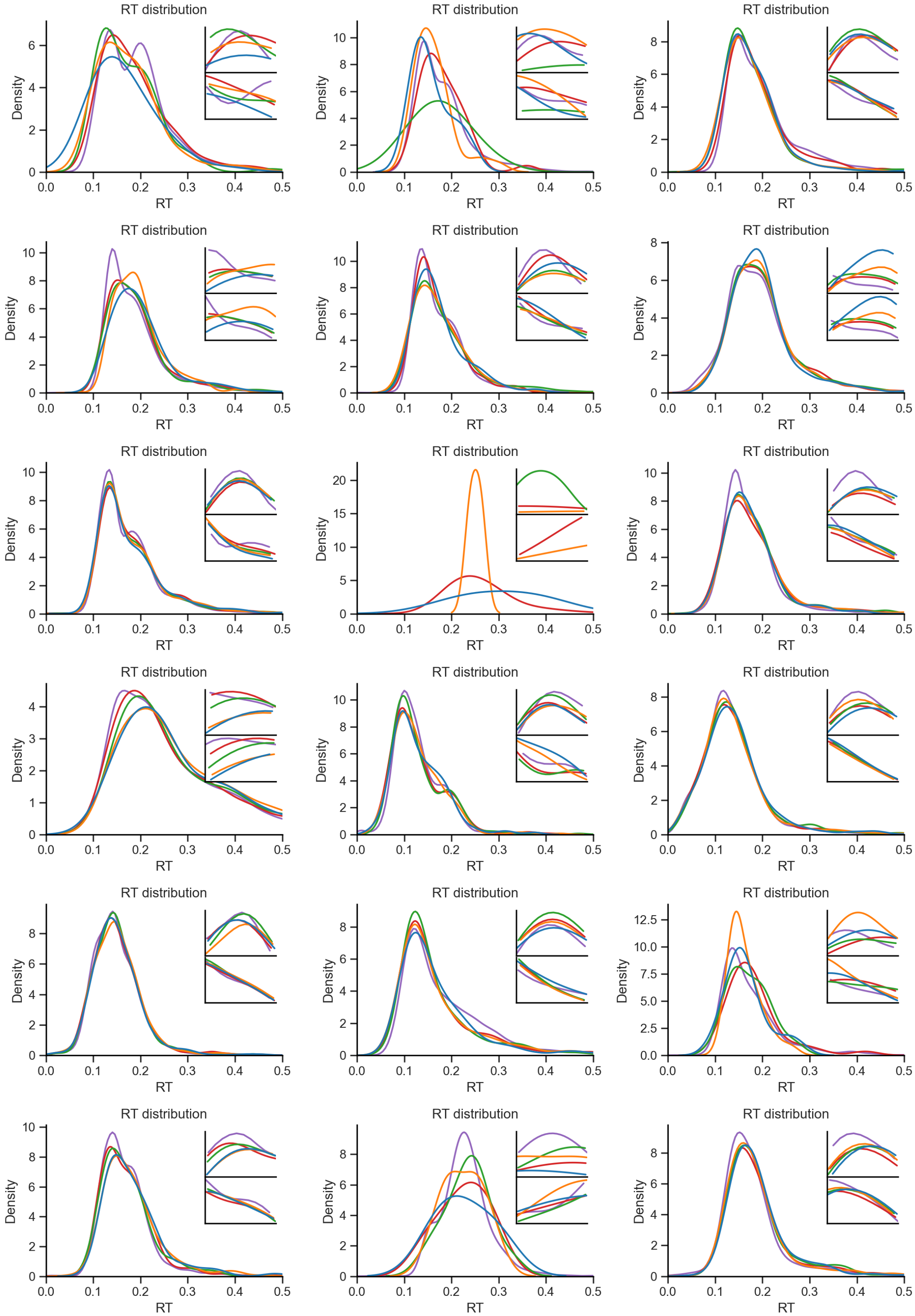
```
ILD 8: peak RT = 0.189 s
ILD 4: peak RT = 0.193 s
```

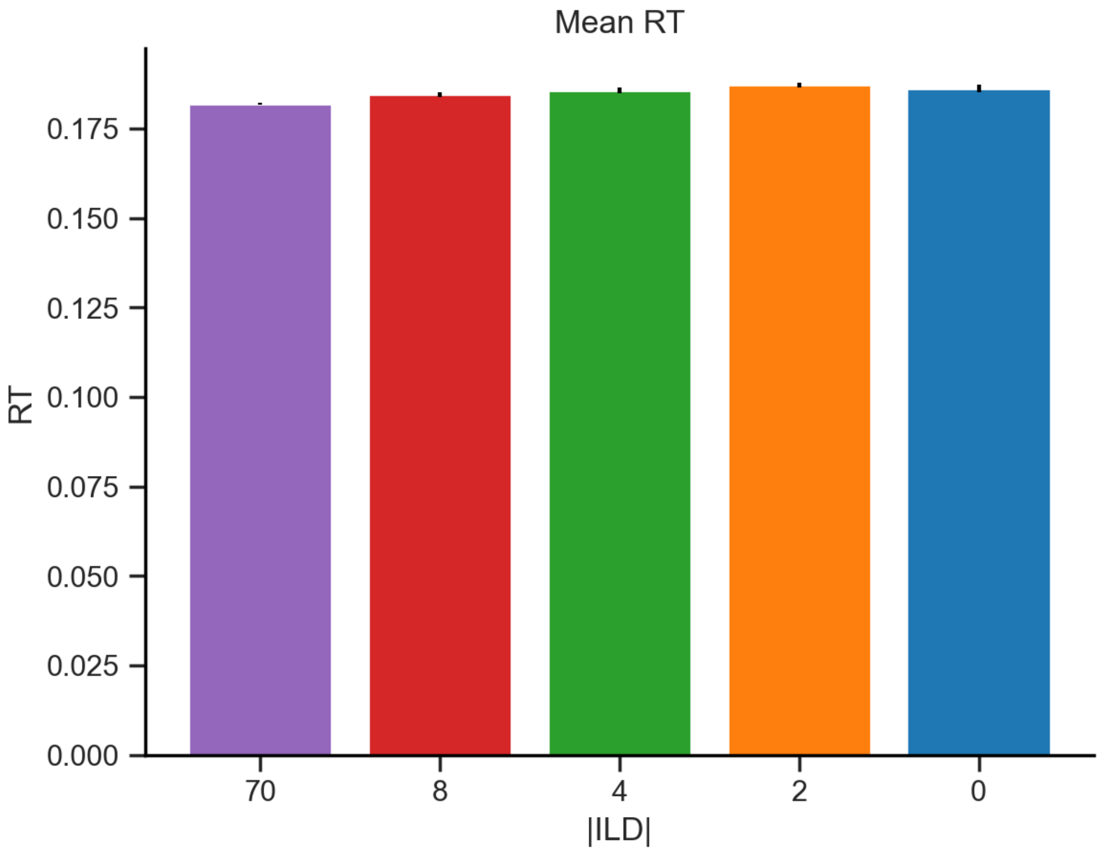
```
C:\Users\Usuario\PycharmProjects\licks\licks.py:401: UserWarning: Dataset has 0
variance; skipping density estimate. Pass `warn_singular=False` to disable this
warning.
  # sns.histplot(df_ild[var], stat='density', element='step', fill=False,
kde=False, color=color, label=ild)
```

```
ILD 2: peak RT = 0.210 s
ILD 0: peak RT = 0.211 s
Mean RT = 0.194 s
ILD 70: peak RT = 0.100 s
ILD 8: peak RT = 0.096 s
ILD 4: peak RT = 0.099 s
ILD 2: peak RT = 0.100 s
ILD 0: peak RT = 0.097 s
Mean RT = 0.099 s
ILD 70: peak RT = 0.118 s
ILD 8: peak RT = 0.119 s
ILD 4: peak RT = 0.124 s
ILD 2: peak RT = 0.117 s
ILD 0: peak RT = 0.125 s
Mean RT = 0.121 s
ILD 70: peak RT = 0.141 s
ILD 8: peak RT = 0.136 s
ILD 4: peak RT = 0.142 s
ILD 2: peak RT = 0.143 s
ILD 0: peak RT = 0.138 s
Mean RT = 0.140 s
ILD 70: peak RT = 0.119 s
ILD 8: peak RT = 0.123 s
ILD 4: peak RT = 0.124 s
ILD 2: peak RT = 0.122 s
ILD 0: peak RT = 0.124 s
Mean RT = 0.122 s
ILD 70: peak RT = 0.135 s
ILD 8: peak RT = 0.162 s
ILD 4: peak RT = 0.146 s
ILD 2: peak RT = 0.144 s
ILD 0: peak RT = 0.152 s
Mean RT = 0.148 s
ILD 70: peak RT = 0.141 s
ILD 8: peak RT = 0.135 s
ILD 4: peak RT = 0.142 s
ILD 2: peak RT = 0.150 s
ILD 0: peak RT = 0.149 s
Mean RT = 0.143 s
ILD 70: peak RT = 0.227 s
```

```
ILD 8: peak RT = 0.241 s
ILD 4: peak RT = 0.243 s
ILD 2: peak RT = 0.208 s
ILD 0: peak RT = 0.212 s
Mean RT = 0.226 s
ILD 70: peak RT = 0.152 s
ILD 8: peak RT = 0.156 s
ILD 4: peak RT = 0.159 s
ILD 2: peak RT = 0.160 s
ILD 0: peak RT = 0.164 s
Mean RT = 0.158 s
```

```
C:\Users\Usuario\PycharmProjects\licks\licks.py:550: UserWarning: This figure
includes Axes that are not compatible with tight_layout, so results might be
incorrect.
  plt.title(title)
```


RT distribution
Density
RT
0.0
0.1
0.2
0.3
0.4
0.5

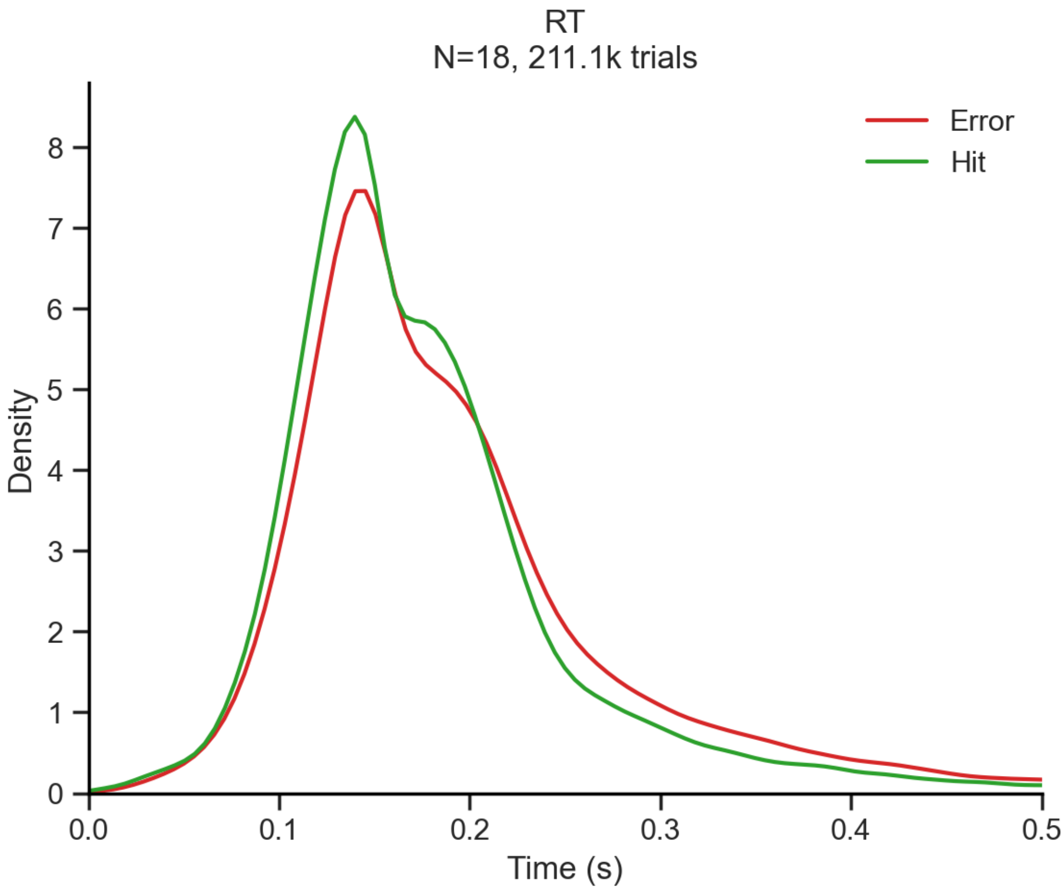


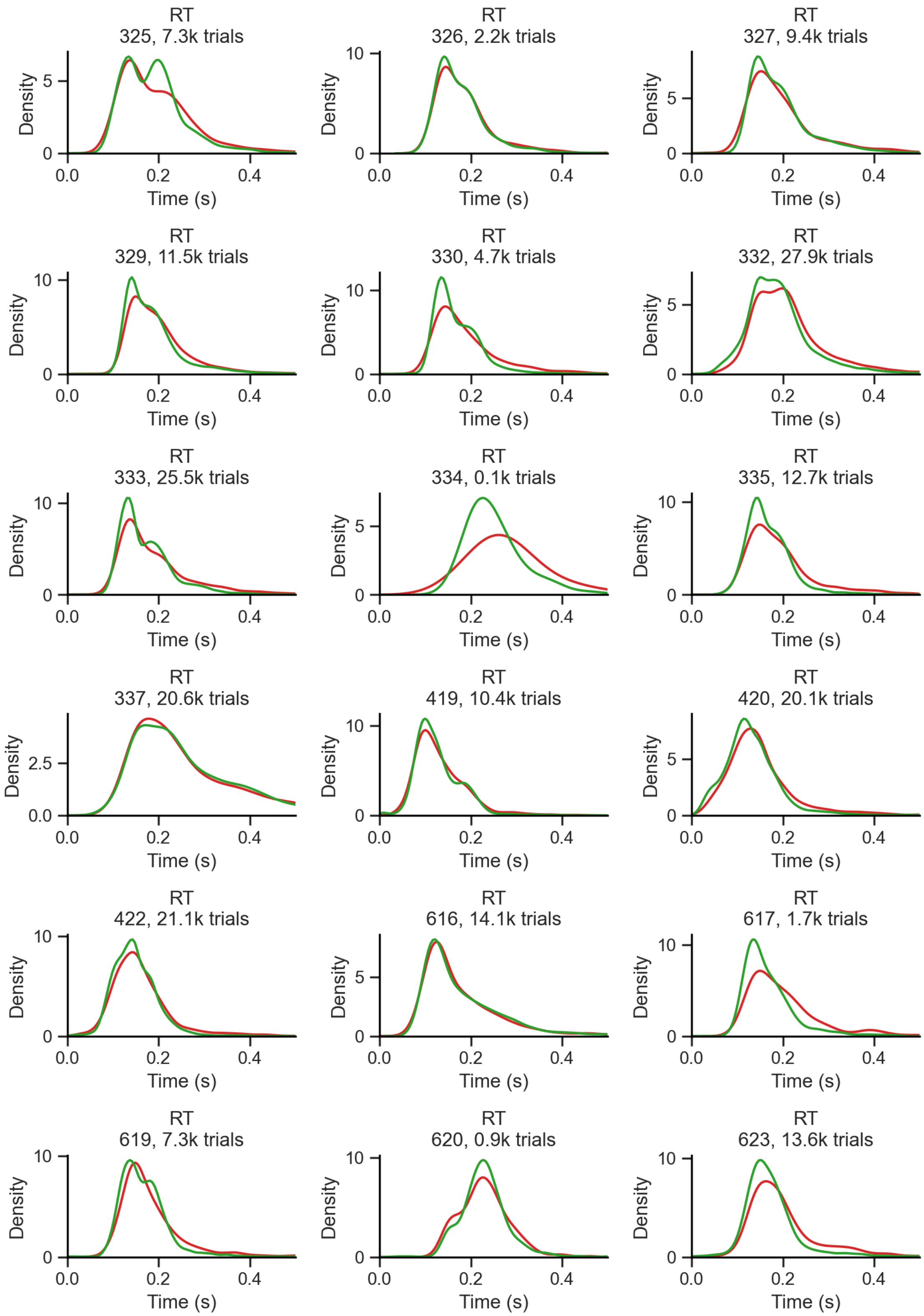
The ILD alone does not seem to explain the **existence** of bimodality in the RT distribution, as the bimodality is present across all ILD levels (for subjects that did show bimodality in their RTs).

Nevertheless, zooming in the peak of the distribution (0.1-0.2 s, first inset), we can see that the RTs are faster for lower ILD levels, which is expected as the evidence is more certain. What is surprising is for this modulation to show up despite the fact that mice were head-fixed and the stimulus was fixed duration. In addition, zooming in the second peak of the distribution (0.15-0.2 s, second inset), we can see that the second peak is more pronounced for lower ILD levels, suggesting that the bimodality is actually modulated by ILD level.

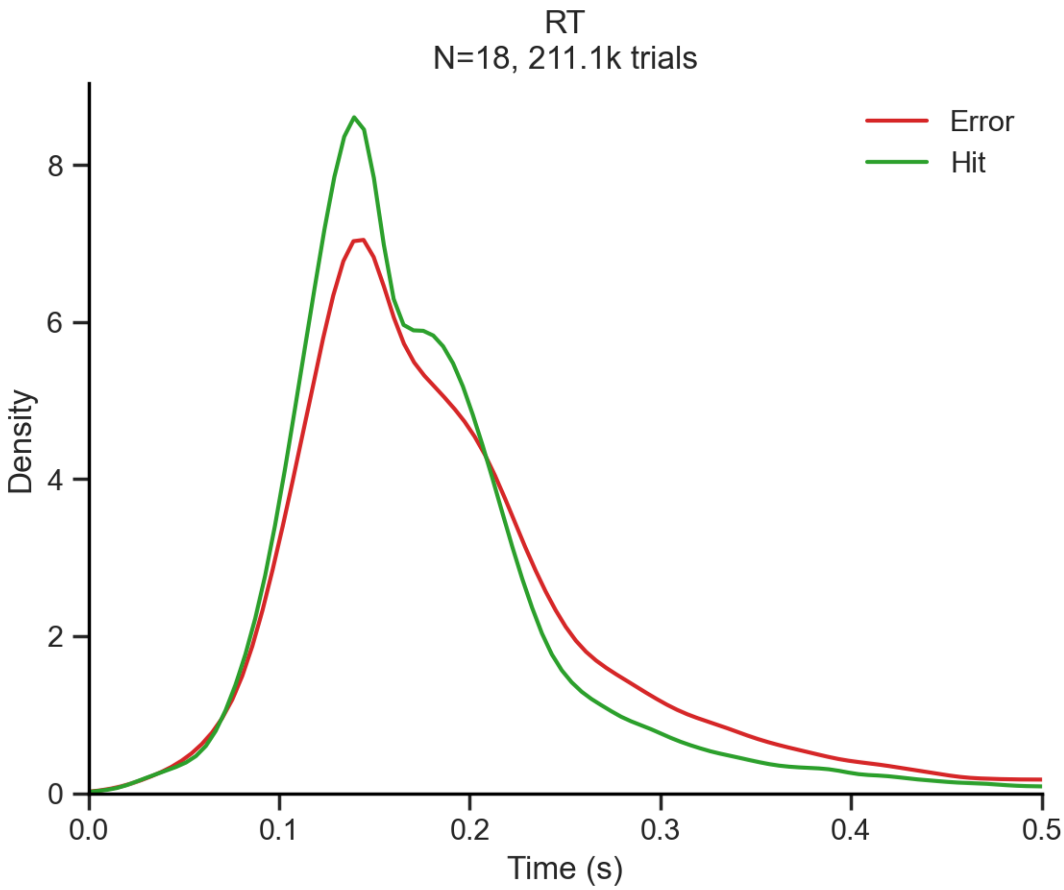
Another hypothesis is that the bimodality is driven by differences in behavioral variables such as outcome, previous outcome, choice, previous choice, repeat trial, stimulus, or session half. To examine this, we can plot a RT distribution per behavioral variable split in 2 conditions.

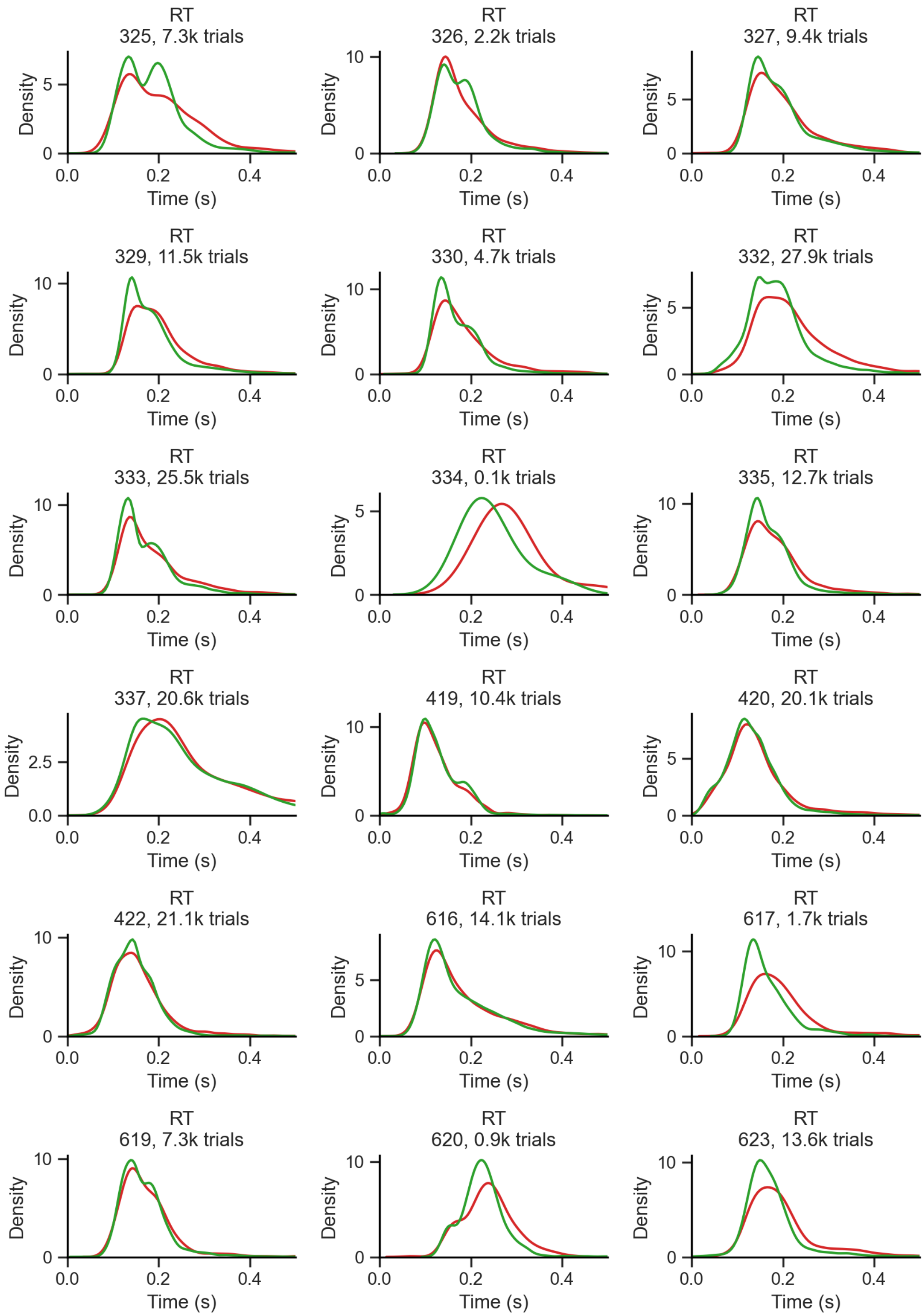
Outcome


RT
N=18, 211.1k trials
Error
Hit
Density
Time (s)
0
1
2
3
4
5
6
7
8
0.0
0.1
0.2
0.3
0.4
0.5

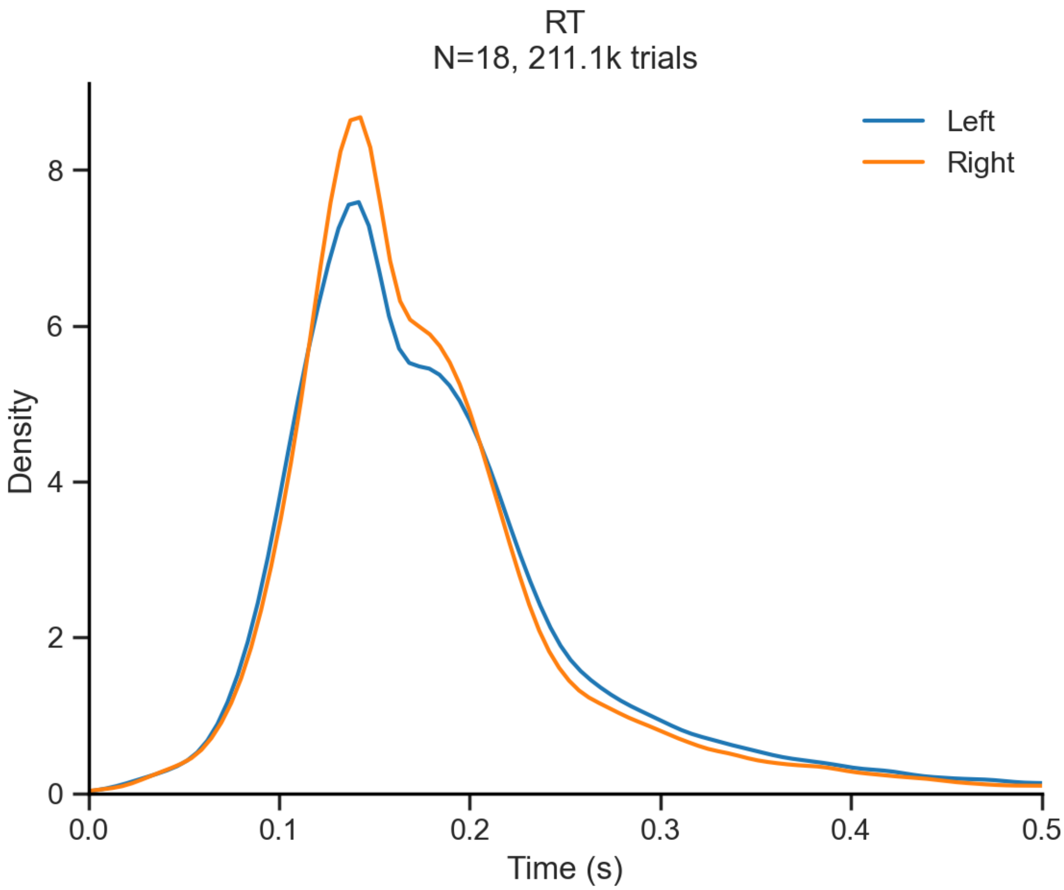

RT
325, 7.3k trials
RT
326, 2.2k trials
RT
327, 9.4k trials
RT
329, 11.5k trials
RT
330, 4.7k trials
RT
332, 27.9k trials
RT
333, 25.5k trials
RT
334, 0.1k trials
RT
335, 12.7k trials
RT
337, 20.6k trials
RT
419, 10.4k trials
RT
420, 20.1k trials
RT
422, 21.1k trials
RT
616, 14.1k trials
RT
617, 1.7k trials
RT
619, 7.3k trials
RT
620, 0.9k trials
RT
623, 13.6k trials
Density
Time (s)
0.0
0.2
0.4
0
5
10
2.5

Previous outcome

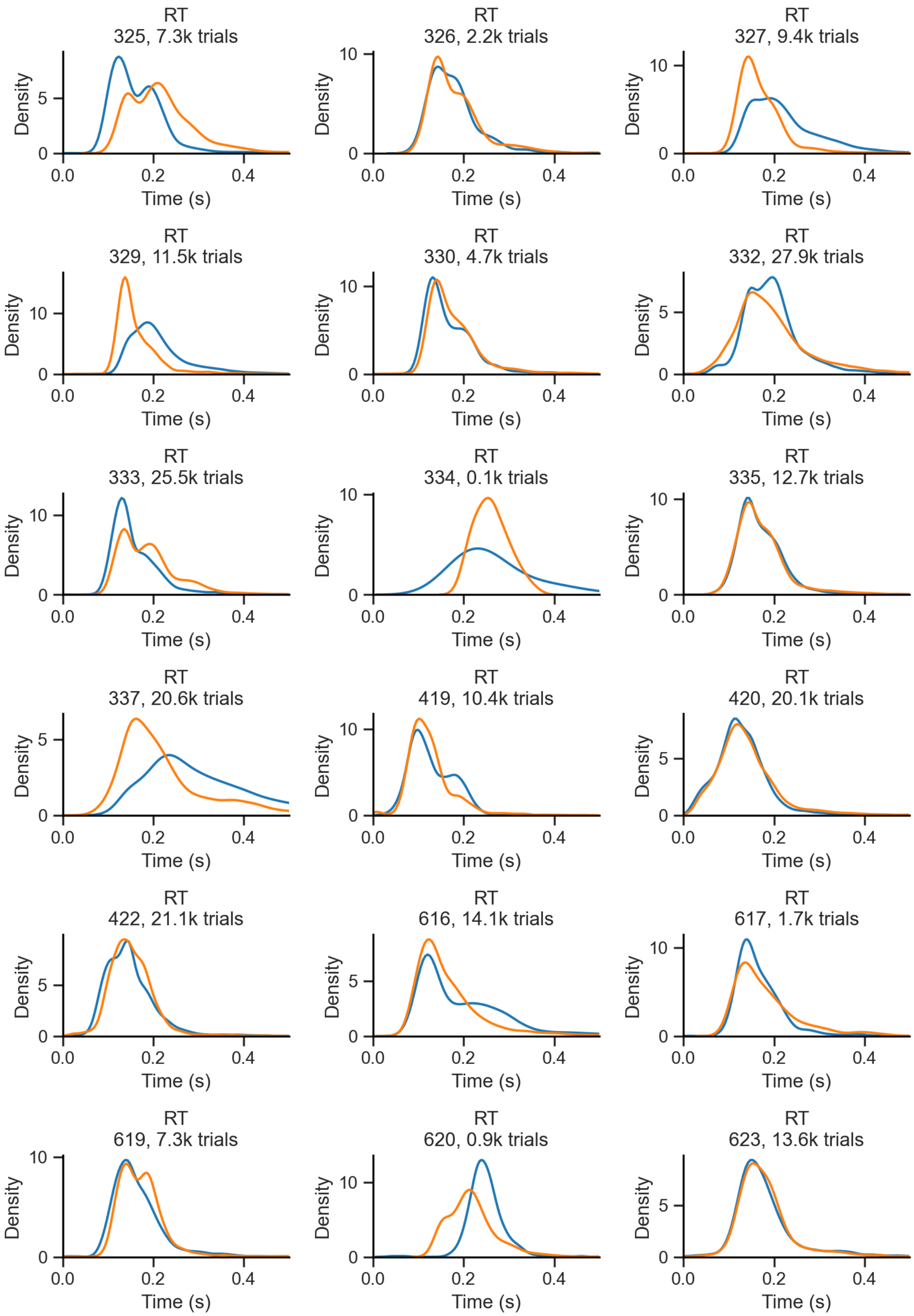



RT
325, 7.3k trials
RT
326, 2.2k trials
RT
327, 9.4k trials
RT
329, 11.5k trials
RT
330, 4.7k trials
RT
332, 27.9k trials
RT
333, 25.5k trials
RT
334, 0.1k trials
RT
335, 12.7k trials
RT
337, 20.6k trials
RT
419, 10.4k trials
RT
420, 20.1k trials
RT
422, 21.1k trials
RT
616, 14.1k trials
RT
617, 1.7k trials
RT
619, 7.3k trials
RT
620, 0.9k trials
RT
623, 13.6k trials
Density
Time (s)
0.0
0.2
0.4

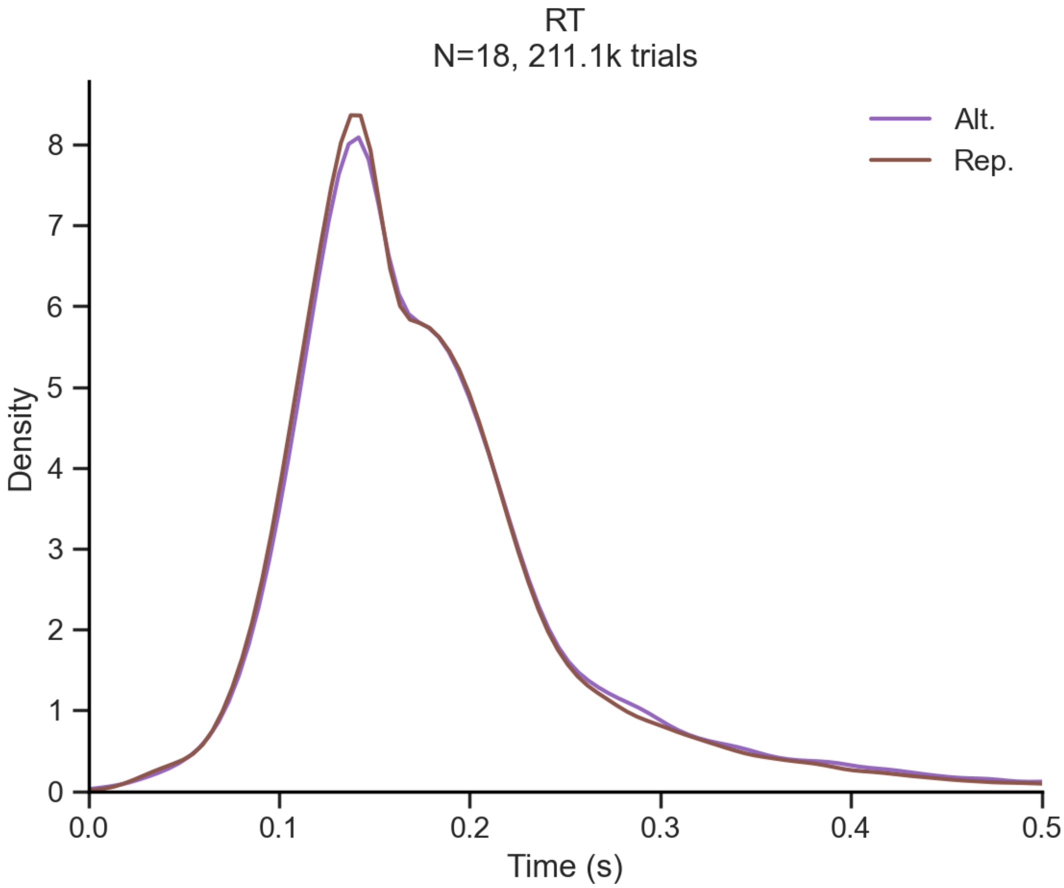
Choice

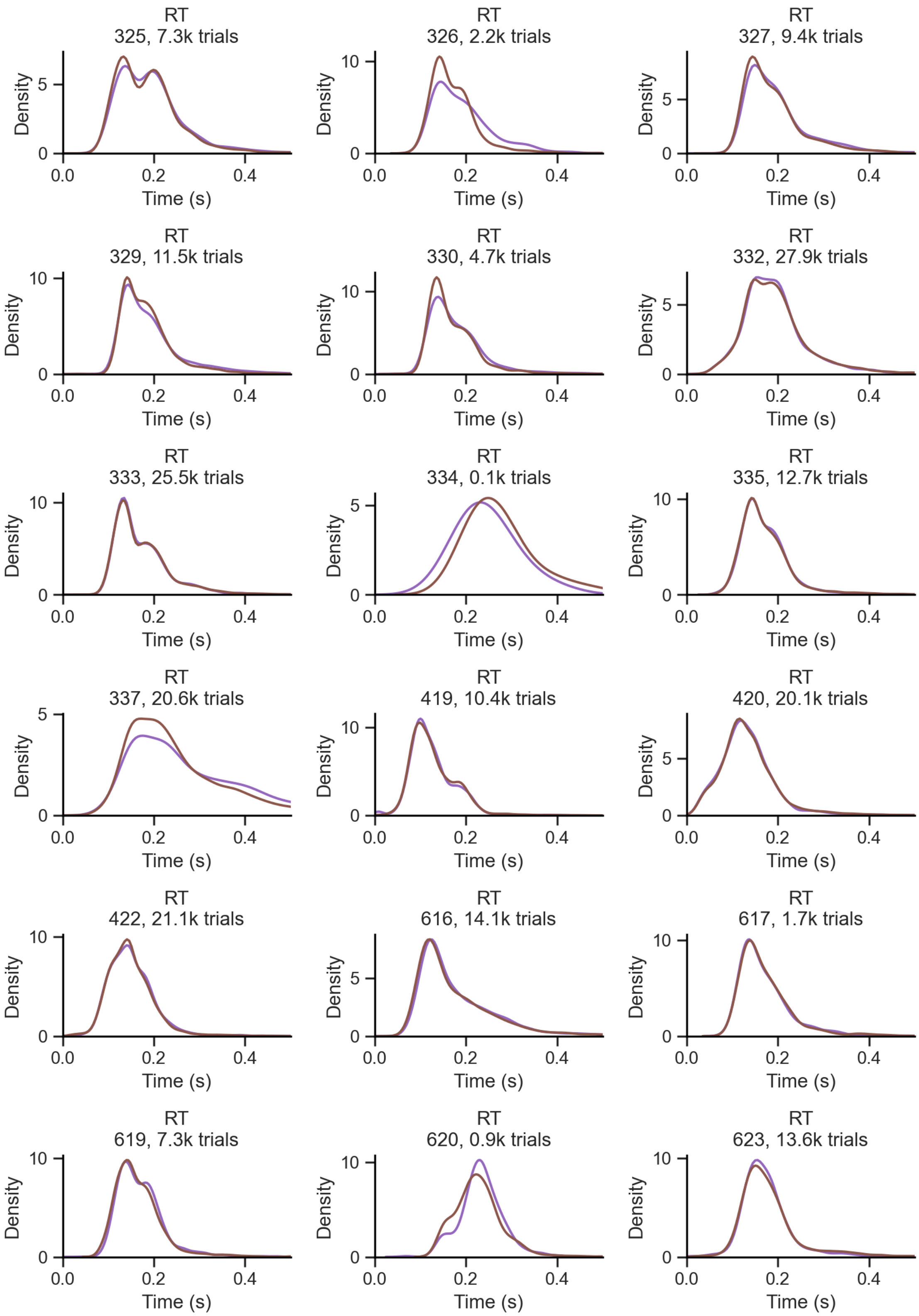


RT
N=18, 211.1k trials
Left
Right
Density
Time (s)
0
2
4
6
8
0.0
0.1
0.2
0.3
0.4
0.5

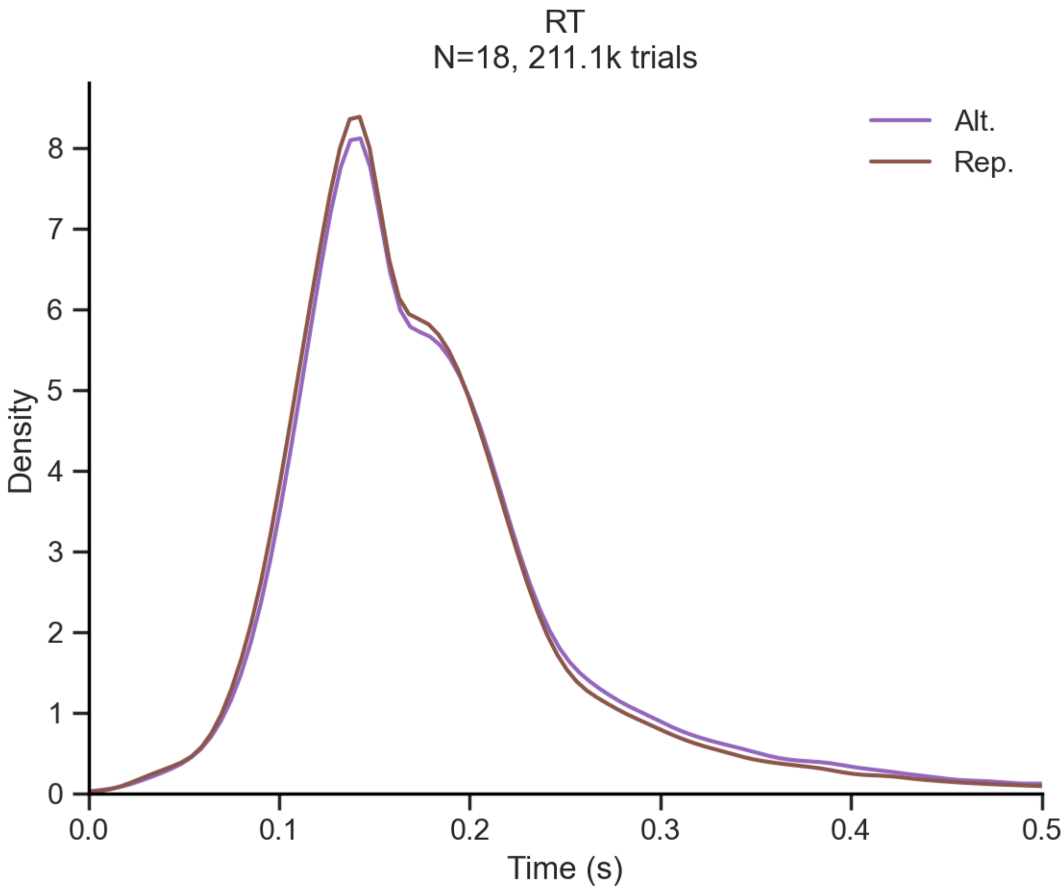

RT
325, 7.3k trials
RT
326, 2.2k trials
RT
327, 9.4k trials
RT
329, 11.5k trials
RT
330, 4.7k trials
RT
332, 27.9k trials
RT
333, 25.5k trials
RT
334, 0.1k trials
RT
335, 12.7k trials
RT
337, 20.6k trials
RT
419, 10.4k trials
RT
420, 20.1k trials
RT
422, 21.1k trials
RT
616, 14.1k trials
RT
617, 1.7k trials
RT
619, 7.3k trials
RT
620, 0.9k trials
RT
623, 13.6k trials
Density
Time (s)

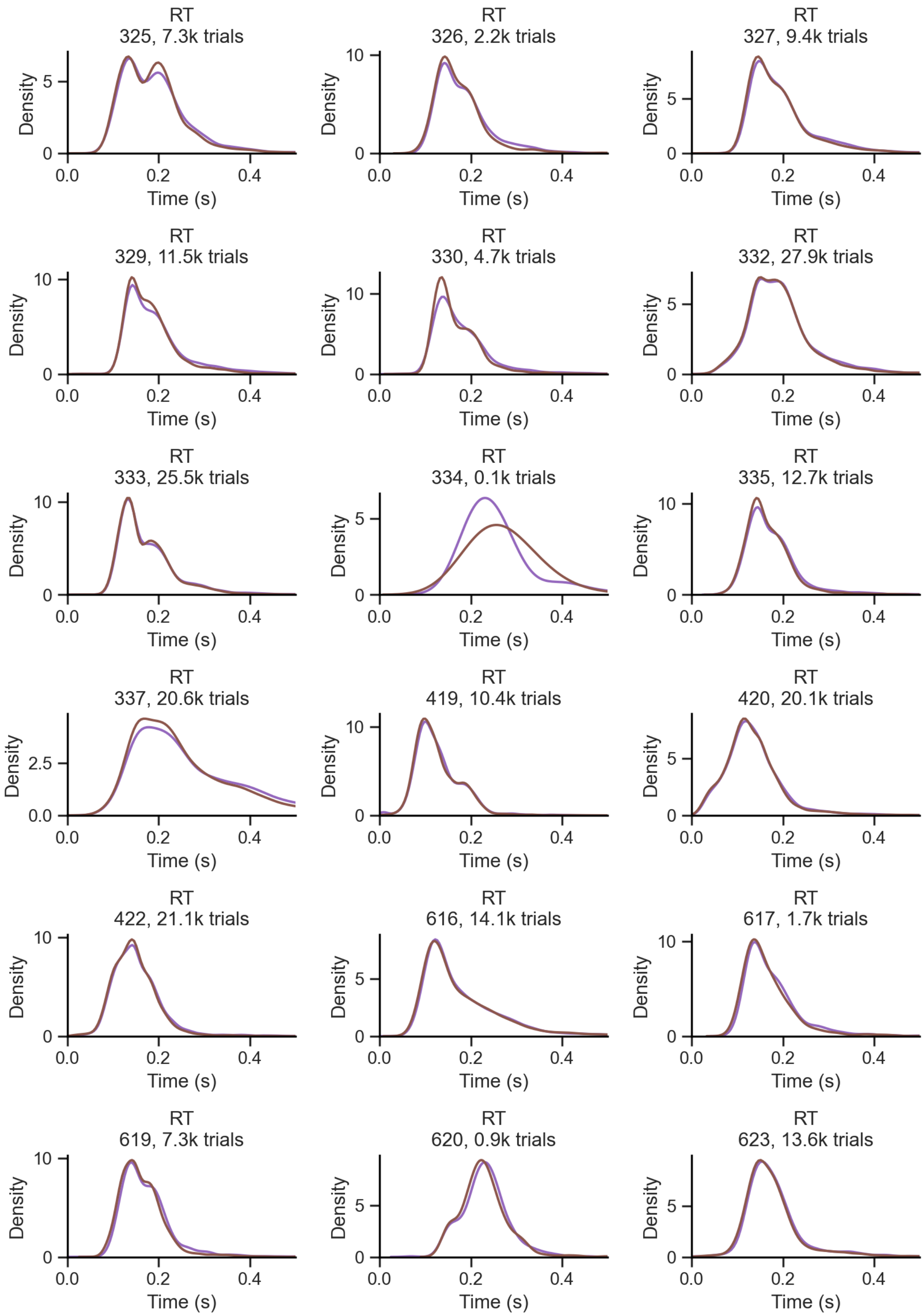
Repeat choice



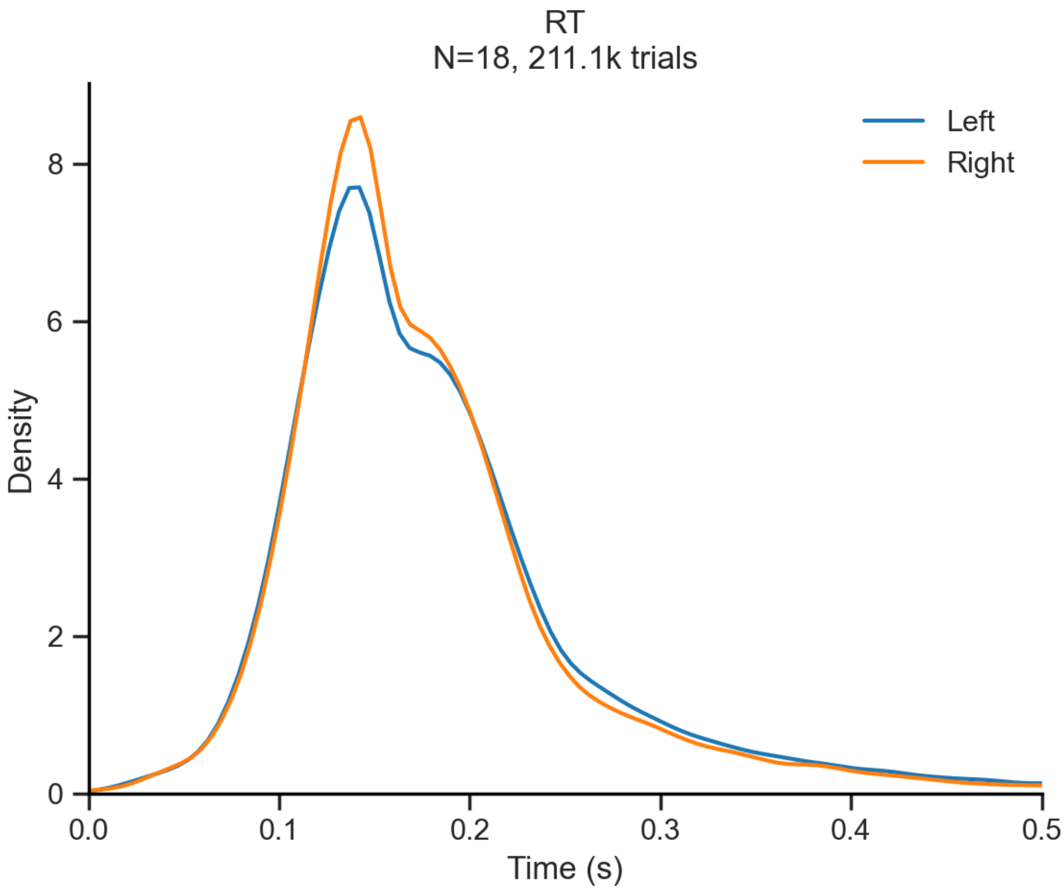

RT
325, 7.3k trials
RT
326, 2.2k trials
RT
327, 9.4k trials
RT
329, 11.5k trials
RT
330, 4.7k trials
RT
332, 27.9k trials
RT
333, 25.5k trials
RT
334, 0.1k trials
RT
335, 12.7k trials
RT
337, 20.6k trials
RT
419, 10.4k trials
RT
420, 20.1k trials
RT
422, 21.1k trials
RT
616, 14.1k trials
RT
617, 1.7k trials
RT
619, 7.3k trials
RT
620, 0.9k trials
RT
623, 13.6k trials
Density
Time (s)

Repeat trial Trials in which the stimulus (rewarded) side coincides with the previous choice

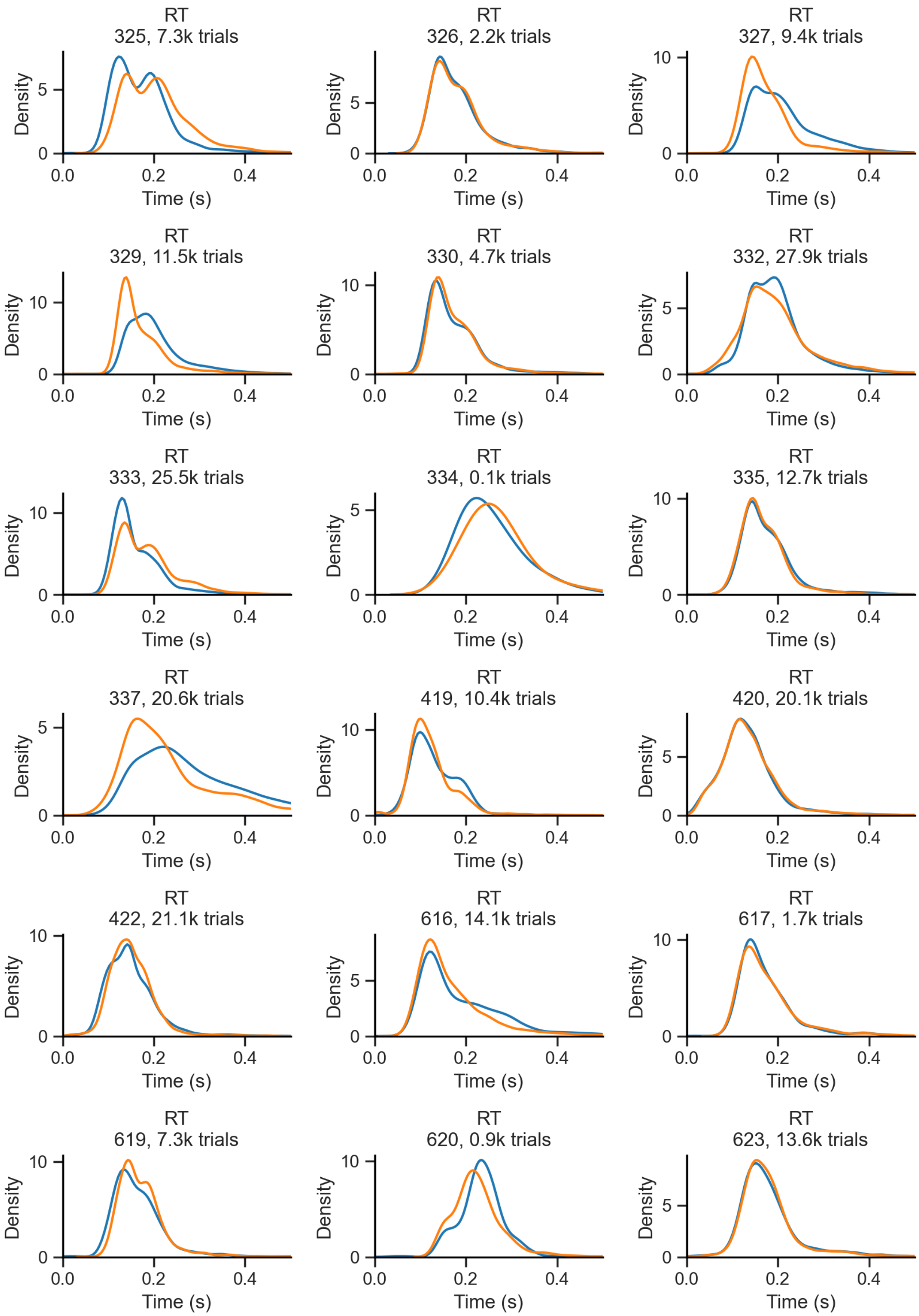



RT
325, 7.3k trials
RT
326, 2.2k trials
RT
327, 9.4k trials
RT
329, 11.5k trials
RT
330, 4.7k trials
RT
332, 27.9k trials
RT
333, 25.5k trials
RT
334, 0.1k trials
RT
335, 12.7k trials
RT
337, 20.6k trials
RT
419, 10.4k trials
RT
420, 20.1k trials
RT
422, 21.1k trials
RT
616, 14.1k trials
RT
617, 1.7k trials
RT
619, 7.3k trials
RT
620, 0.9k trials
RT
623, 13.6k trials
Density
Time (s)

Stimulus

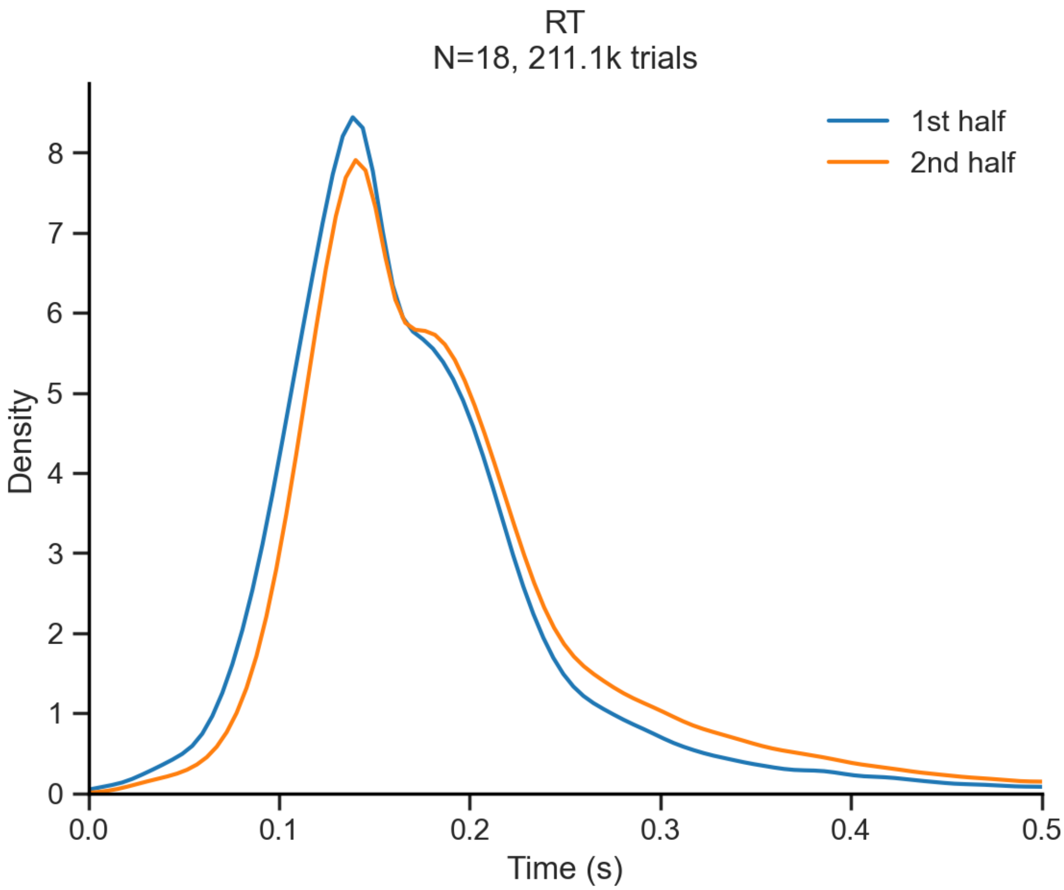


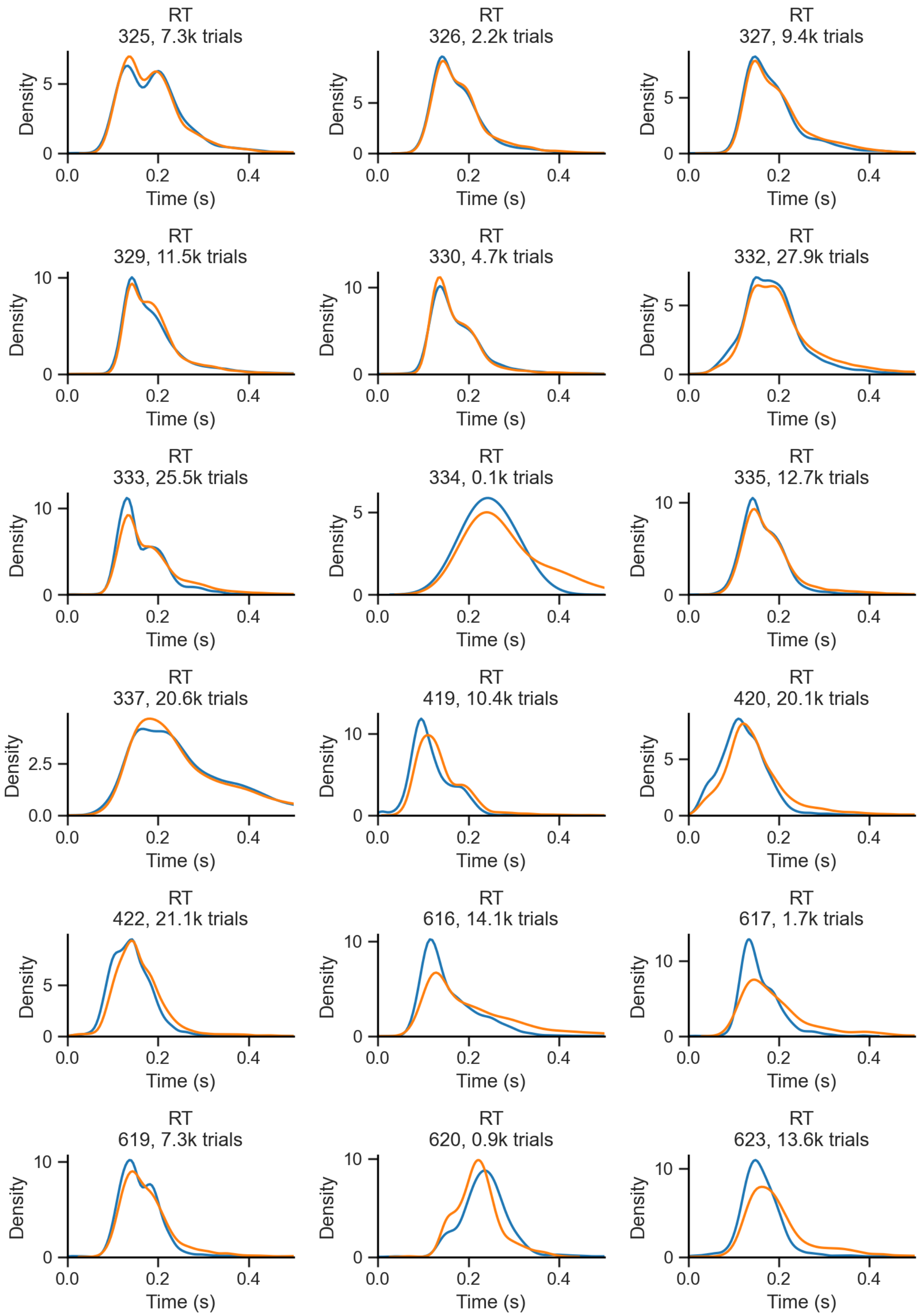
RT
N=18, 211.1k trials
Left
Right
Density
Time (s)
0
2
4
6
8
0.0
0.1
0.2
0.3
0.4
0.5


RT
325, 7.3k trials
RT
326, 2.2k trials
RT
327, 9.4k trials
RT
329, 11.5k trials
RT
330, 4.7k trials
RT
332, 27.9k trials
RT
333, 25.5k trials
RT
334, 0.1k trials
RT
335, 12.7k trials
RT
337, 20.6k trials
RT
419, 10.4k trials
RT
420, 20.1k trials
RT
422, 21.1k trials
RT
616, 14.1k trials
RT
617, 1.7k trials
RT
619, 7.3k trials
RT
620, 0.9k trials
RT
623, 13.6k trials
Density
Time (s)
0.0
0.2
0.4
0
5
10

This bimodality can not be explained by differences in behavioral variables such as outcome, choice, stimulus, repeat, or previous outcome.

1st vs 2nd session half




RT
325, 7.3k trials
RT
326, 2.2k trials
RT
327, 9.4k trials
RT
329, 11.5k trials
RT
330, 4.7k trials
RT
332, 27.9k trials
RT
333, 25.5k trials
RT
334, 0.1k trials
RT
335, 12.7k trials
RT
337, 20.6k trials
RT
419, 10.4k trials
RT
420, 20.1k trials
RT
422, 21.1k trials
RT
616, 14.1k trials
RT
617, 1.7k trials
RT
619, 7.3k trials
RT
620, 0.9k trials
RT
623, 13.6k trials
Density
Time (s)

The bimodality is present in both halves of the session, suggesting that it is not driven by a change in the internal state of the animal. The RT distribution is similar in both halves of the session, with a slight shift towards slower RTs in the second half.

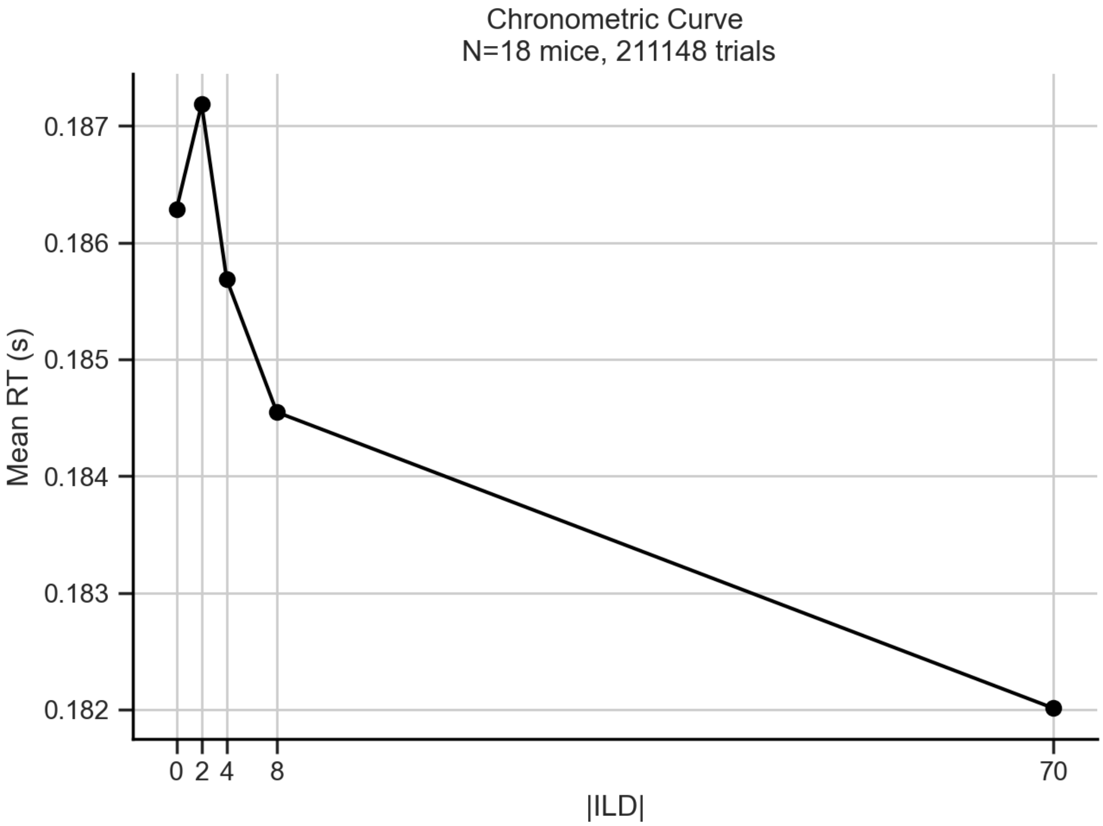
Another possibility is that the bimodality just results from the individual variability in the RTs of each subject. To examine this, we can plot the RT distribution for each subject separately. If the bimodality is driven by individual variability, we would expect to see a unimodal distribution in each subject.

Hidden Markov Model (HMM) Another possibility is that the bimodality reflects the a change in the internal state of the animal (e.g. motivation, reduced by satiation as time passes) that is not captured by the behavioral variables. To examine this, we can split the data of each session into first and second half and plot the RT distribution for each half. If the bimodality is driven by a change in the internal state of the animal, we would expect to see a difference in the RT distribution between the first and second half of the session.

5.2 Chronometric Curves

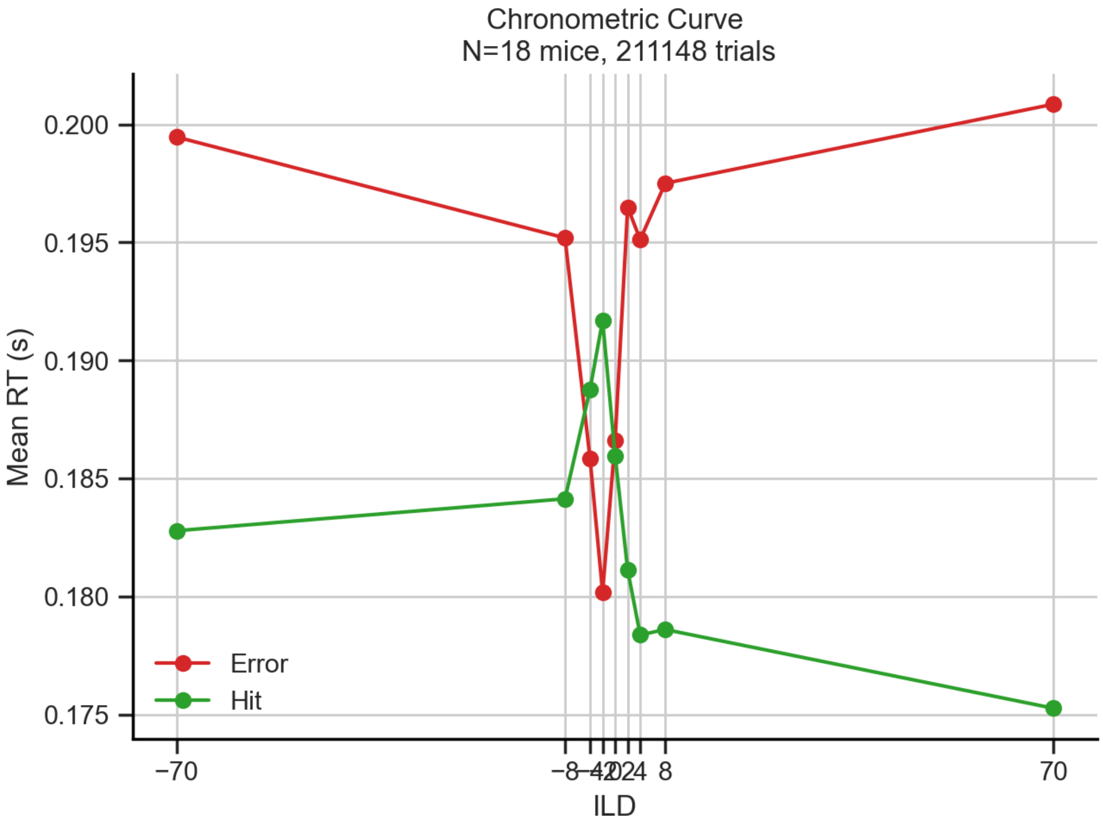
5.2.1 All trials

```
   absILD        RT
0     0.0  0.186289
1     2.0  0.187186
2     4.0  0.185685
3     8.0  0.184547
4    70.0  0.182010
```

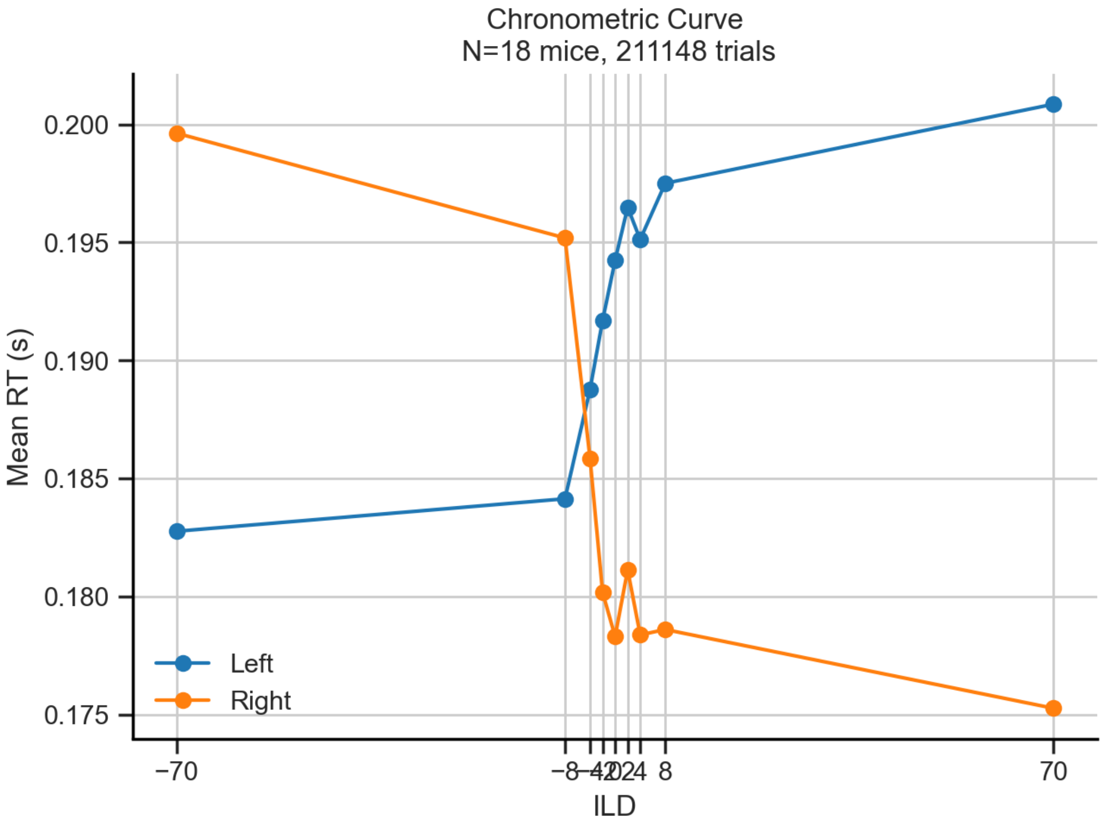


5.2.2 Splits

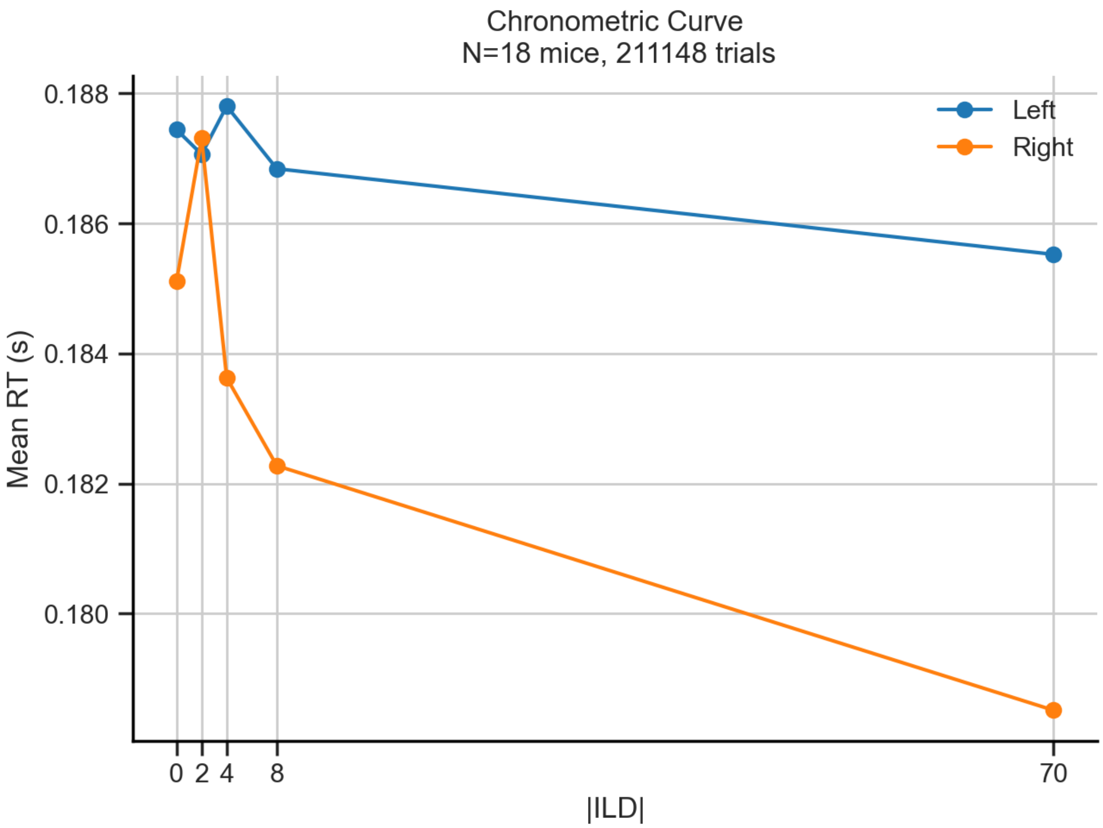
Outcome


Chronometric Curve
N=18 mice, 211148 trials
Mean RT (s)
0.200
0.195
0.190
0.185
0.180
0.175
Error
Hit
−70
−8
−4
−2
0
2
4
8
70
ILD

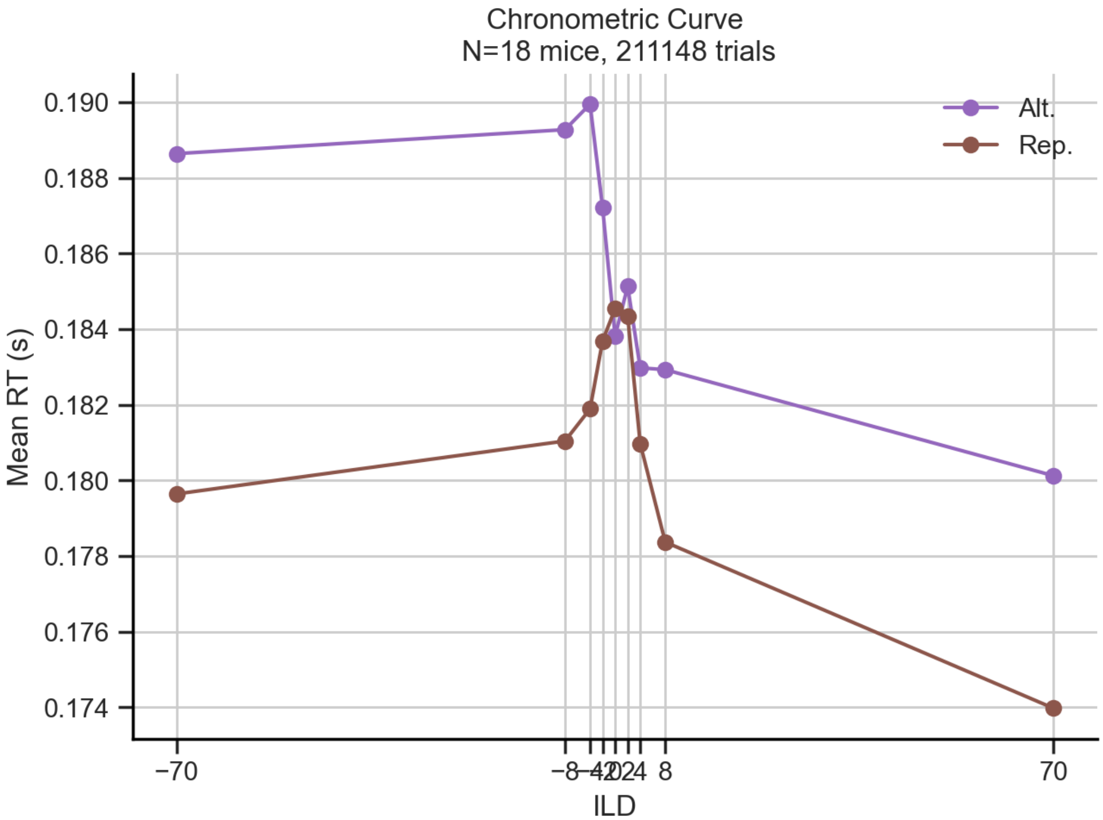
Choice


Chronometric Curve
N=18 mice, 211148 trials
Mean RT (s)
0.200
0.195
0.190
0.185
0.180
0.175
Left
Right
−70
−8
−4
0
2
4
8
70
ILD

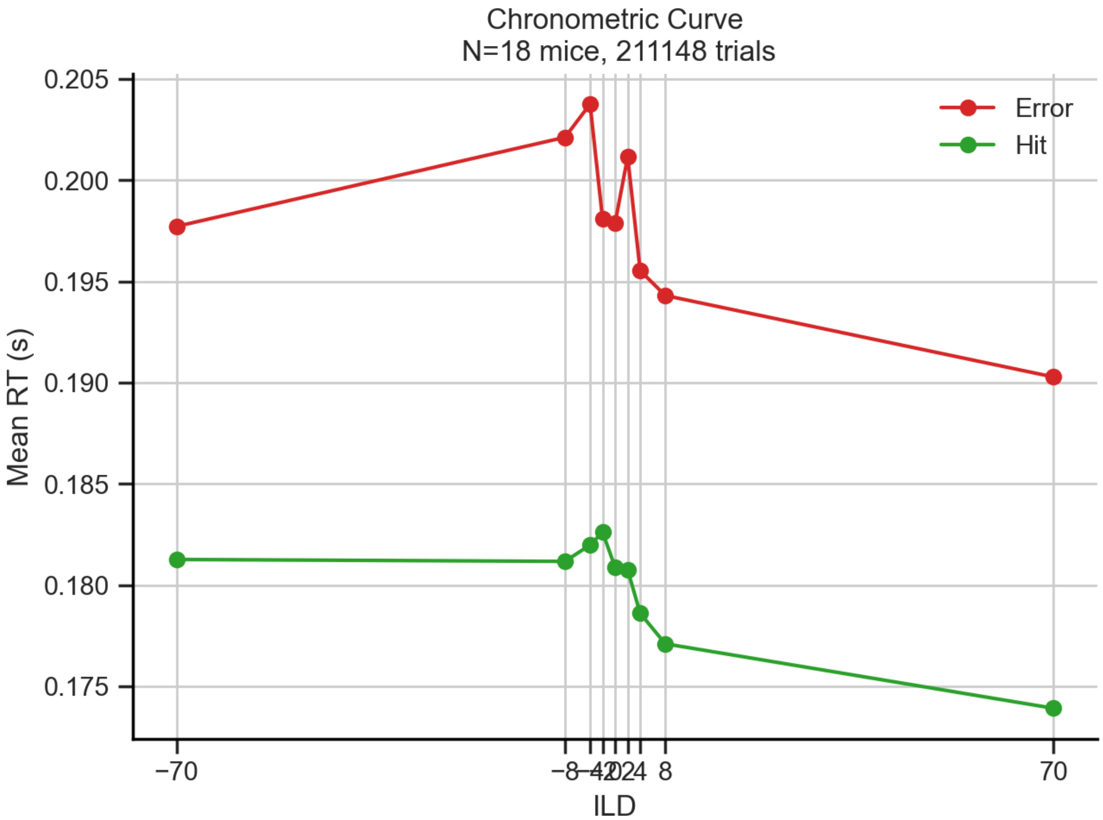
Stimulus


Chronometric Curve
N=18 mice, 211148 trials
Left
Right
Mean RT (s)
0.188
0.186
0.184
0.182
0.180
0
2
4
8
70
|ILD|

Repeat



Previous outcome



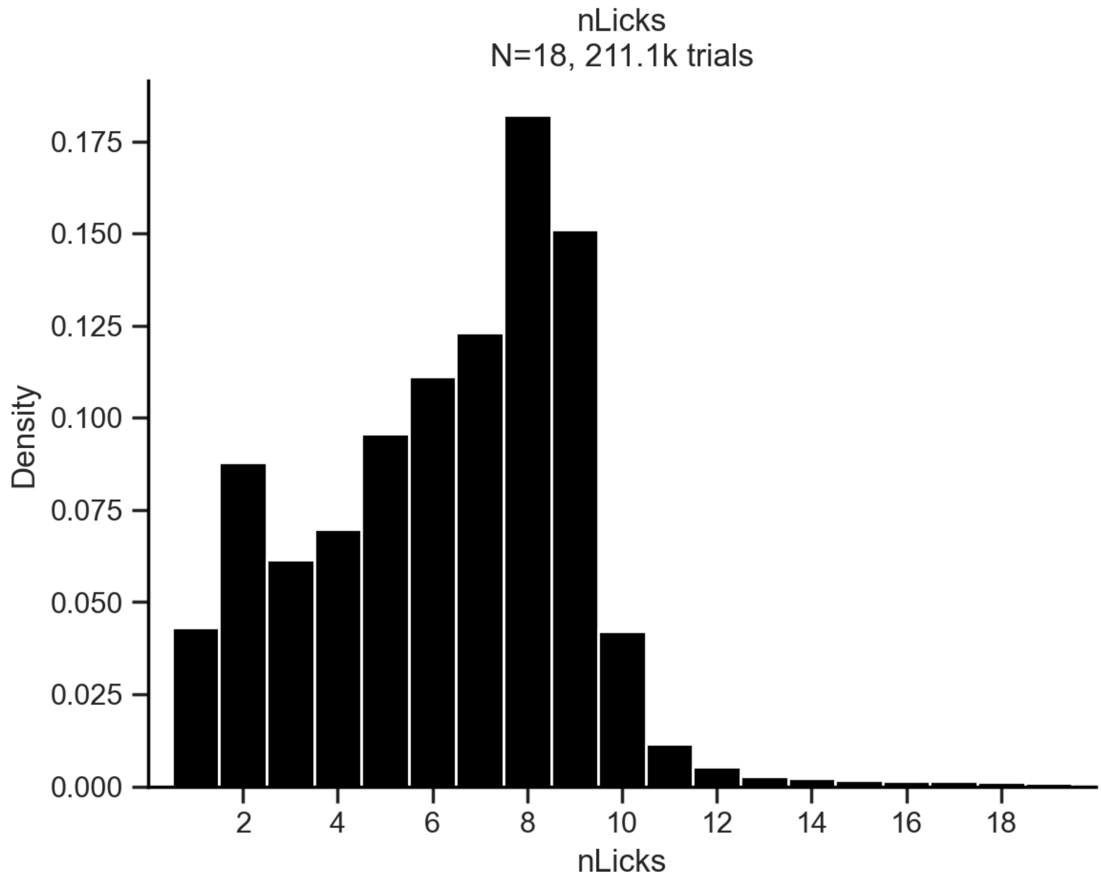
6 Lick Rate

As the number of licks depends on the time the lickport was available (motor in), which ultimately depends on the outcome (correct vs error), a time window to count the licks needs to be established. As the time to collect water in correct (rewarded) trials was always 1 s, and the timeout period following error trials varied from 2-5 s, the chosen time window to count the number of licks was 1 s. This is very convenient for the lick rate, as it is defined as

$$\text{Lick rate} = \frac{N_{\text{licks}}}{\text{time (s)}}$$

so that N licks = Lick Rate

6.1 All trials



We can observe a first peak at 2 licks (error trials), and an sawtooth increasing second peak with max at 8 licks (correct trials). Let's split the data into several variables (like with RT) to observe how they modulate the lick rate.

As there was no water (reward) coming out of the spouts in error trials, it makes sense that the distribution of lick rate peaks at 0. It is a needed control, but from now on the analysis will focus on **correct trials only**

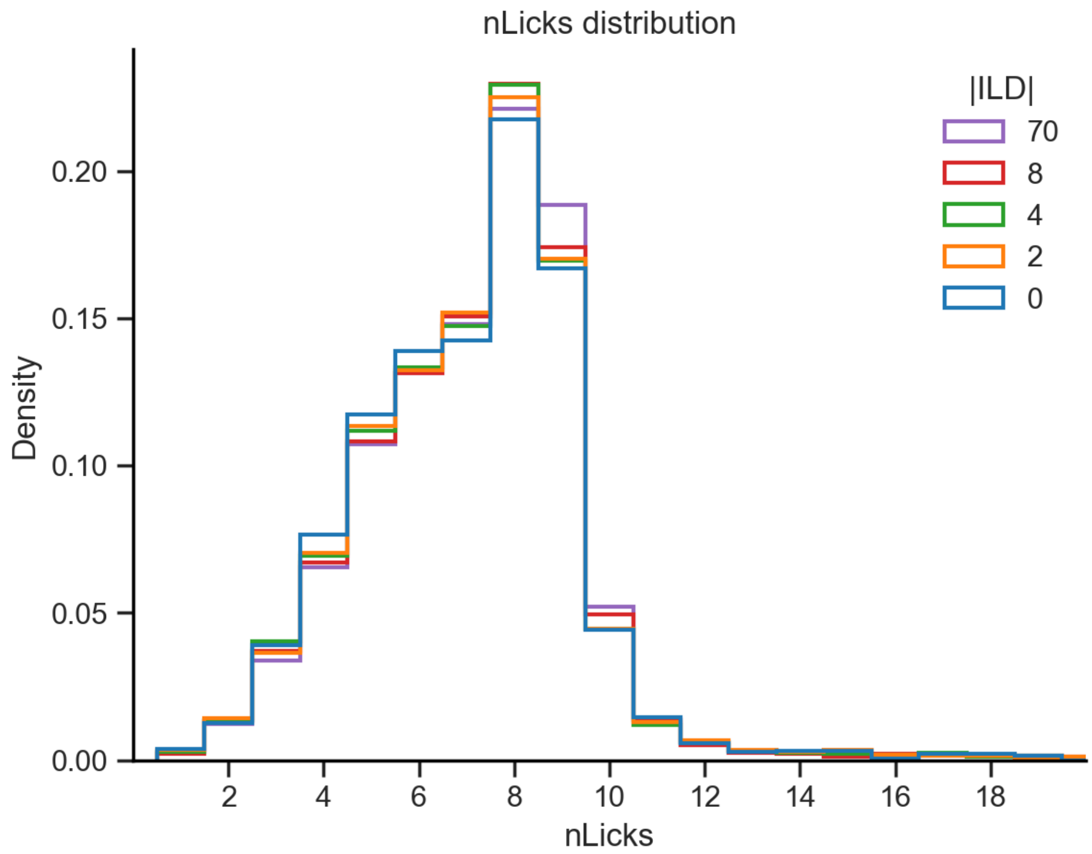
6.2 Splits

6.2.1 ILD

```
Mean nLicks = nan s
```

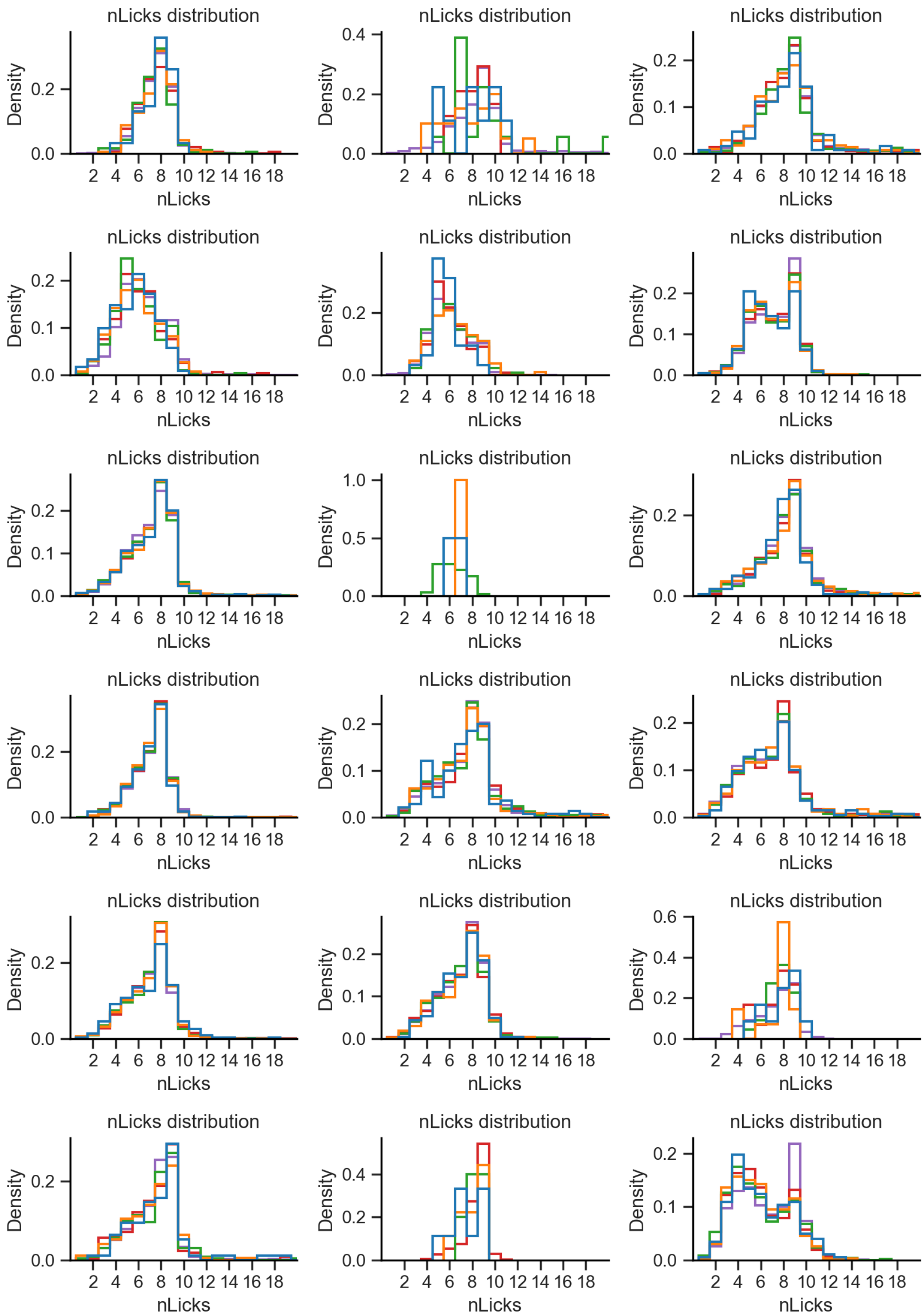
```
C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\numpy\core\fromnumeric.py:3504: RuntimeWarning: Mean of empty slice.
  return _methods._mean(a, axis=axis, dtype=dtype,
C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\numpy\core\_methods.py:129: RuntimeWarning: invalid value encountered
```

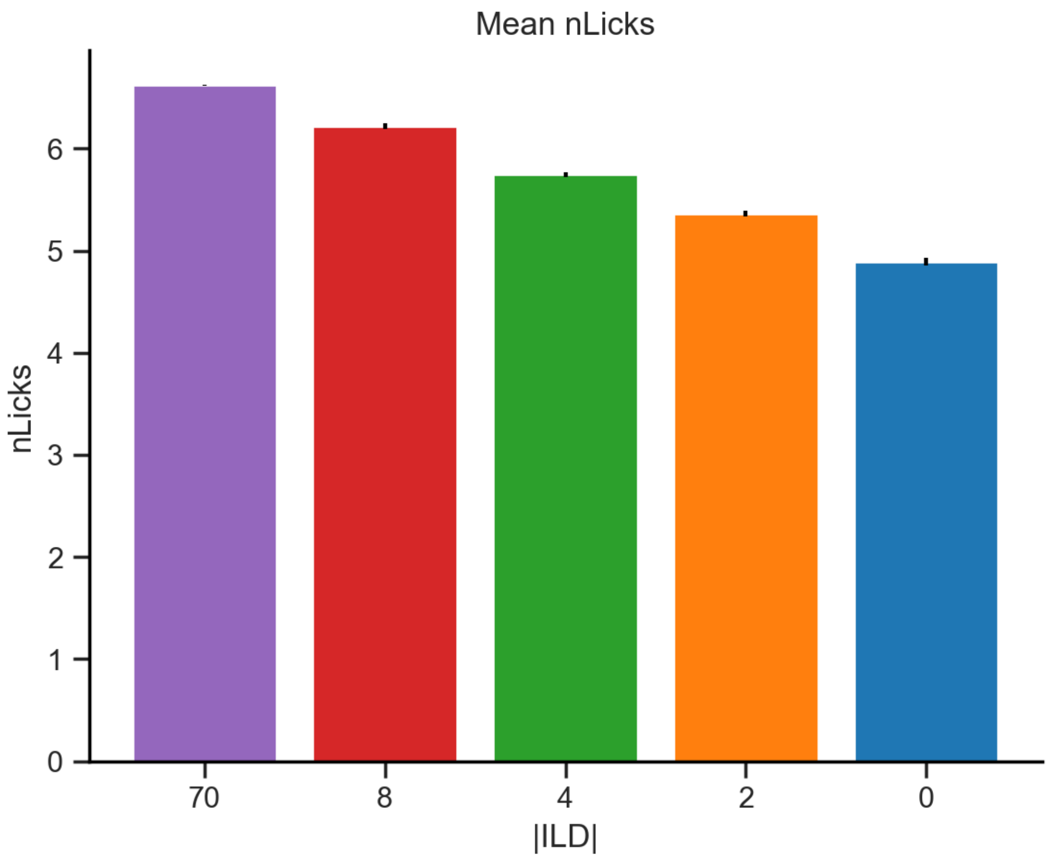
```
in scalar divide
  ret = ret.dtype.type(ret / rcount)
```




```
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
```

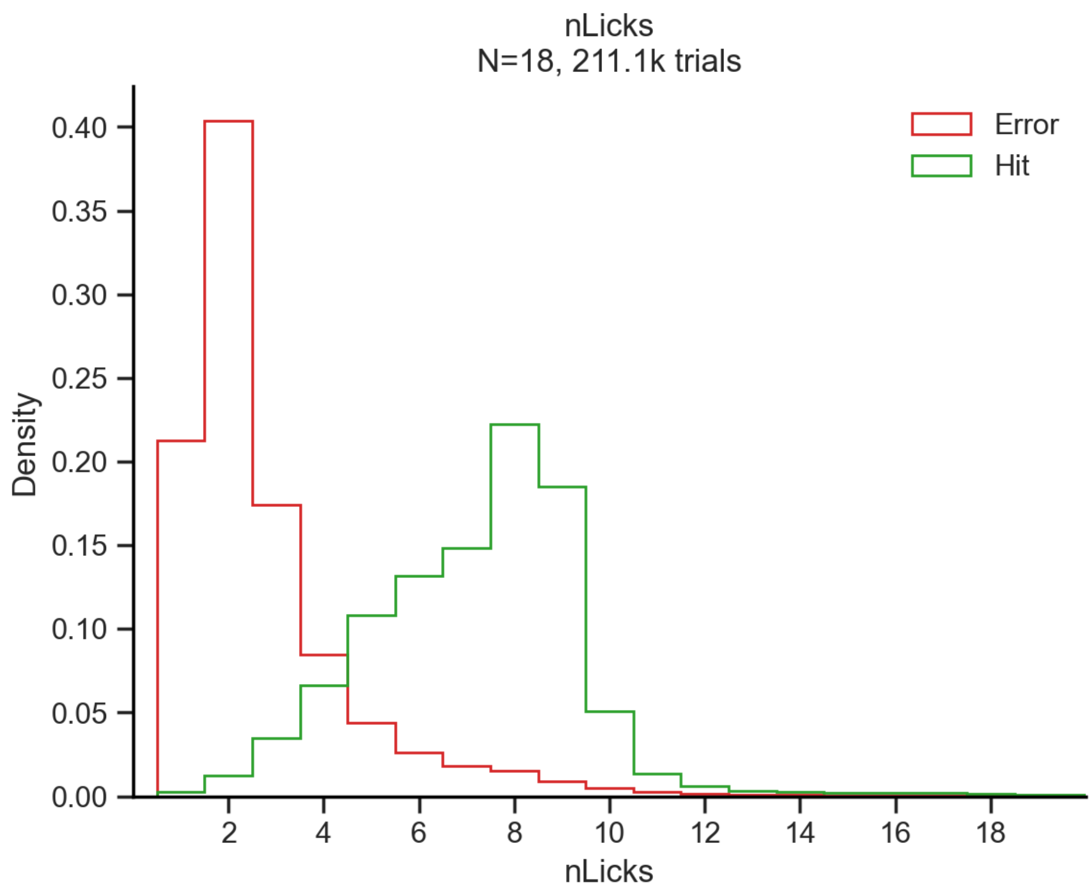
```
Mean nLicks = nan s
```



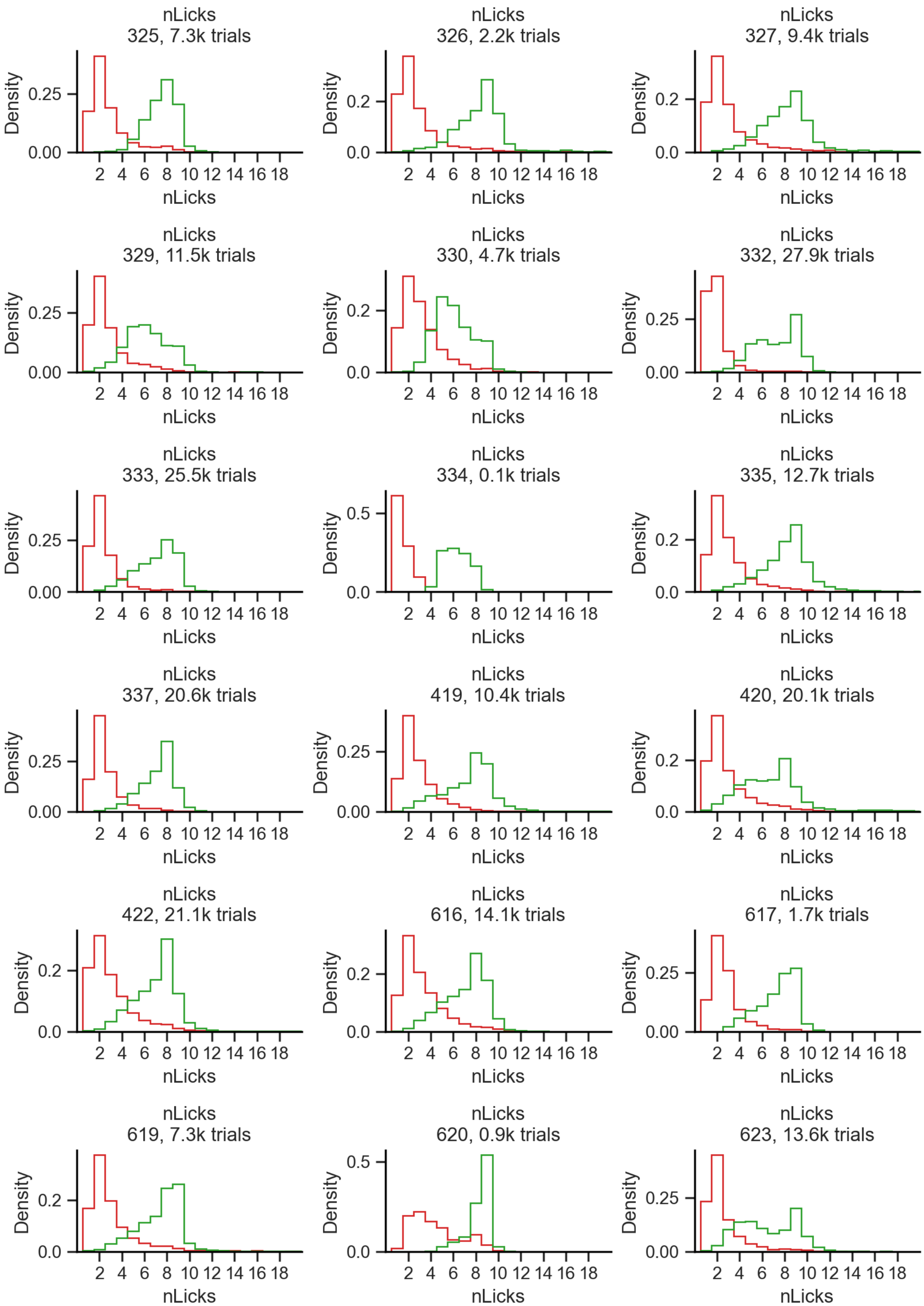


There is a modulation in the lick rate by ILD in correct trials. The higher the ILD the higher the lick rate.

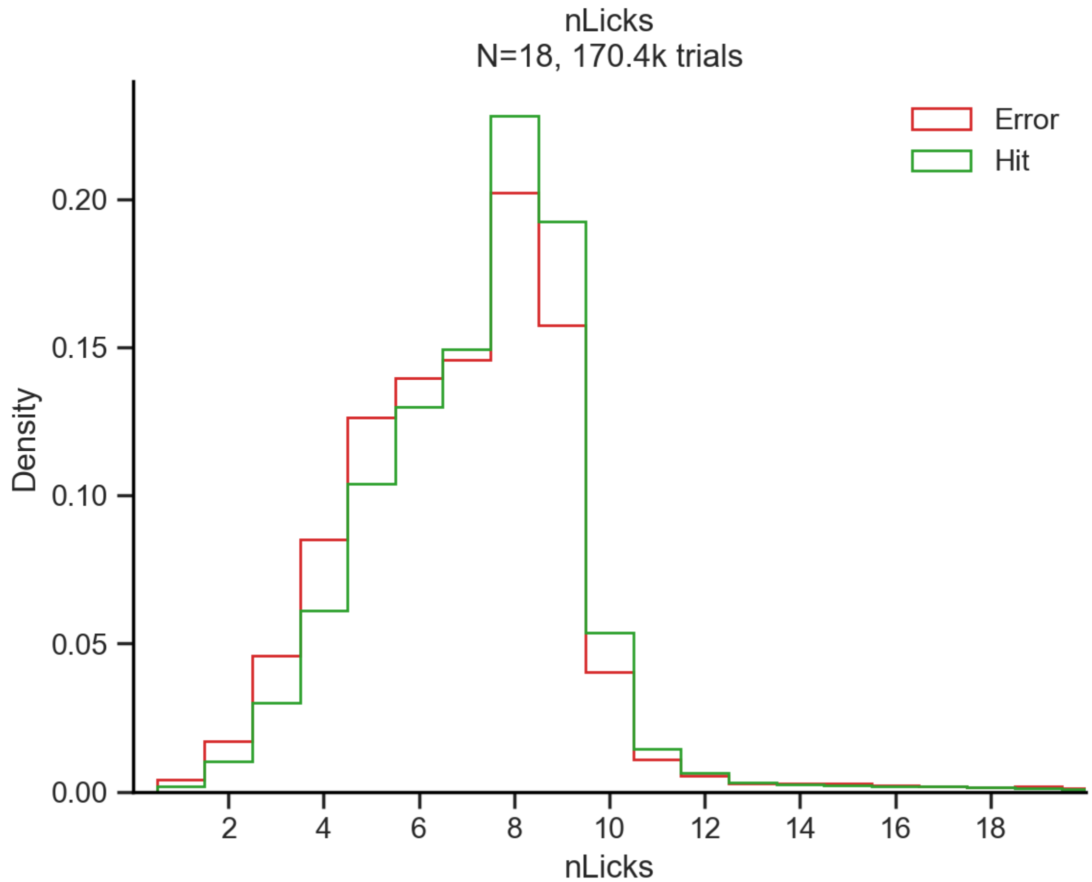
6.2.2 Outcome

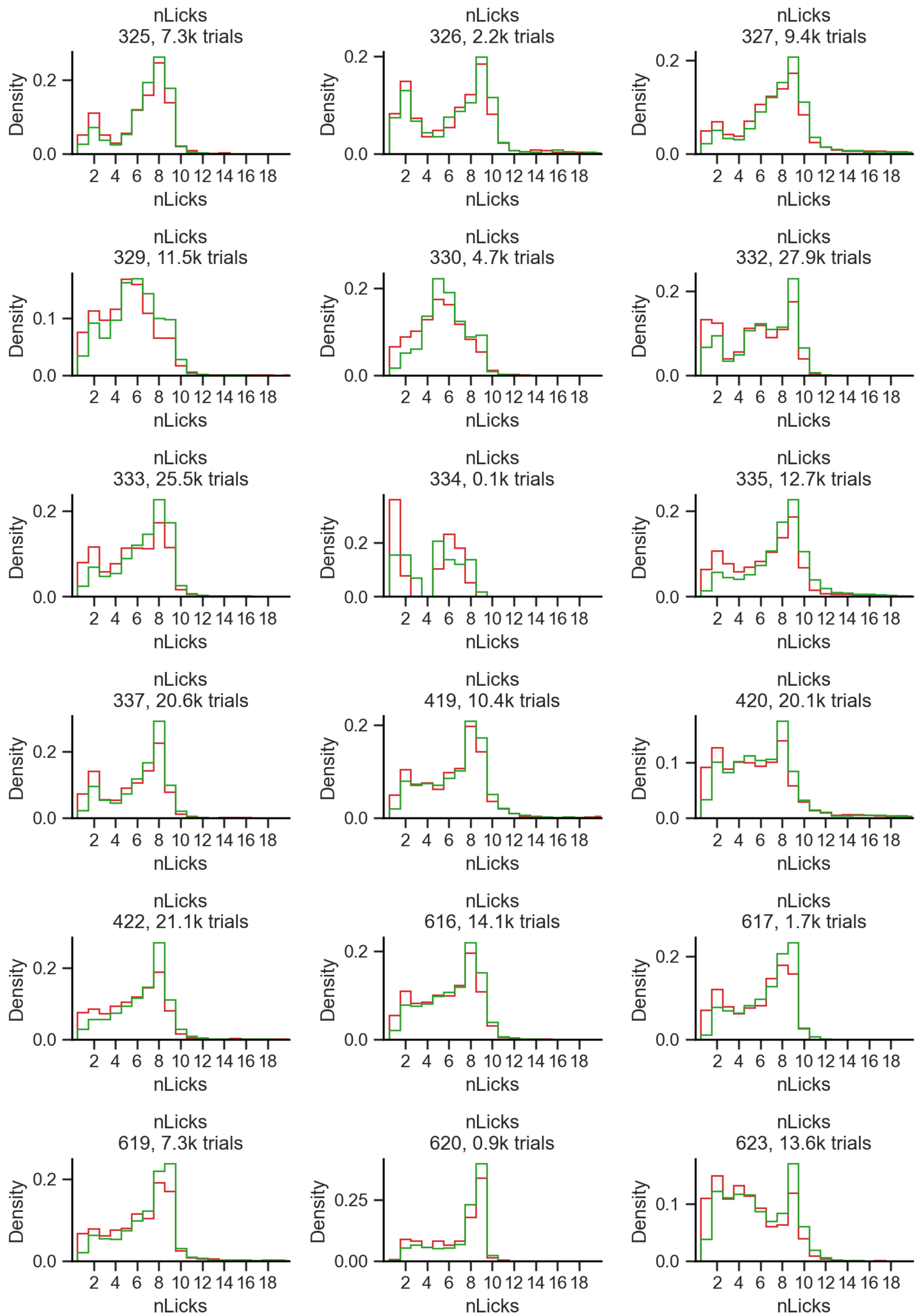


nLicks
N=18, 211.1k trials
Error
Hit
Density
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
2
4
6
8
10
12
14
16
18
nLicks

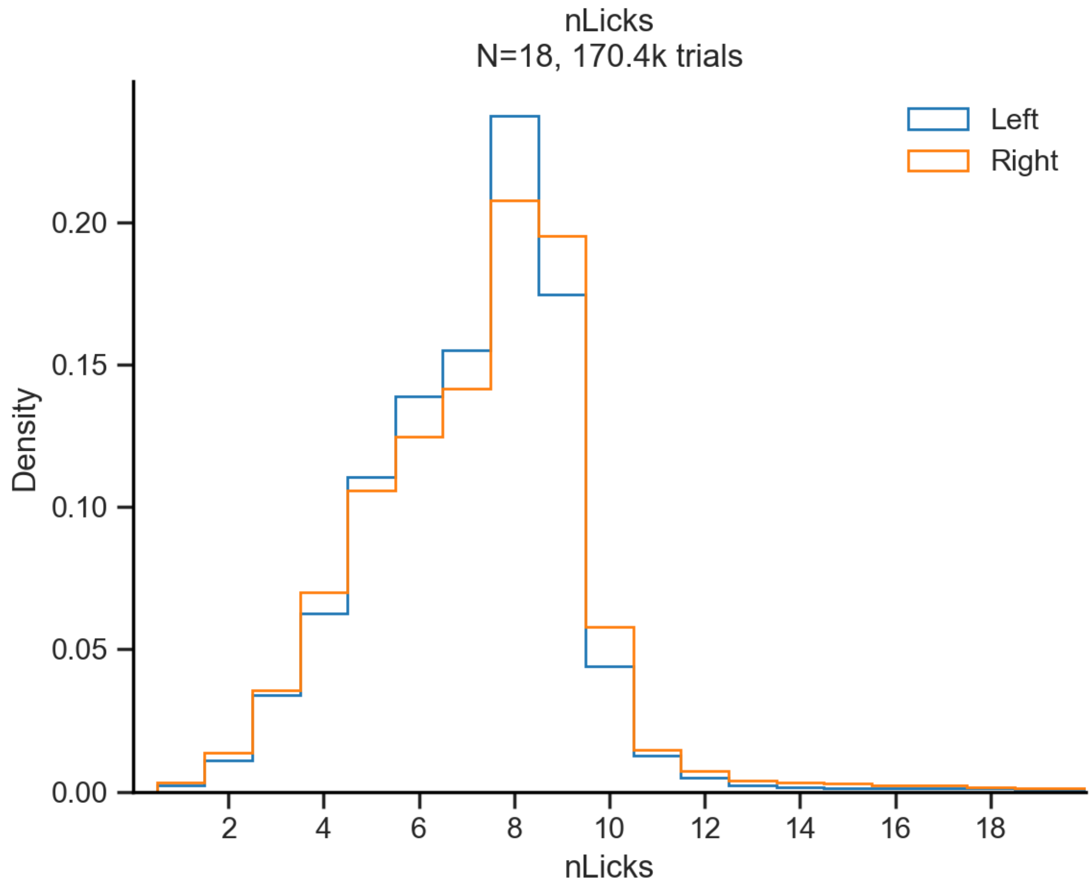

nLicks
325, 7.3k trials
nLicks
326, 2.2k trials
nLicks
327, 9.4k trials
nLicks
329, 11.5k trials
nLicks
330, 4.7k trials
nLicks
332, 27.9k trials
nLicks
333, 25.5k trials
nLicks
334, 0.1k trials
nLicks
335, 12.7k trials
nLicks
337, 20.6k trials
nLicks
419, 10.4k trials
nLicks
420, 20.1k trials
nLicks
422, 21.1k trials
nLicks
616, 14.1k trials
nLicks
617, 1.7k trials
nLicks
619, 7.3k trials
nLicks
620, 0.9k trials
nLicks
623, 13.6k trials
Density
nLicks
2 4 6 8 10 12 14 16 18

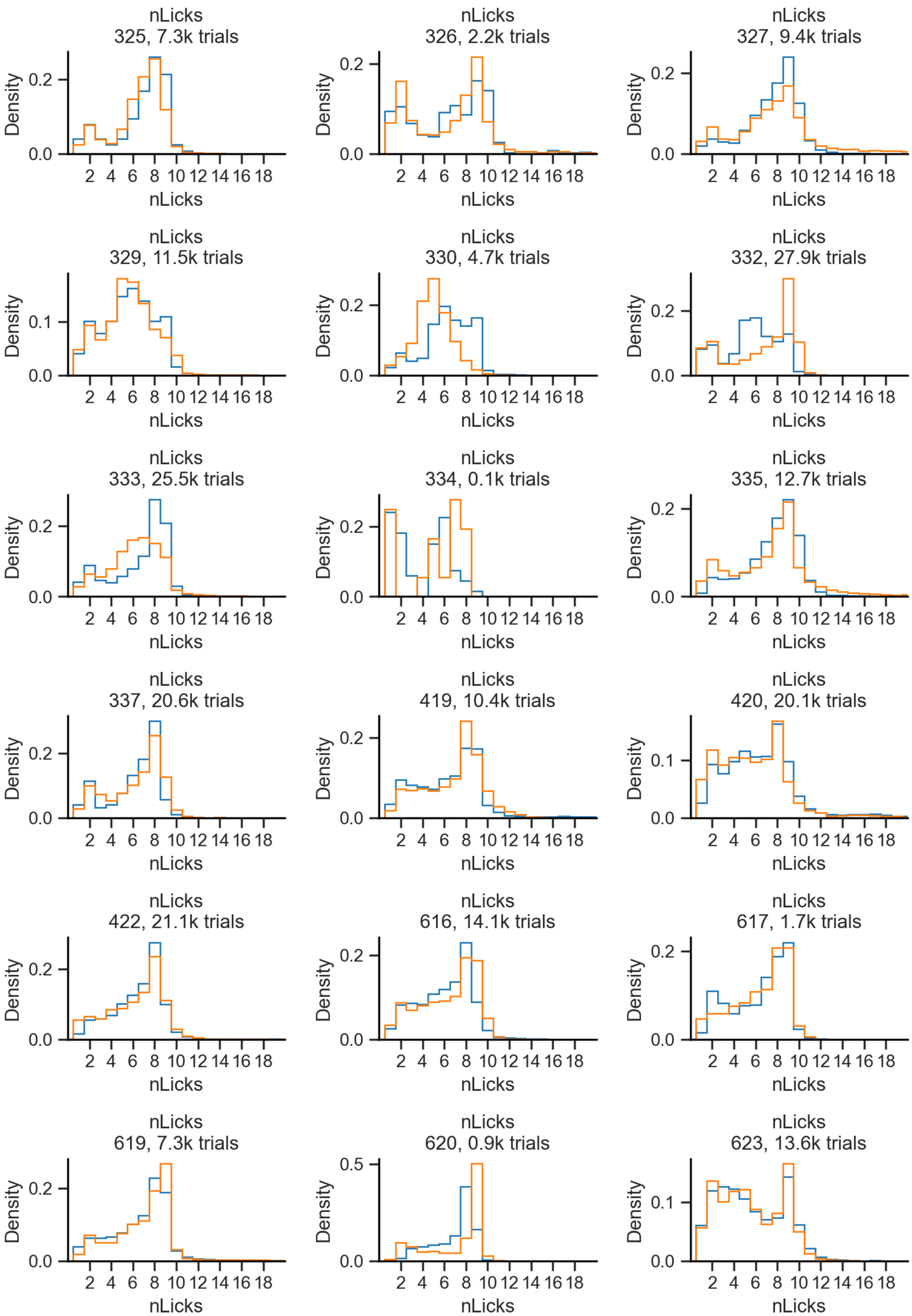
6.2.3 Previous outcome



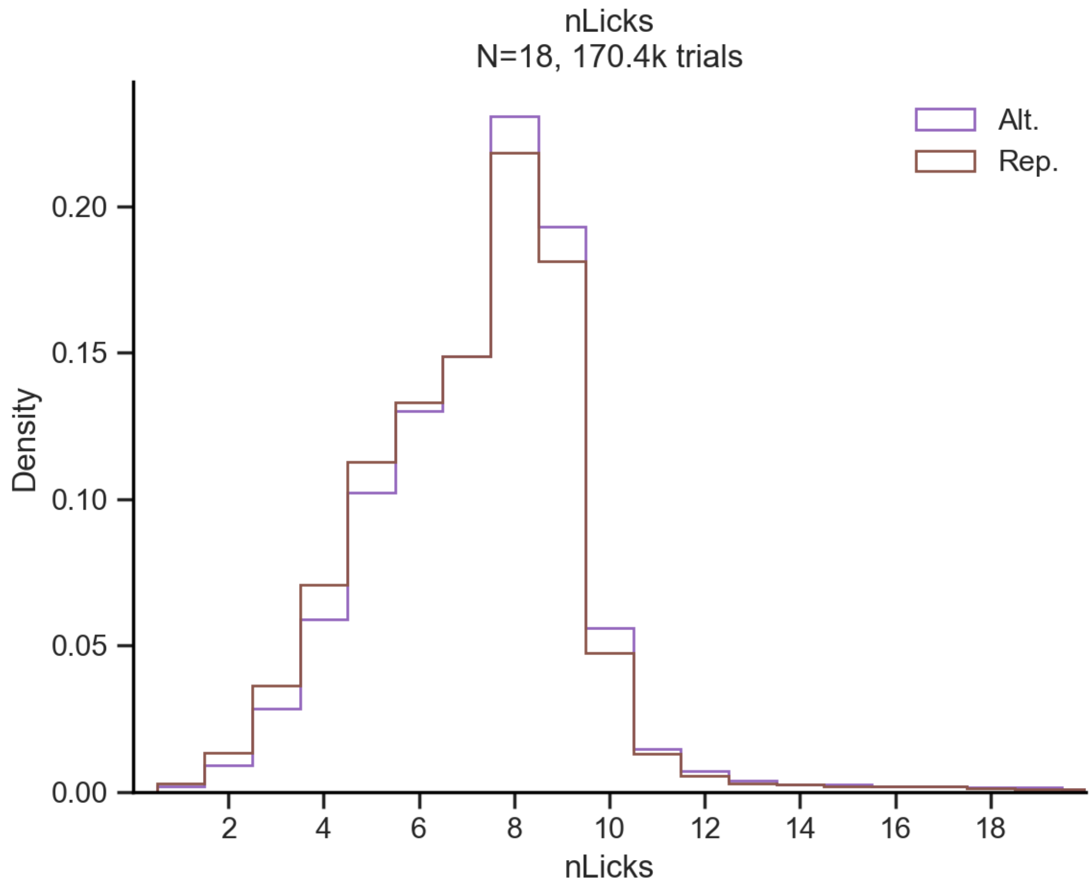

nLicks
325, 7.3k trials
nLicks
326, 2.2k trials
nLicks
327, 9.4k trials
nLicks
329, 11.5k trials
nLicks
330, 4.7k trials
nLicks
332, 27.9k trials
nLicks
333, 25.5k trials
nLicks
334, 0.1k trials
nLicks
335, 12.7k trials
nLicks
337, 20.6k trials
nLicks
419, 10.4k trials
nLicks
420, 20.1k trials
nLicks
422, 21.1k trials
nLicks
616, 14.1k trials
nLicks
617, 1.7k trials
nLicks
619, 7.3k trials
nLicks
620, 0.9k trials
nLicks
623, 13.6k trials
Density
nLicks
2 4 6 8 10 12 14 16 18

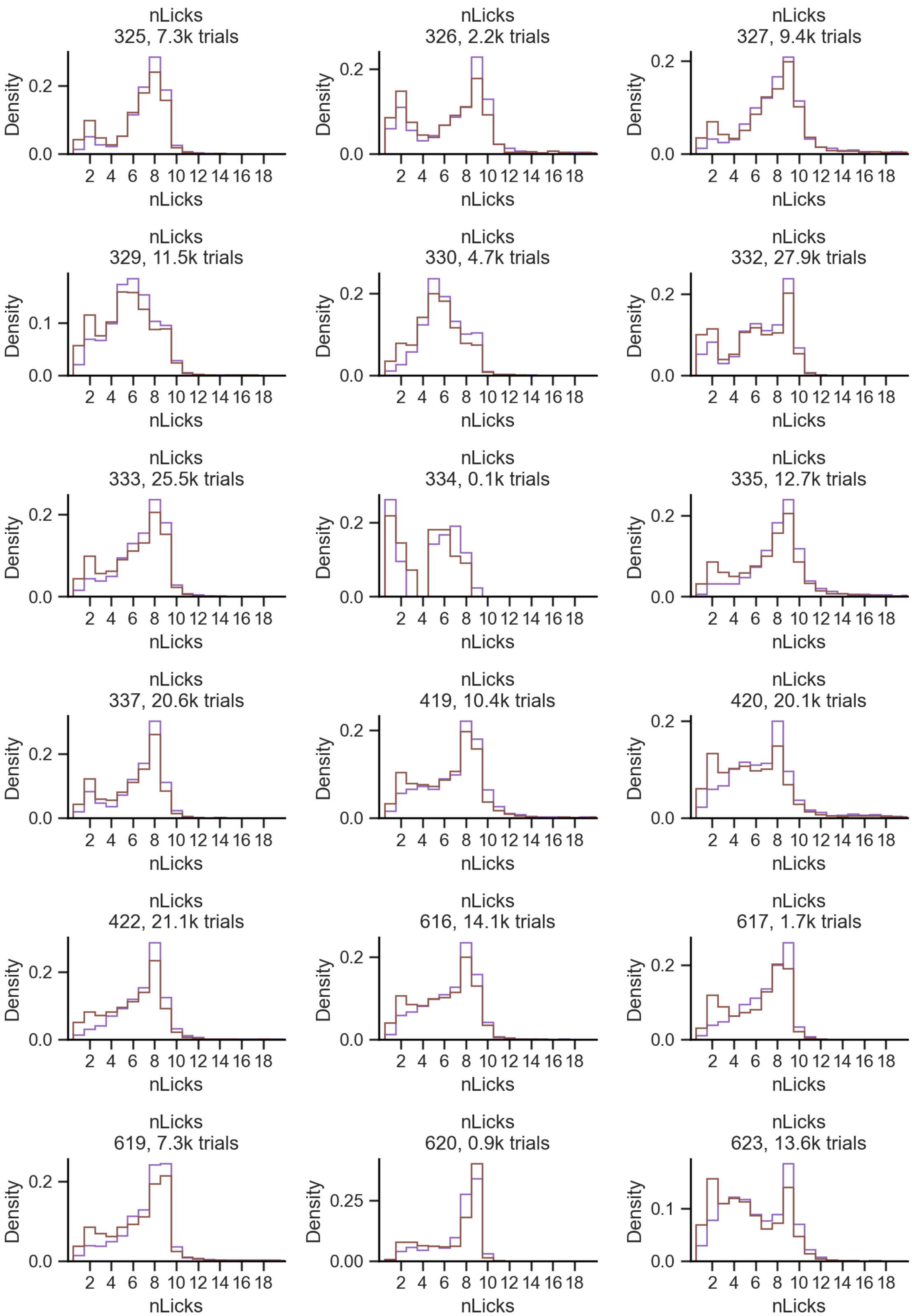
6.2.4 Choice


nLicks
N=18, 170.4k trials
Left
Right
Density
0.00
0.05
0.10
0.15
0.20
2
4
6
8
10
12
14
16
18
nLicks

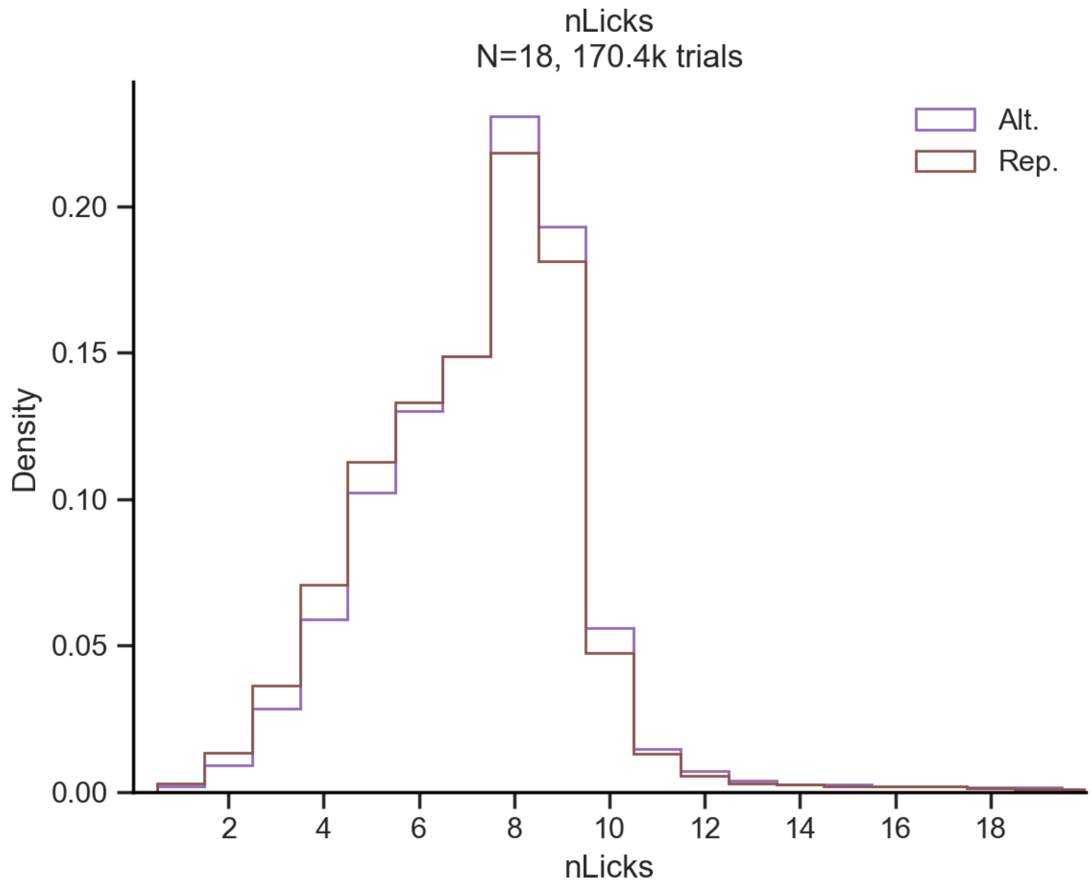

nLicks
325, 7.3k trials
nLicks
326, 2.2k trials
nLicks
327, 9.4k trials
nLicks
329, 11.5k trials
nLicks
330, 4.7k trials
nLicks
332, 27.9k trials
nLicks
333, 25.5k trials
nLicks
334, 0.1k trials
nLicks
335, 12.7k trials
nLicks
337, 20.6k trials
nLicks
419, 10.4k trials
nLicks
420, 20.1k trials
nLicks
422, 21.1k trials
nLicks
616, 14.1k trials
nLicks
617, 1.7k trials
nLicks
619, 7.3k trials
nLicks
620, 0.9k trials
nLicks
623, 13.6k trials
Density
nLicks

6.2.5 Repeat choice

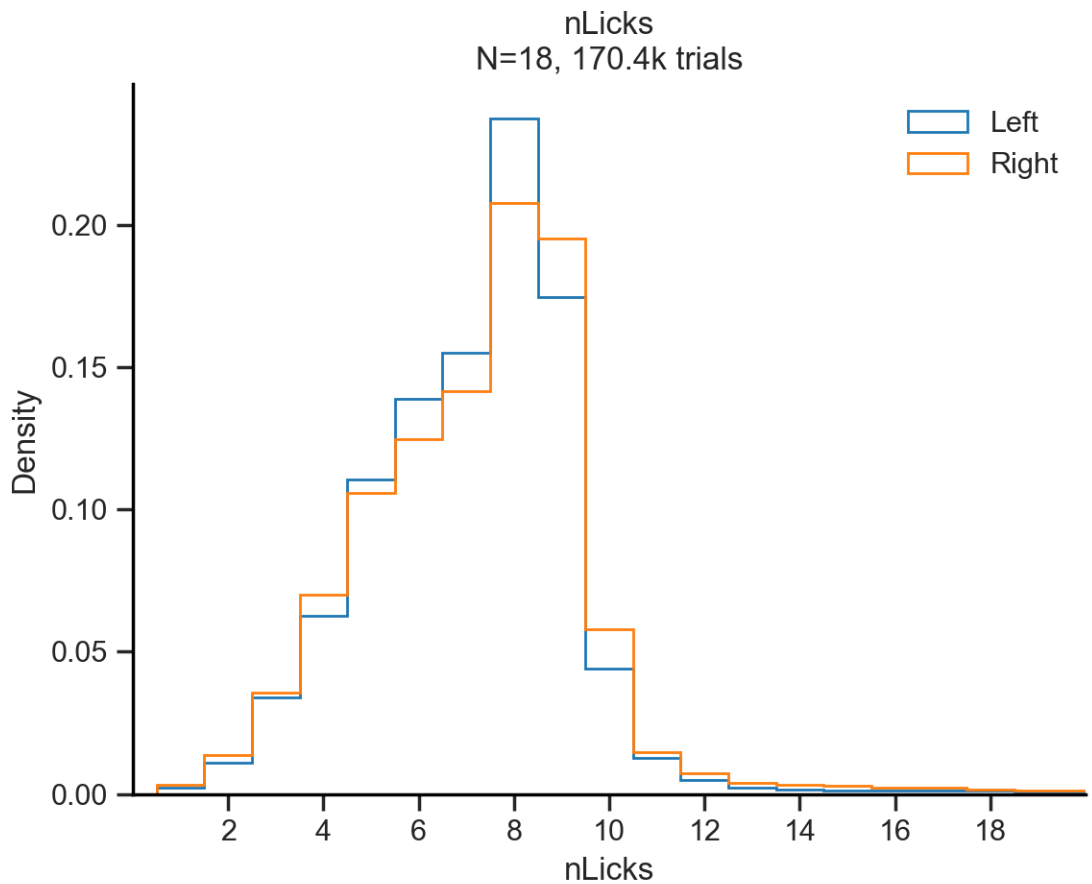


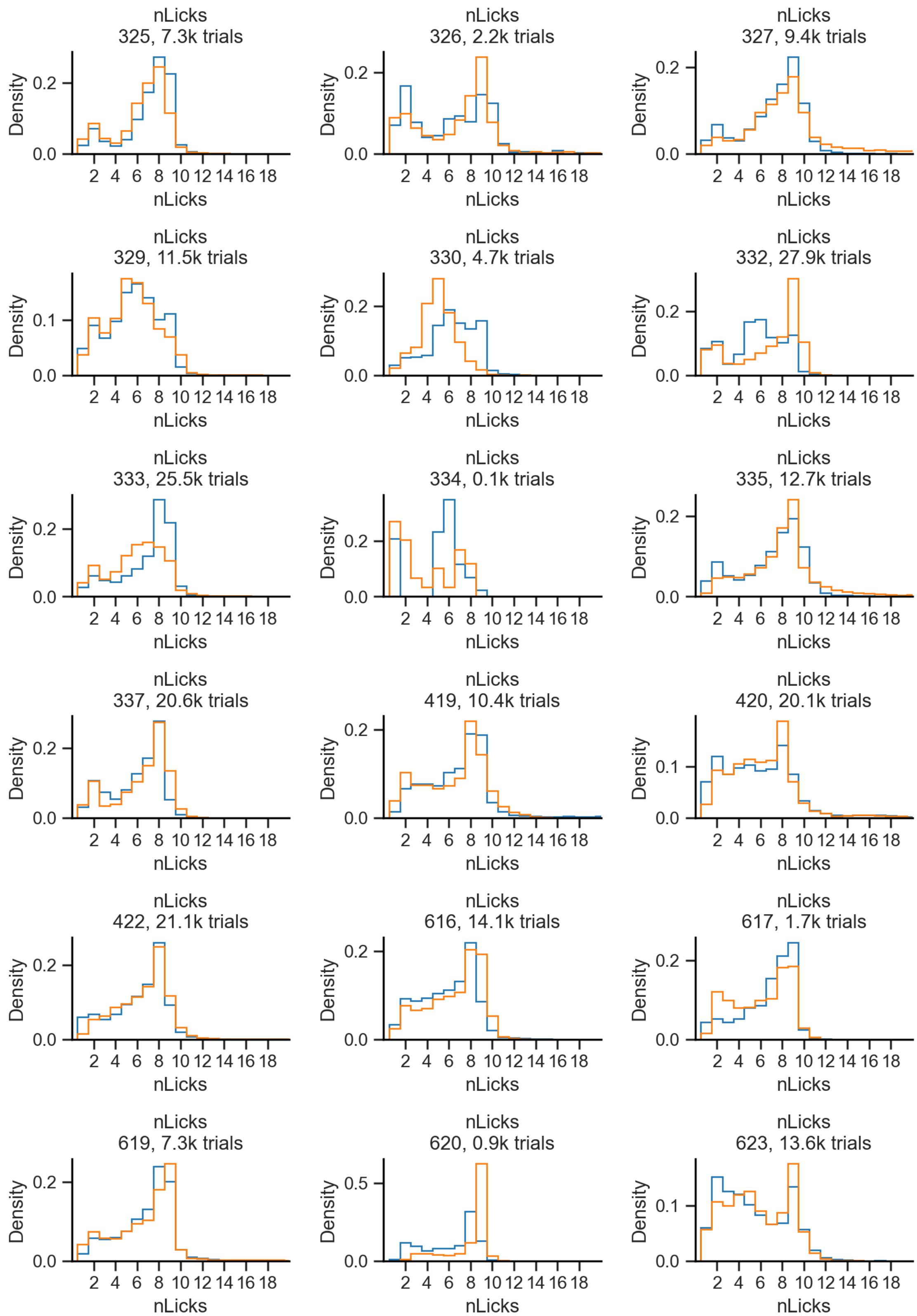

nLicks
325, 7.3k trials
nLicks
326, 2.2k trials
nLicks
327, 9.4k trials
nLicks
329, 11.5k trials
nLicks
330, 4.7k trials
nLicks
332, 27.9k trials
nLicks
333, 25.5k trials
nLicks
334, 0.1k trials
nLicks
335, 12.7k trials
nLicks
337, 20.6k trials
nLicks
419, 10.4k trials
nLicks
420, 20.1k trials
nLicks
422, 21.1k trials
nLicks
616, 14.1k trials
nLicks
617, 1.7k trials
nLicks
619, 7.3k trials
nLicks
620, 0.9k trials
nLicks
623, 13.6k trials
Density
nLicks
2 4 6 8 10 12 14 16 18

6.2.6 Repeat trial

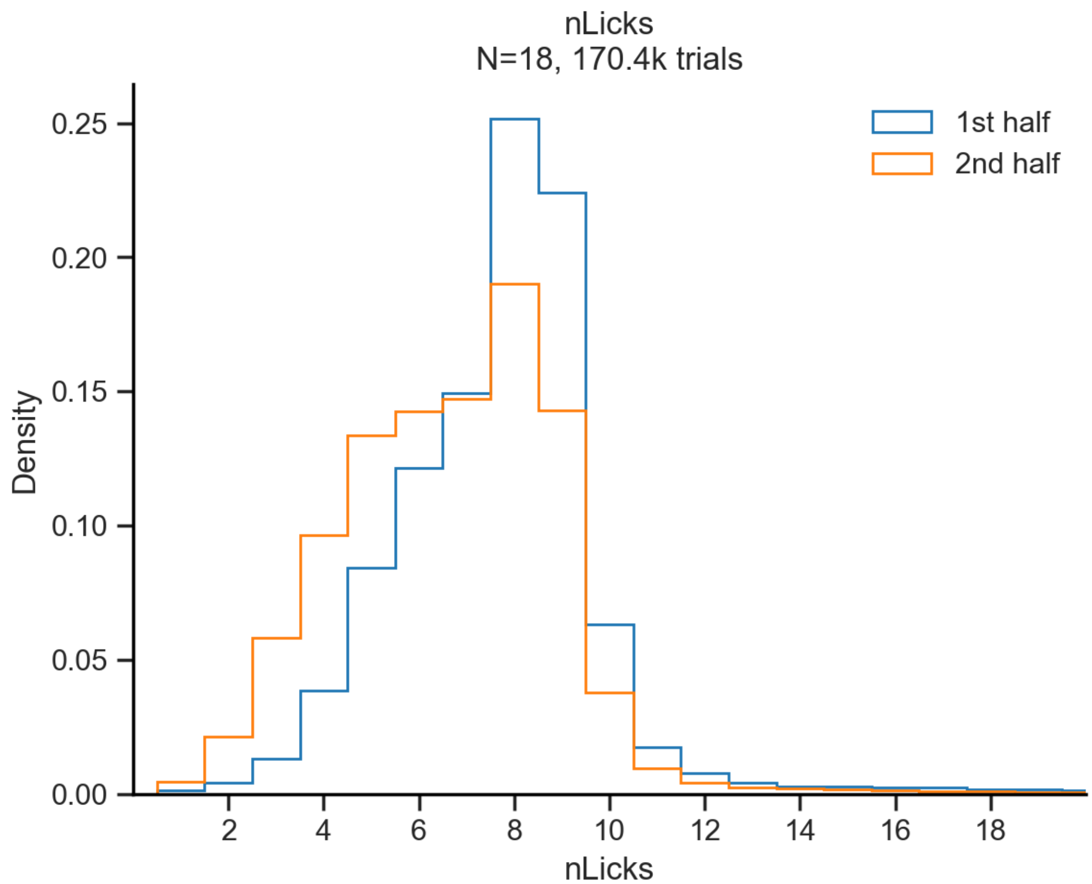


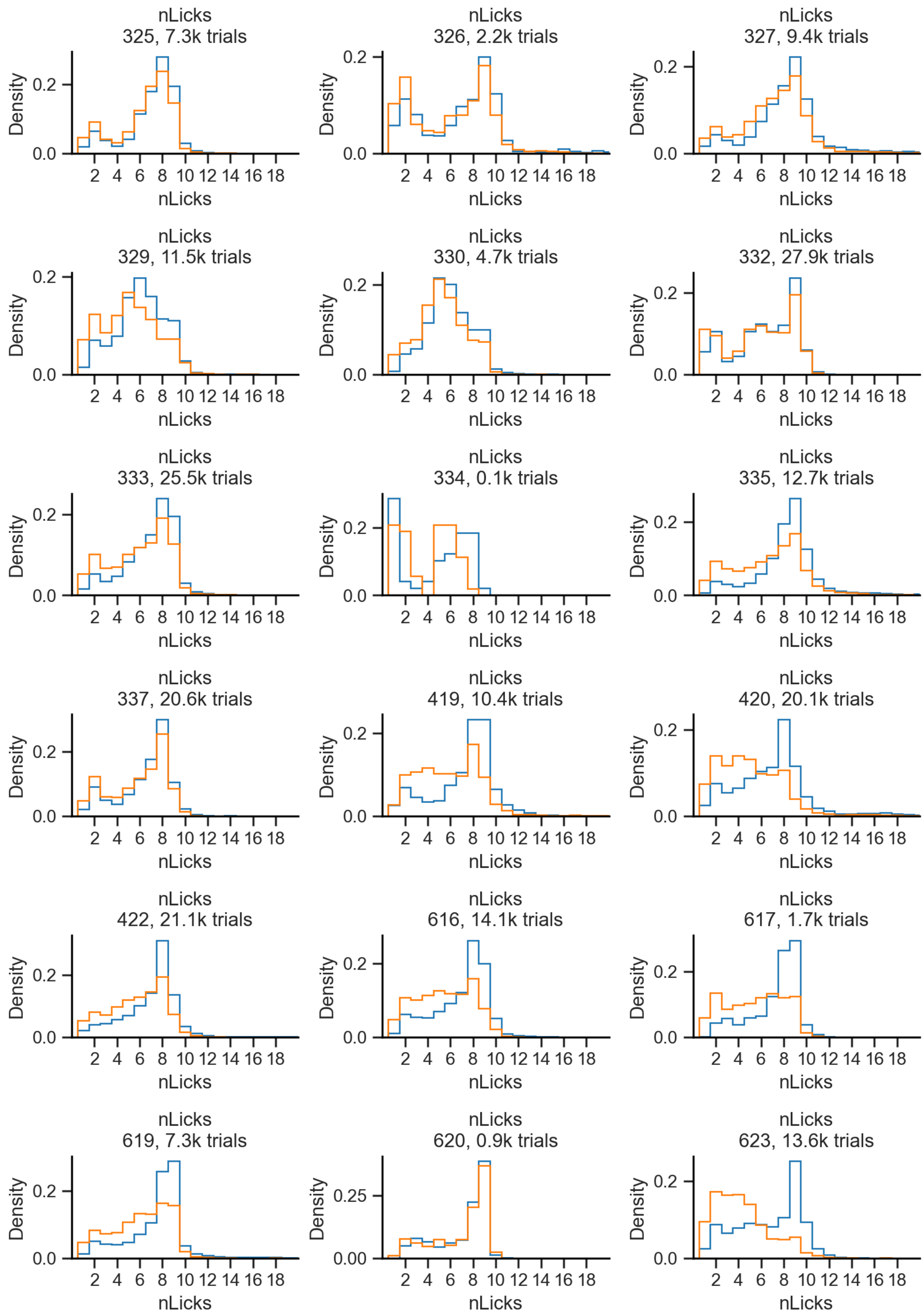
6.2.7 Stimulus




nLicks
325, 7.3k trials
nLicks
326, 2.2k trials
nLicks
327, 9.4k trials
nLicks
329, 11.5k trials
nLicks
330, 4.7k trials
nLicks
332, 27.9k trials
nLicks
333, 25.5k trials
nLicks
334, 0.1k trials
nLicks
335, 12.7k trials
nLicks
337, 20.6k trials
nLicks
419, 10.4k trials
nLicks
420, 20.1k trials
nLicks
422, 21.1k trials
nLicks
616, 14.1k trials
nLicks
617, 1.7k trials
nLicks
619, 7.3k trials
nLicks
620, 0.9k trials
nLicks
623, 13.6k trials
Density
nLicks
2 4 6 8 10 12 14 16 18

6.2.8 1st vs 2nd session half



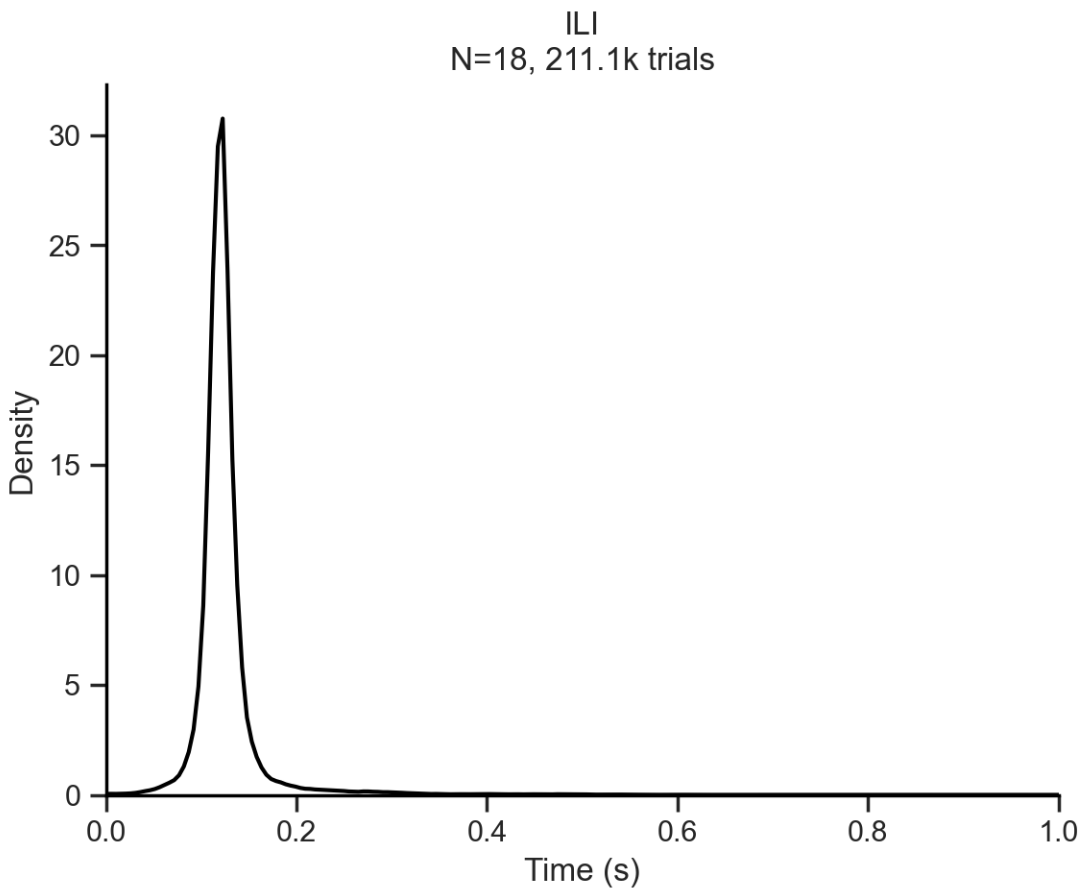

nLicks
325, 7.3k trials
nLicks
326, 2.2k trials
nLicks
327, 9.4k trials
nLicks
329, 11.5k trials
nLicks
330, 4.7k trials
nLicks
332, 27.9k trials
nLicks
333, 25.5k trials
nLicks
334, 0.1k trials
nLicks
335, 12.7k trials
nLicks
337, 20.6k trials
nLicks
419, 10.4k trials
nLicks
420, 20.1k trials
nLicks
422, 21.1k trials
nLicks
616, 14.1k trials
nLicks
617, 1.7k trials
nLicks
619, 7.3k trials
nLicks
620, 0.9k trials
nLicks
623, 13.6k trials
Density
nLicks

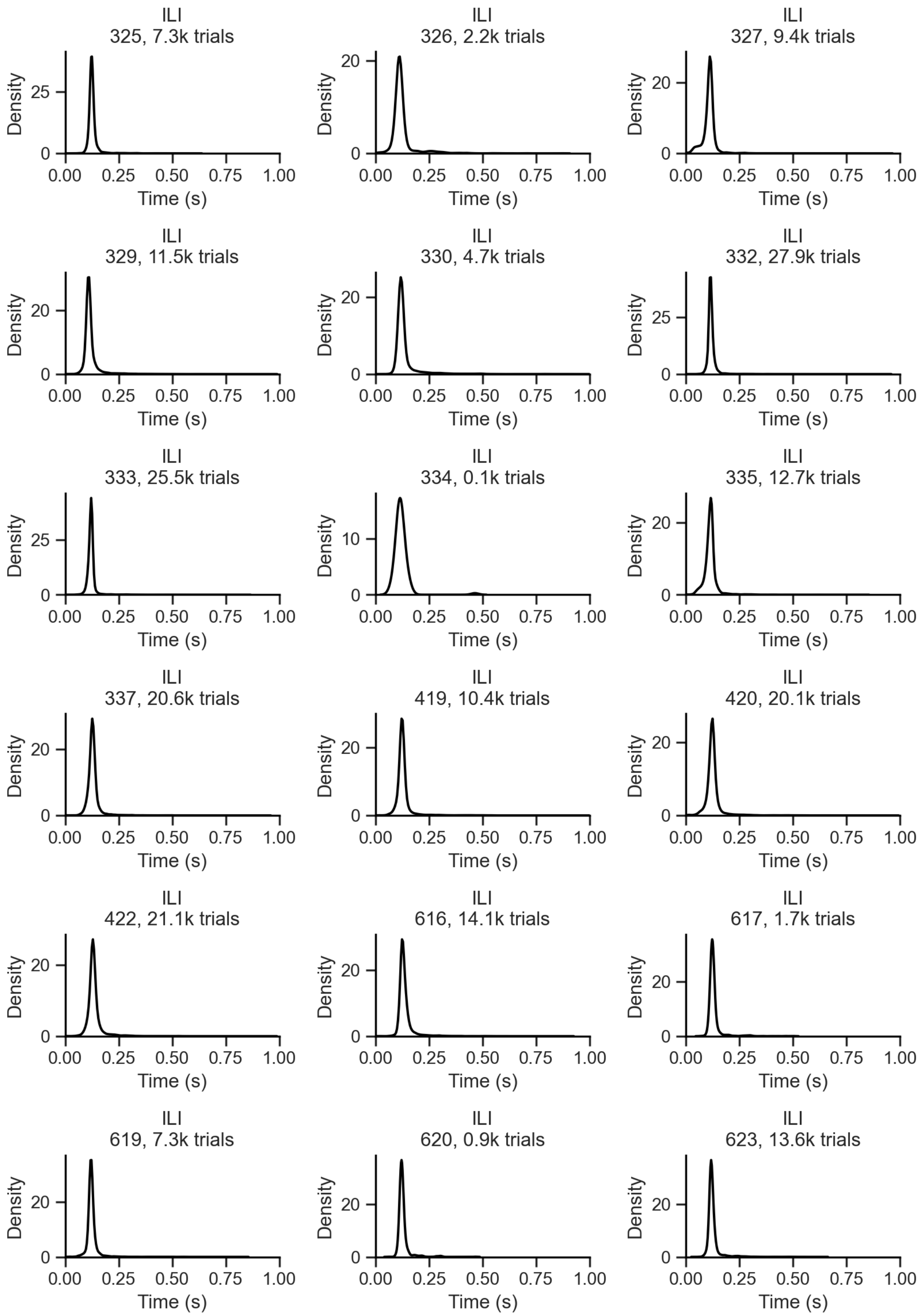
6.2.9 Hidden Markov Model (HMM)

7 Inter Lick Interval (ILI)

$$\text{Lick rate (Hz)} \approx \frac{1}{\text{Mean ILI (s)}}$$

7.0.1 All trials

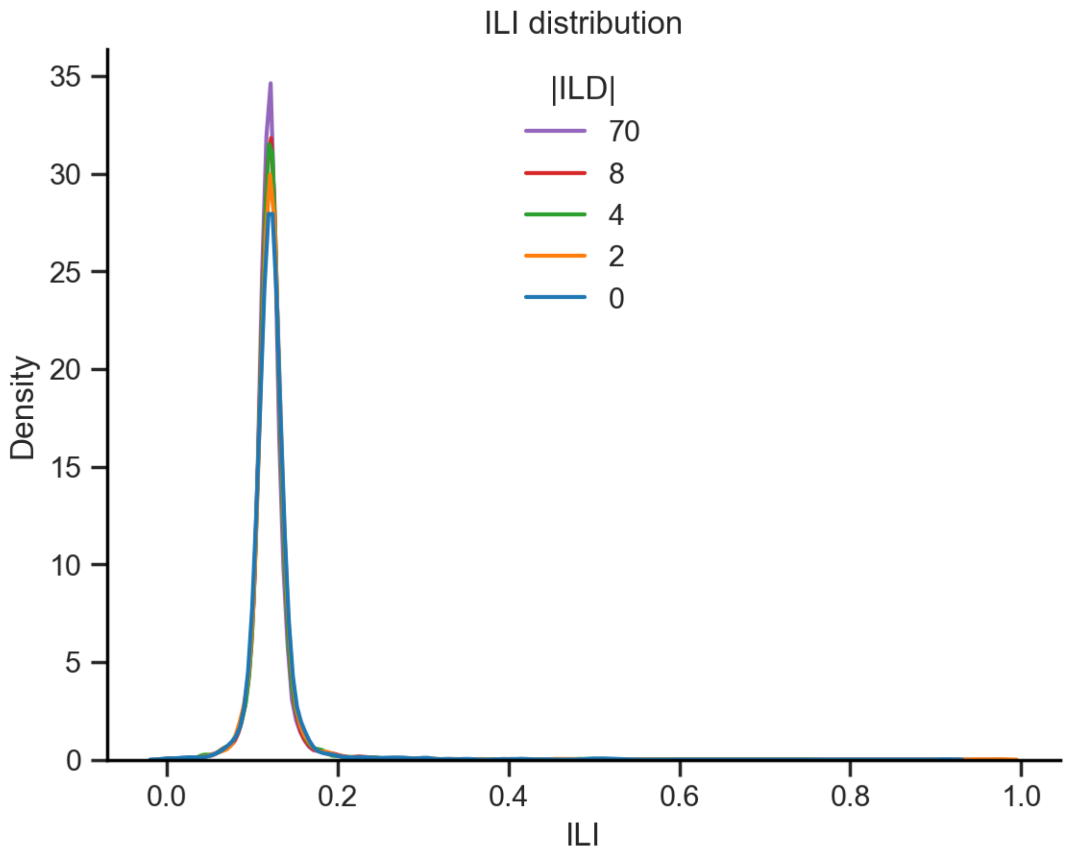



ILI
325, 7.3k trials
Density
0
25
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
326, 2.2k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
327, 9.4k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
329, 11.5k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
330, 4.7k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
332, 27.9k trials
Density
0
25
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
333, 25.5k trials
Density
0
25
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
334, 0.1k trials
Density
0
10
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
335, 12.7k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
337, 20.6k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
419, 10.4k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
420, 20.1k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
422, 21.1k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
616, 14.1k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
617, 1.7k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
619, 7.3k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
620, 0.9k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)
ILI
623, 13.6k trials
Density
0
20
0.00 0.25 0.50 0.75 1.00
Time (s)

7.0.2 Splits

ILD

```
ILD 70: peak ILI = 0.122 s
ILD 8: peak ILI = 0.122 s
ILD 4: peak ILI = 0.120 s
ILD 2: peak ILI = 0.121 s
ILD 0: peak ILI = 0.119 s
Mean ILI = 0.121 s
```



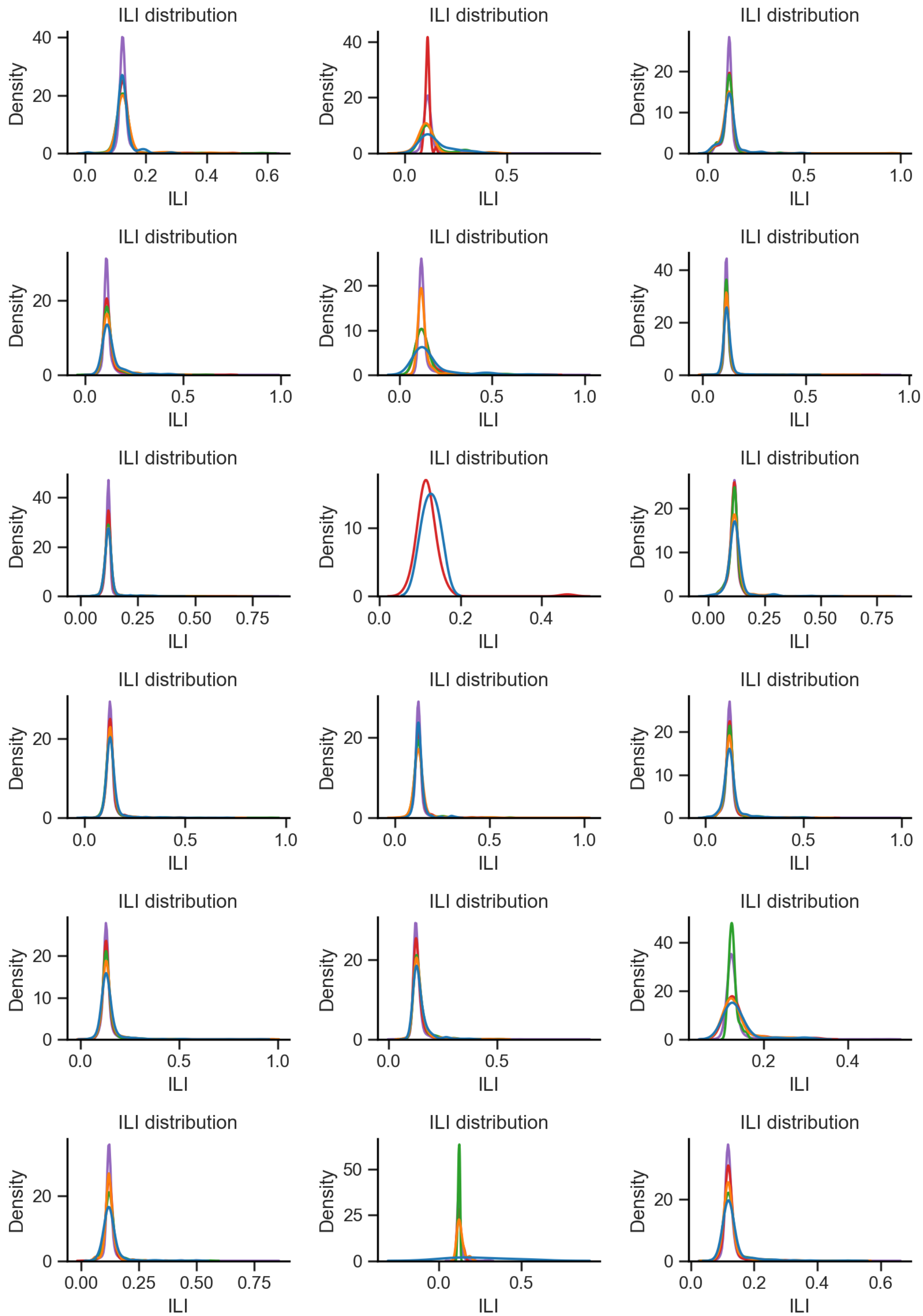

```
ILD 70: peak ILI = 0.122 s
ILD 8: peak ILI = 0.123 s
ILD 4: peak ILI = 0.125 s
ILD 2: peak ILI = 0.125 s
ILD 0: peak ILI = 0.122 s
Mean ILI = 0.123 s
ILD 70: peak ILI = 0.112 s
ILD 8: peak ILI = 0.112 s
ILD 4: peak ILI = 0.107 s
ILD 2: peak ILI = 0.104 s
```

```
ILD 0: peak ILI = 0.113 s
Mean ILI = 0.110 s
ILD 70: peak ILI = 0.112 s
ILD 8: peak ILI = 0.114 s
ILD 4: peak ILI = 0.112 s
ILD 2: peak ILI = 0.111 s
ILD 0: peak ILI = 0.114 s
Mean ILI = 0.113 s
ILD 70: peak ILI = 0.107 s
ILD 8: peak ILI = 0.110 s
ILD 4: peak ILI = 0.110 s
ILD 2: peak ILI = 0.110 s
ILD 0: peak ILI = 0.112 s
Mean ILI = 0.110 s
ILD 70: peak ILI = 0.118 s
ILD 8: peak ILI = 0.118 s
ILD 4: peak ILI = 0.119 s
ILD 2: peak ILI = 0.117 s
ILD 0: peak ILI = 0.122 s
Mean ILI = 0.119 s
ILD 70: peak ILI = 0.116 s
ILD 8: peak ILI = 0.115 s
ILD 4: peak ILI = 0.114 s
ILD 2: peak ILI = 0.114 s
ILD 0: peak ILI = 0.116 s
Mean ILI = 0.115 s
ILD 70: peak ILI = 0.122 s
ILD 8: peak ILI = 0.121 s
ILD 4: peak ILI = 0.121 s
ILD 2: peak ILI = 0.119 s
ILD 0: peak ILI = 0.122 s
Mean ILI = 0.121 s
ILD 70: peak ILI = 0.113 s
ILD 8: peak ILI = 0.113 s
ILD 2: peak ILI = 0.113 s
ILD 0: peak ILI = 0.126 s
Mean ILI = 0.117 s
ILD 70: peak ILI = 0.115 s
ILD 8: peak ILI = 0.116 s
ILD 4: peak ILI = 0.116 s
ILD 2: peak ILI = 0.117 s
ILD 0: peak ILI = 0.118 s
Mean ILI = 0.117 s

C:\Users\Usuario\PycharmProjects\licks\licks.py:400: UserWarning: Dataset has 0
variance; skipping density estimate. Pass `warn_singular=False` to disable this
warning.
  sns.kdeplot(df_ild[var], color=color, label=ild)
```

```
C:\Users\Usuario\PycharmProjects\licks\licks.py:400: UserWarning: Dataset has 0
variance; skipping density estimate. Pass `warn_singular=False` to disable this
warning.
  sns.kdeplot(df_ild[var], color=color, label=ild)
ILD 70: peak ILI = 0.126 s
ILD 8: peak ILI = 0.128 s
ILD 4: peak ILI = 0.129 s
ILD 2: peak ILI = 0.129 s
ILD 0: peak ILI = 0.128 s
Mean ILI = 0.128 s
ILD 70: peak ILI = 0.124 s
ILD 8: peak ILI = 0.124 s
ILD 4: peak ILI = 0.124 s
ILD 2: peak ILI = 0.123 s
ILD 0: peak ILI = 0.124 s
Mean ILI = 0.124 s
ILD 70: peak ILI = 0.123 s
ILD 8: peak ILI = 0.124 s
ILD 4: peak ILI = 0.125 s
ILD 2: peak ILI = 0.122 s
ILD 0: peak ILI = 0.122 s
Mean ILI = 0.123 s
ILD 70: peak ILI = 0.128 s
ILD 8: peak ILI = 0.130 s
ILD 4: peak ILI = 0.130 s
ILD 2: peak ILI = 0.128 s
ILD 0: peak ILI = 0.130 s
Mean ILI = 0.129 s
ILD 70: peak ILI = 0.123 s
ILD 8: peak ILI = 0.126 s
ILD 4: peak ILI = 0.129 s
ILD 2: peak ILI = 0.129 s
ILD 0: peak ILI = 0.128 s
Mean ILI = 0.127 s
ILD 70: peak ILI = 0.122 s
ILD 8: peak ILI = 0.124 s
ILD 4: peak ILI = 0.124 s
ILD 2: peak ILI = 0.121 s
ILD 0: peak ILI = 0.125 s
Mean ILI = 0.123 s
ILD 70: peak ILI = 0.123 s
ILD 8: peak ILI = 0.121 s
ILD 4: peak ILI = 0.122 s
ILD 2: peak ILI = 0.120 s
ILD 0: peak ILI = 0.120 s
Mean ILI = 0.121 s
ILD 70: peak ILI = 0.121 s
```

```
ILD 8: peak ILI = 0.121 s
ILD 4: peak ILI = 0.125 s
ILD 2: peak ILI = 0.123 s
ILD 0: peak ILI = 0.165 s
Mean ILI = 0.131 s
ILD 70: peak ILI = 0.117 s
ILD 8: peak ILI = 0.118 s
ILD 4: peak ILI = 0.119 s
ILD 2: peak ILI = 0.119 s
ILD 0: peak ILI = 0.119 s
Mean ILI = 0.119 s
```


ILI distribution
Density
ILI
ILI distribution
Density
ILI
ILI distribution
Density
ILI
ILI distribution
Density
ILI
ILI distribution
Density
ILI
ILI distribution
Density
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ILI distribution
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ILI distribution
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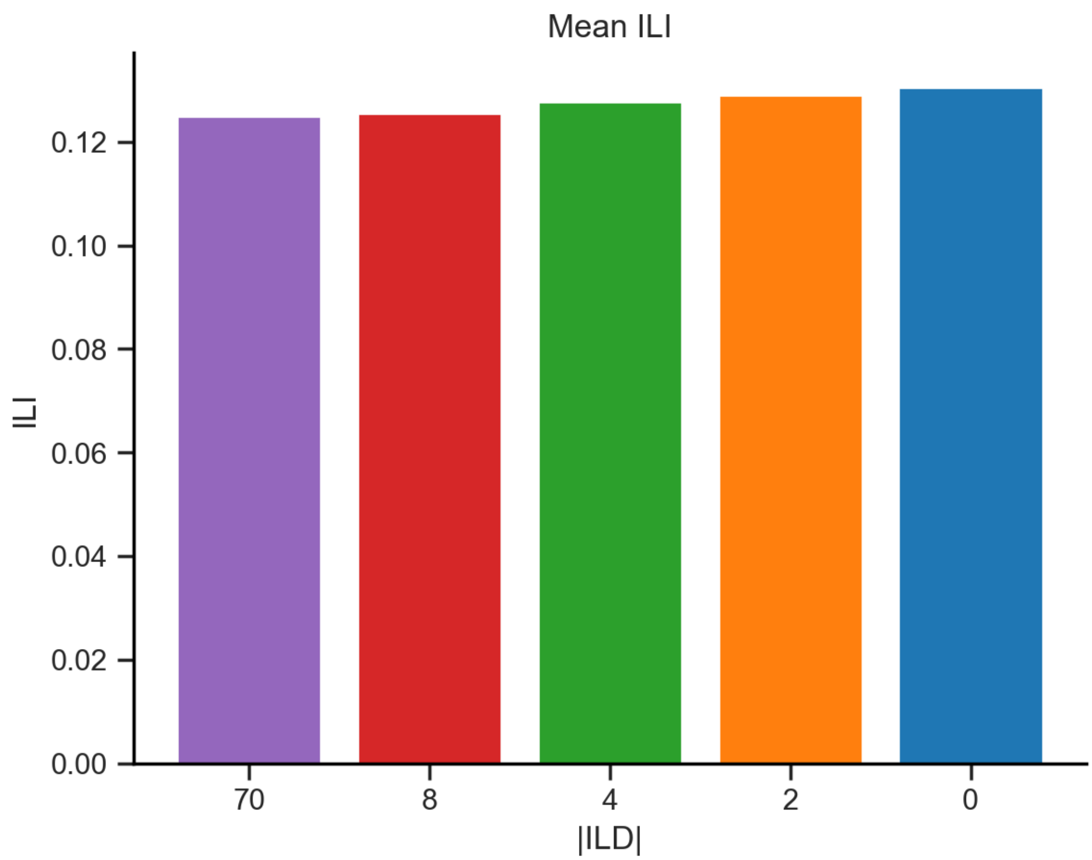
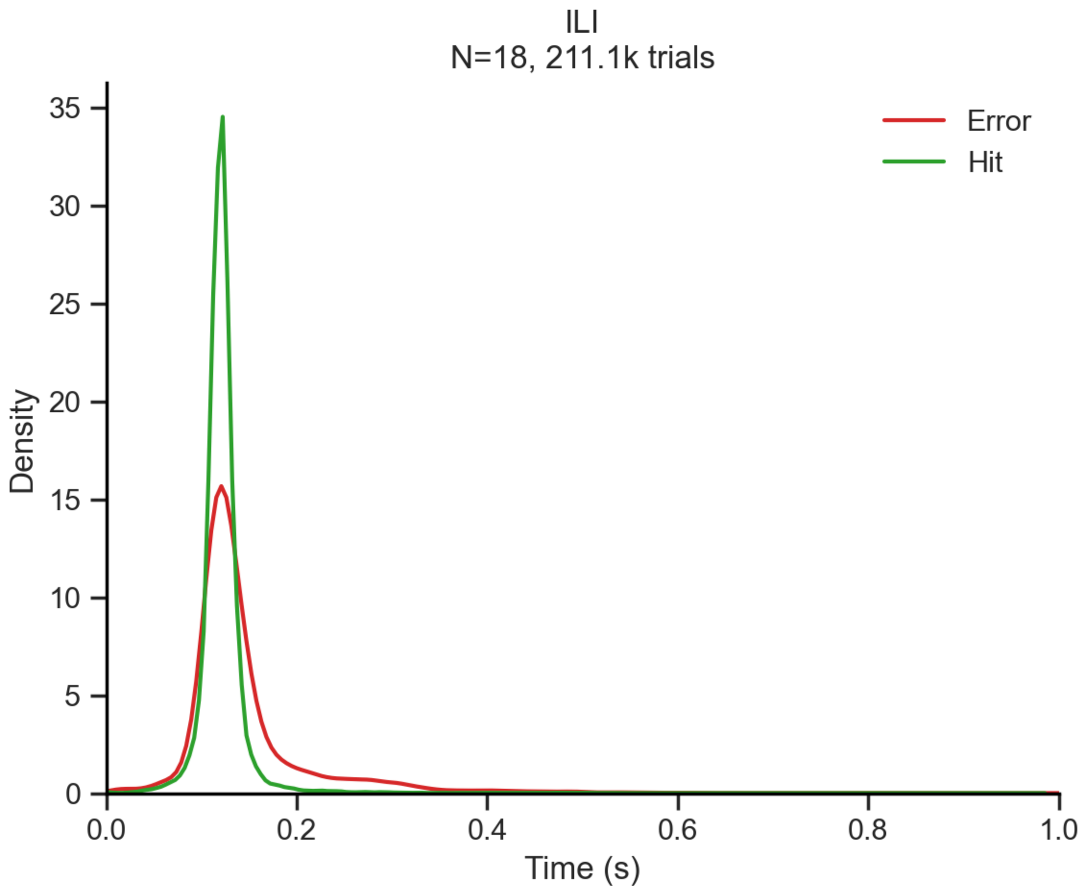
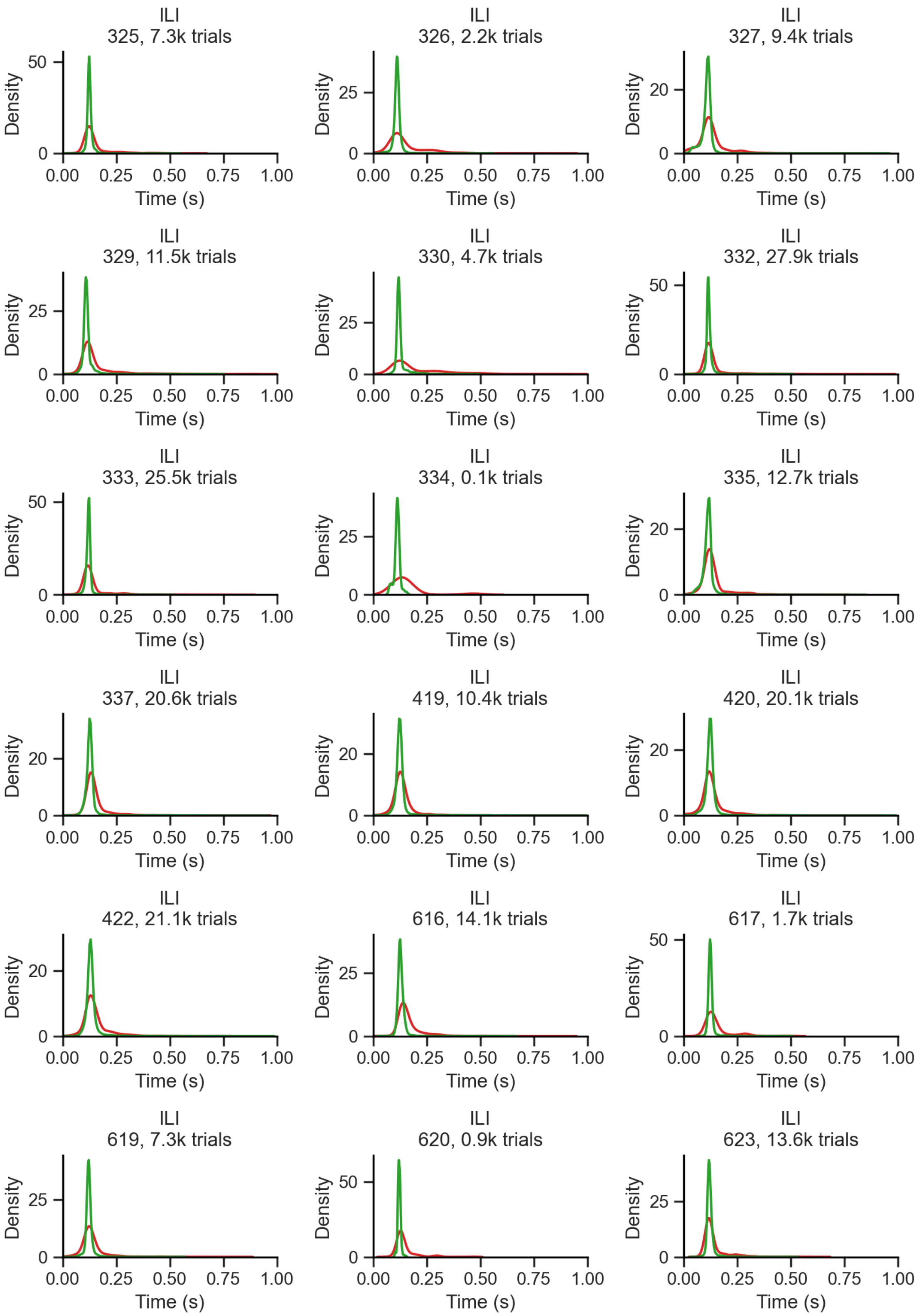
ILI distribution
Density
ILI
ILI distribution
Density
ILI
ILI distribution
Density
ILI

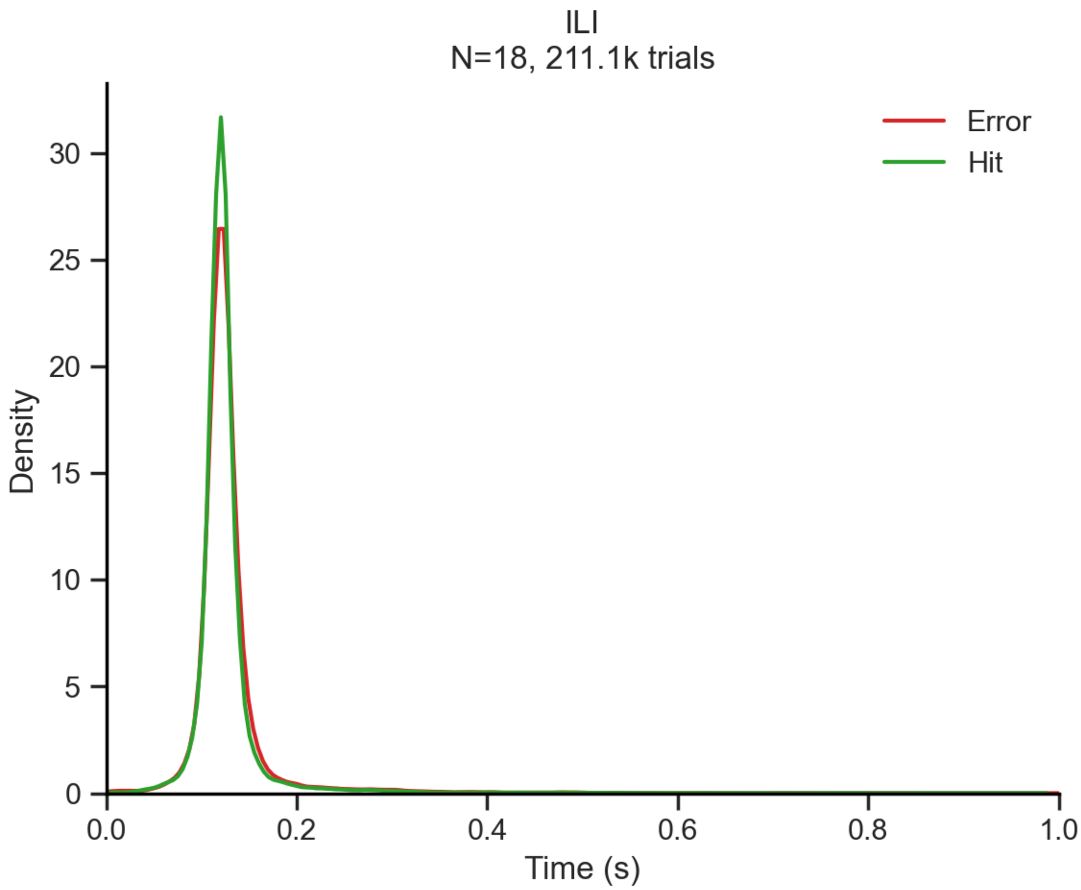


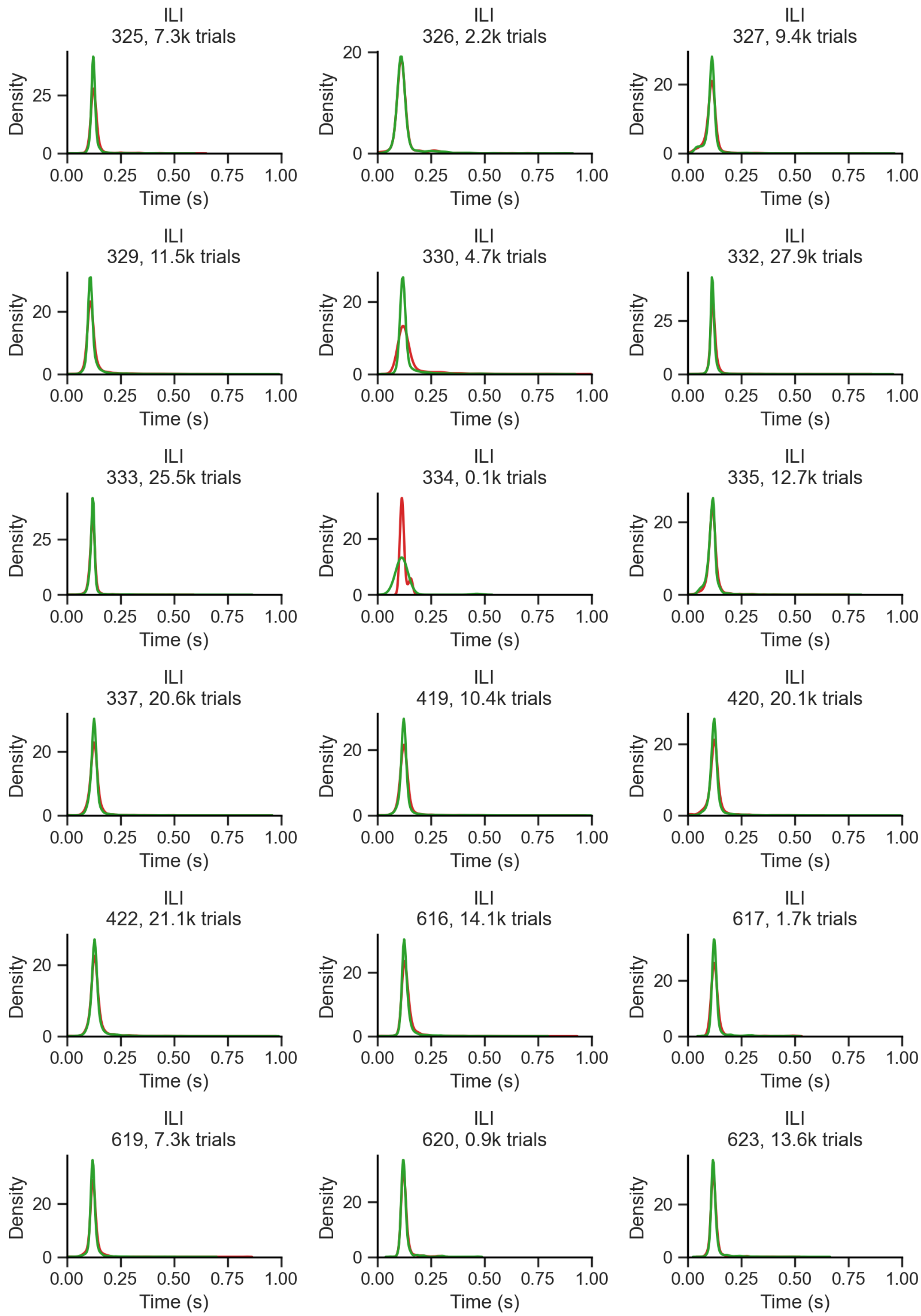
Outcome


ILI
N=18, 211.1k trials
Error
Hit
Density
Time (s)
0
5
10
15
20
25
30
35
0.0
0.2
0.4
0.6
0.8
1.0

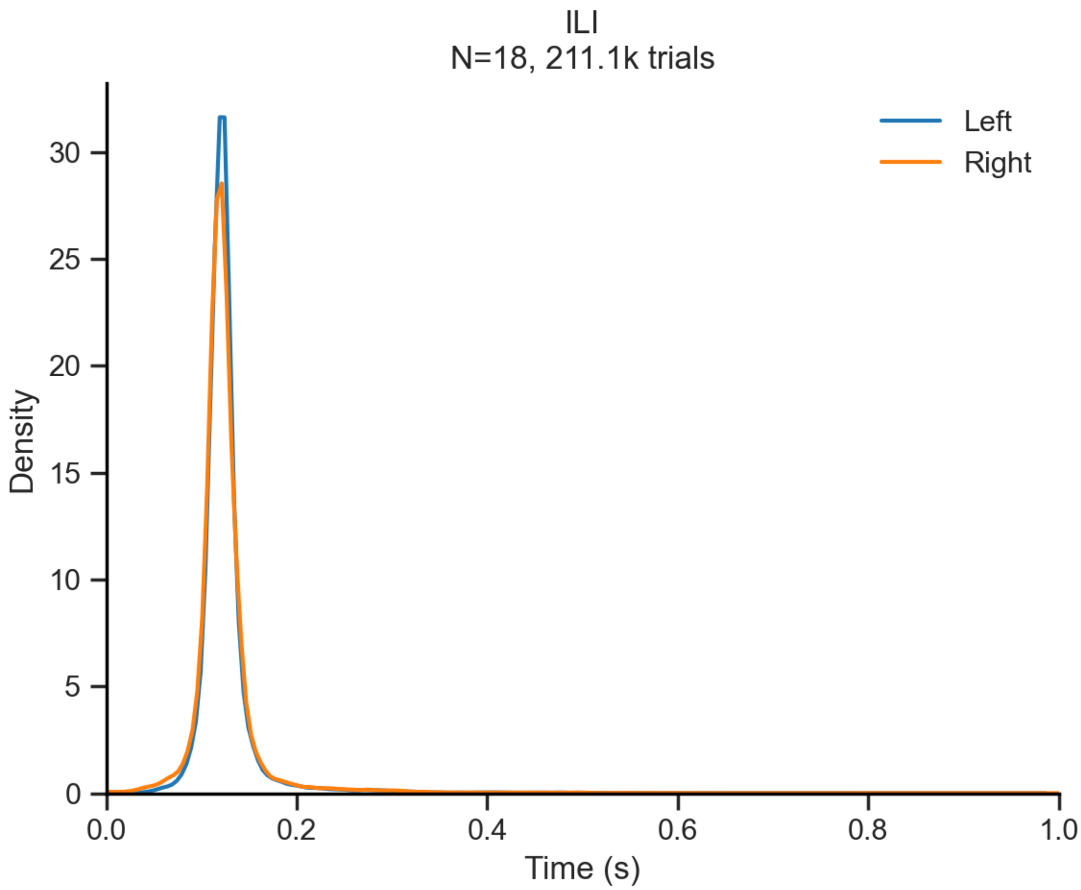

ILI
325, 7.3k trials
ILI
326, 2.2k trials
ILI
327, 9.4k trials
ILI
329, 11.5k trials
ILI
330, 4.7k trials
ILI
332, 27.9k trials
ILI
333, 25.5k trials
ILI
334, 0.1k trials
ILI
335, 12.7k trials
ILI
337, 20.6k trials
ILI
419, 10.4k trials
ILI
420, 20.1k trials
ILI
422, 21.1k trials
ILI
616, 14.1k trials
ILI
617, 1.7k trials
ILI
619, 7.3k trials
ILI
620, 0.9k trials
ILI
623, 13.6k trials
Density
Time (s)
0.00
0.25
0.50
0.75
1.00

Previous Outcome

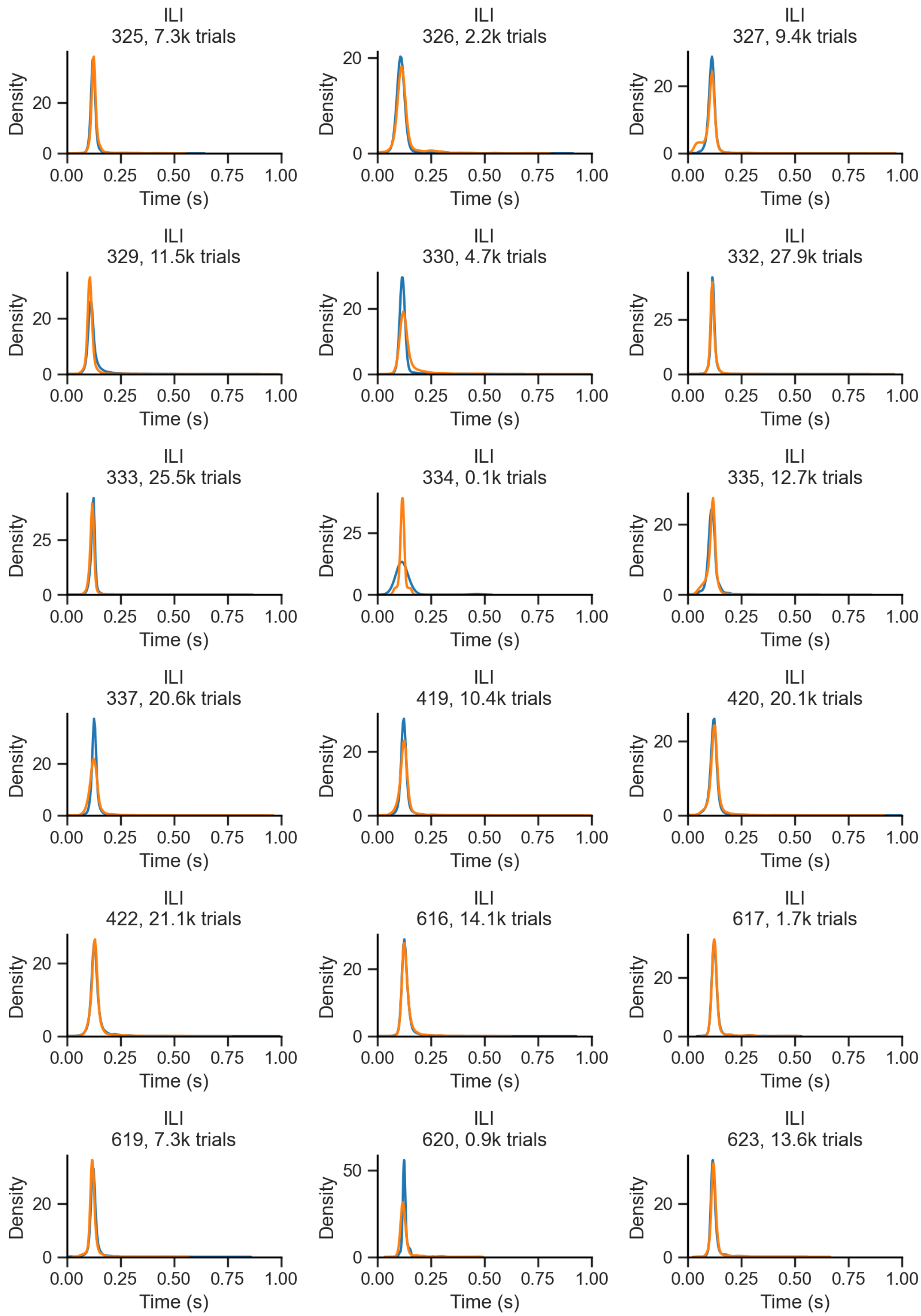



ILI
325, 7.3k trials
ILI
326, 2.2k trials
ILI
327, 9.4k trials
ILI
329, 11.5k trials
ILI
330, 4.7k trials
ILI
332, 27.9k trials
ILI
333, 25.5k trials
ILI
334, 0.1k trials
ILI
335, 12.7k trials
ILI
337, 20.6k trials
ILI
419, 10.4k trials
ILI
420, 20.1k trials
ILI
422, 21.1k trials
ILI
616, 14.1k trials
ILI
617, 1.7k trials
ILI
619, 7.3k trials
ILI
620, 0.9k trials
ILI
623, 13.6k trials
Density
Time (s)
0.00
0.25
0.50
0.75
1.00

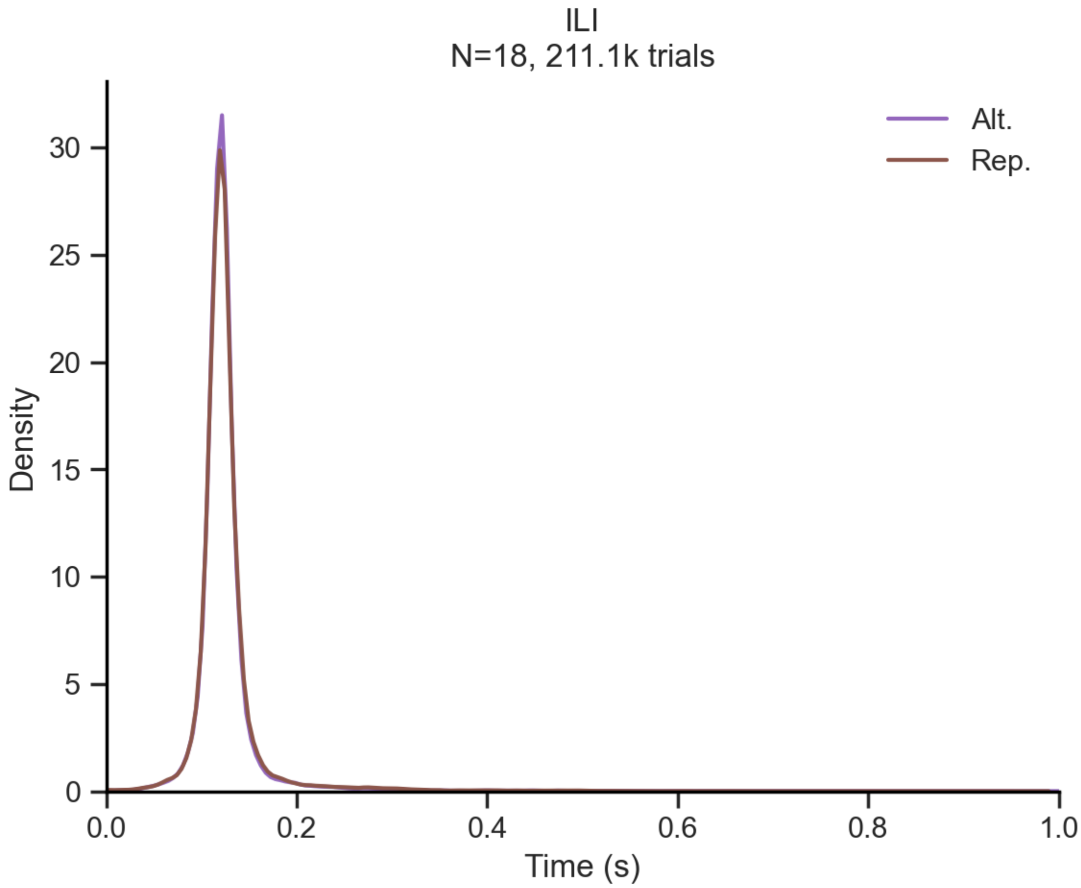
Choice

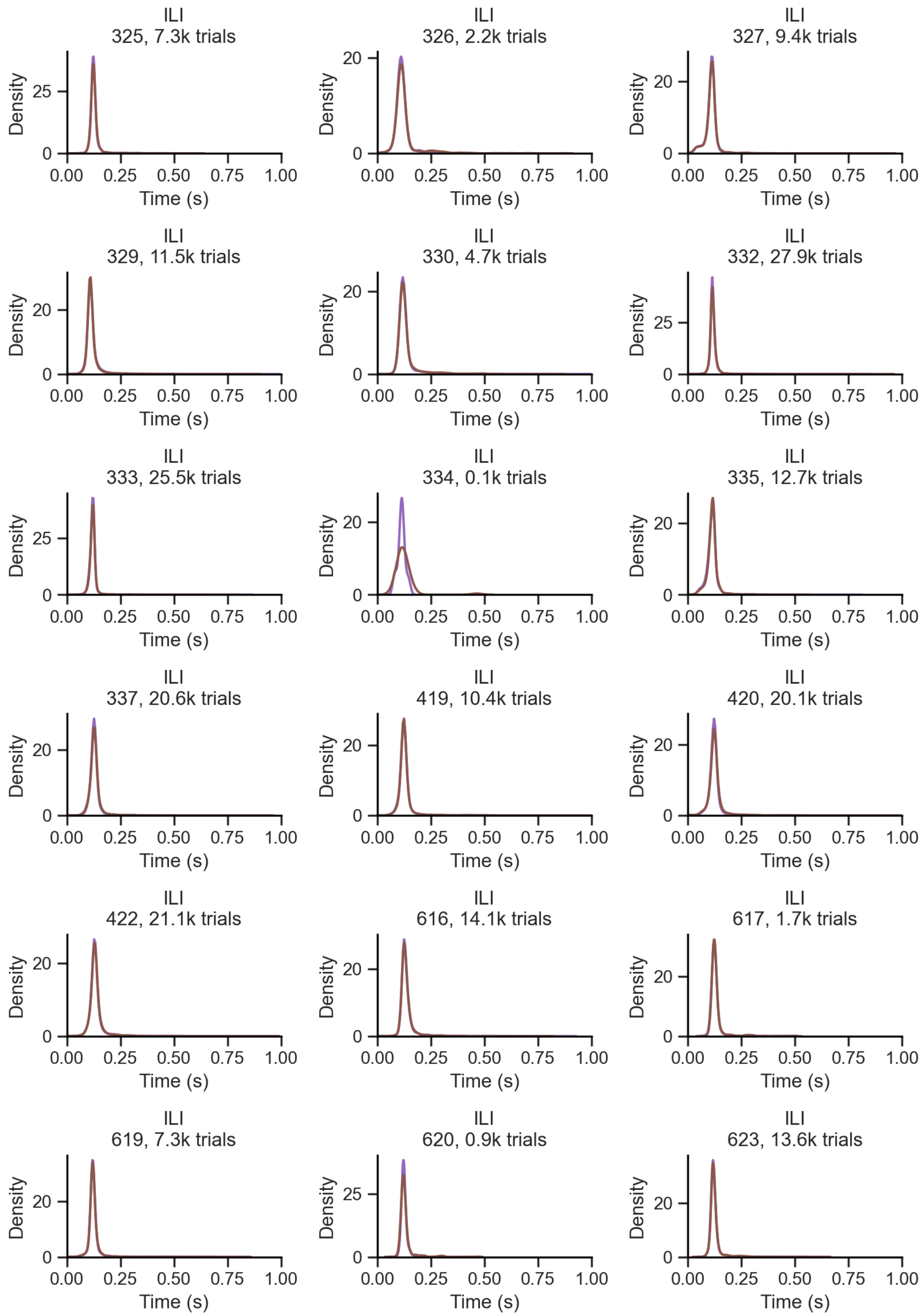


ILI
N=18, 211.1k trials
Left
Right
Density
Time (s)
0
5
10
15
20
25
30
0.0
0.2
0.4
0.6
0.8
1.0

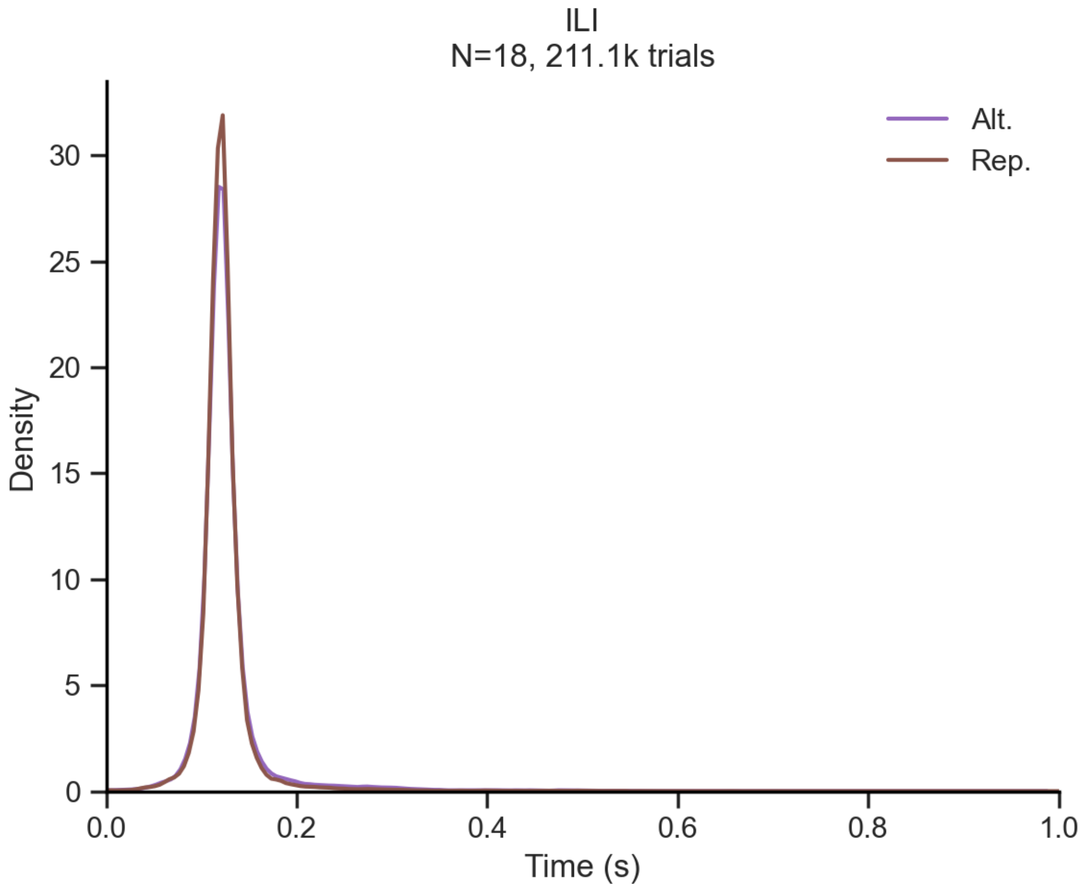

ILI
325, 7.3k trials
ILI
326, 2.2k trials
ILI
327, 9.4k trials
ILI
329, 11.5k trials
ILI
330, 4.7k trials
ILI
332, 27.9k trials
ILI
333, 25.5k trials
ILI
334, 0.1k trials
ILI
335, 12.7k trials
ILI
337, 20.6k trials
ILI
419, 10.4k trials
ILI
420, 20.1k trials
ILI
422, 21.1k trials
ILI
616, 14.1k trials
ILI
617, 1.7k trials
ILI
619, 7.3k trials
ILI
620, 0.9k trials
ILI
623, 13.6k trials
Density
Time (s)
0.00
0.25
0.50
0.75
1.00

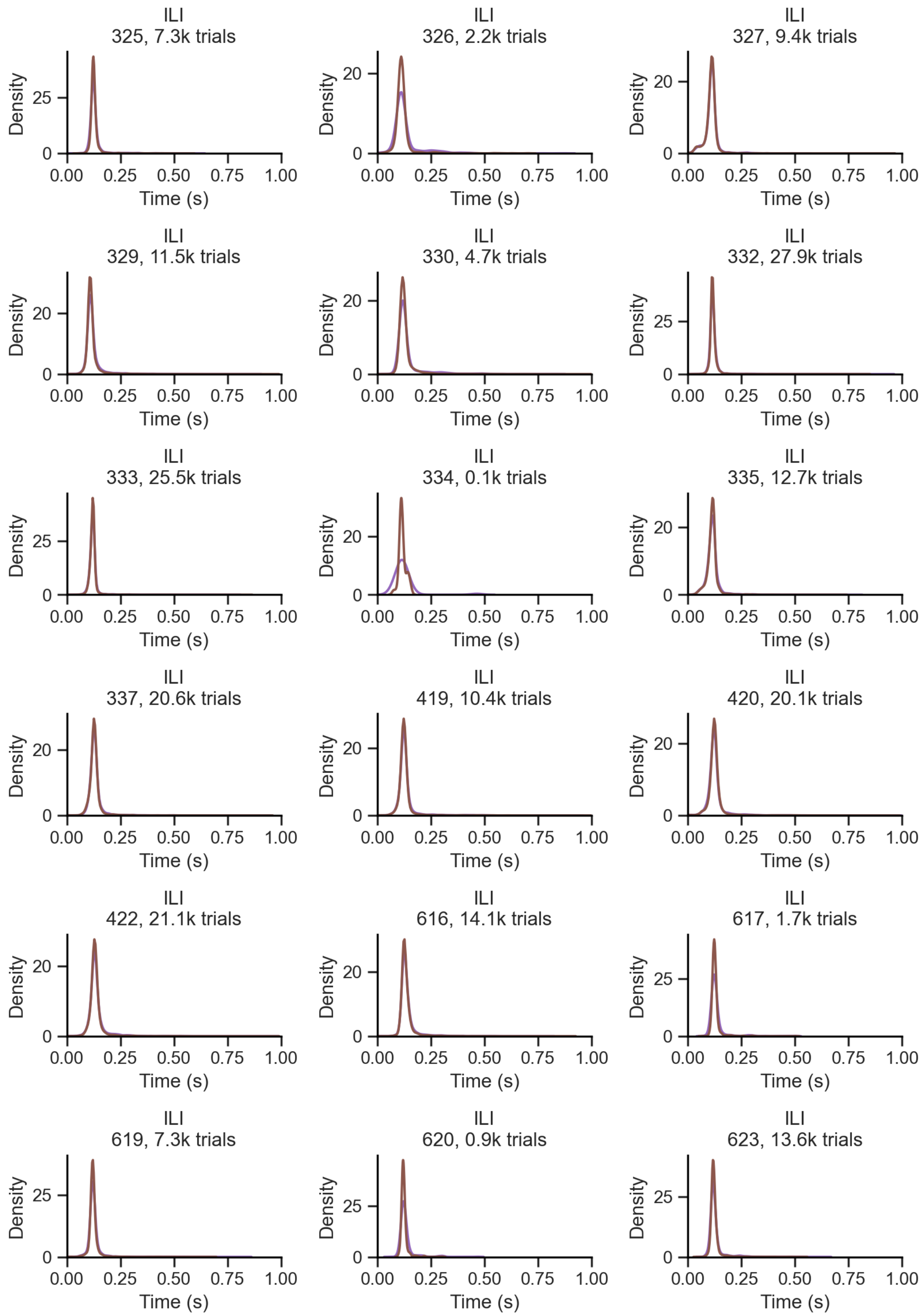
Repeat choice



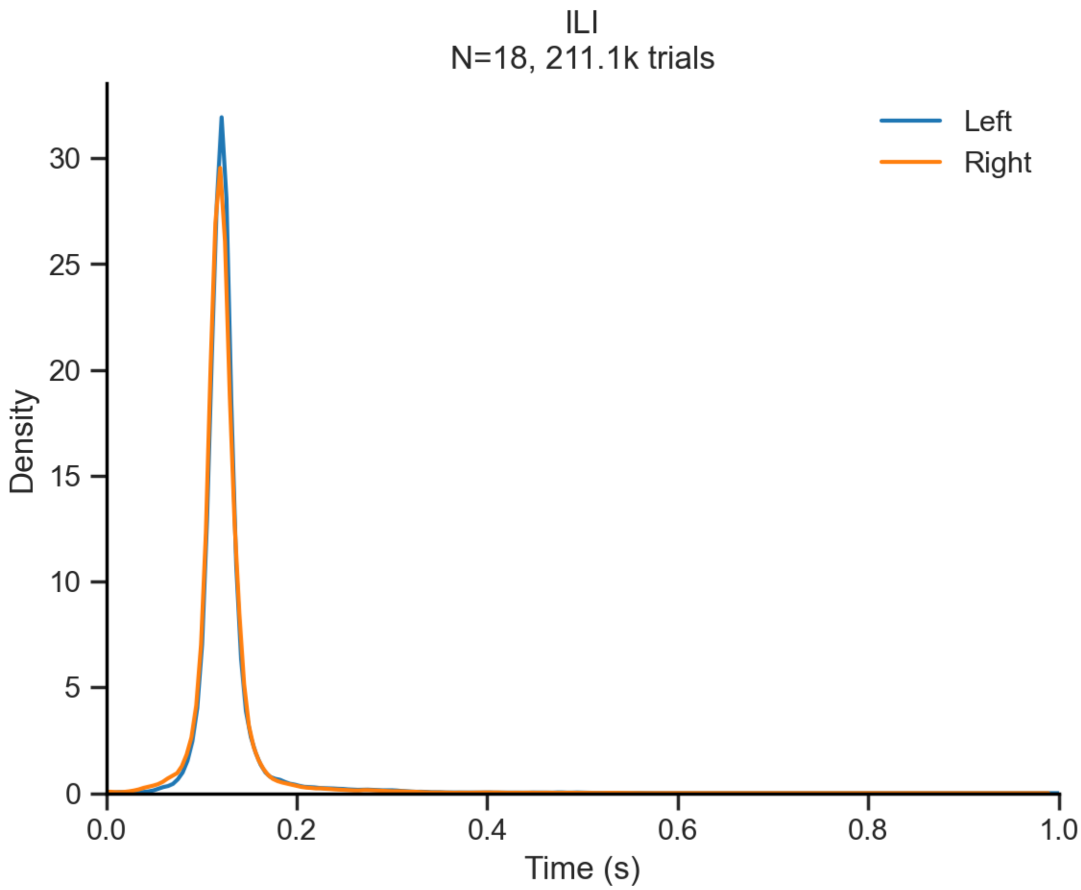

ILI
325, 7.3k trials
ILI
326, 2.2k trials
ILI
327, 9.4k trials
ILI
329, 11.5k trials
ILI
330, 4.7k trials
ILI
332, 27.9k trials
ILI
333, 25.5k trials
ILI
334, 0.1k trials
ILI
335, 12.7k trials
ILI
337, 20.6k trials
ILI
419, 10.4k trials
ILI
420, 20.1k trials
ILI
422, 21.1k trials
ILI
616, 14.1k trials
ILI
617, 1.7k trials
ILI
619, 7.3k trials
ILI
620, 0.9k trials
ILI
623, 13.6k trials
Density
Time (s)

Repeat trial

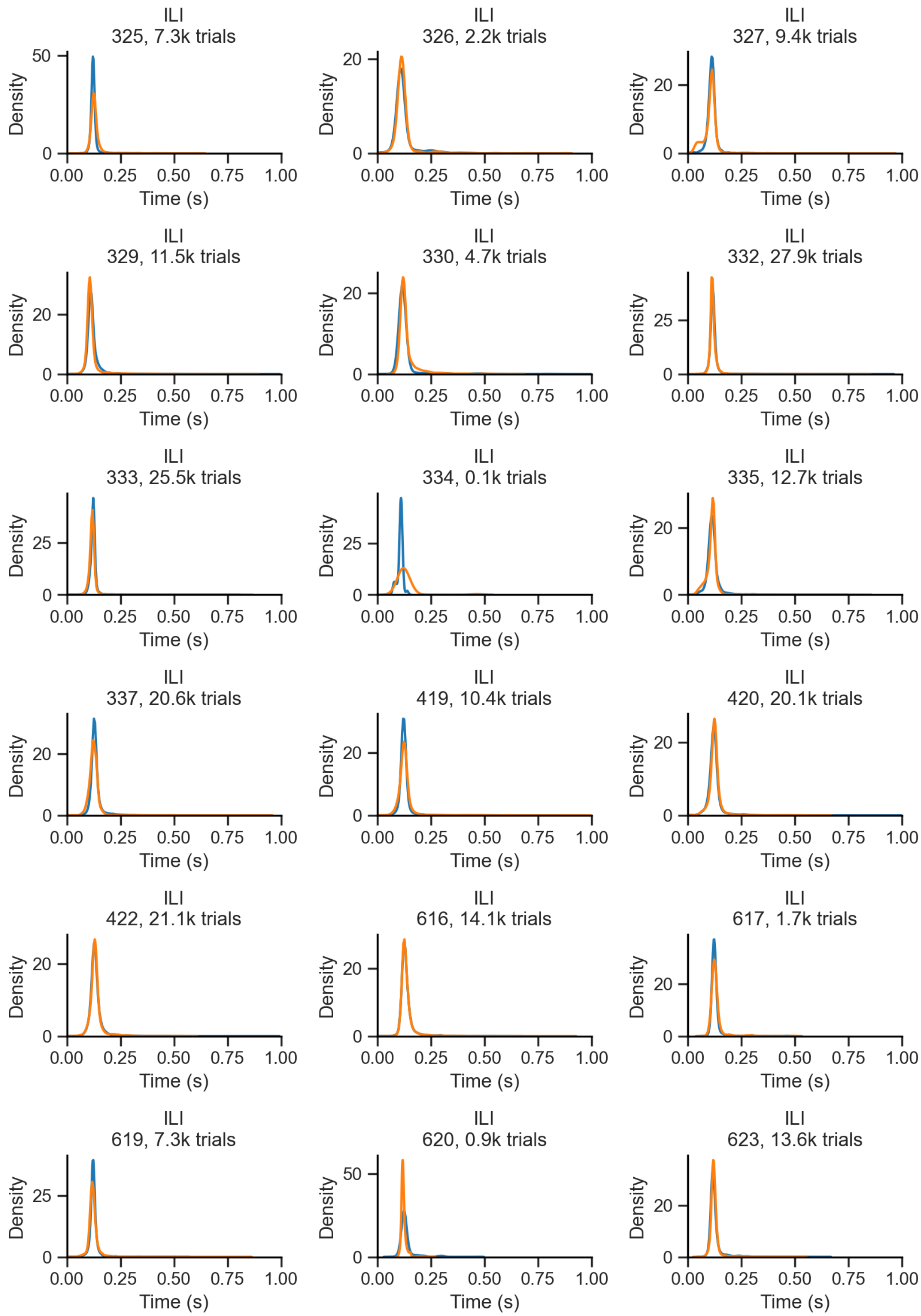



ILI
325, 7.3k trials
ILI
326, 2.2k trials
ILI
327, 9.4k trials
ILI
329, 11.5k trials
ILI
330, 4.7k trials
ILI
332, 27.9k trials
ILI
333, 25.5k trials
ILI
334, 0.1k trials
ILI
335, 12.7k trials
ILI
337, 20.6k trials
ILI
419, 10.4k trials
ILI
420, 20.1k trials
ILI
422, 21.1k trials
ILI
616, 14.1k trials
ILI
617, 1.7k trials
ILI
619, 7.3k trials
ILI
620, 0.9k trials
ILI
623, 13.6k trials
Density
Time (s)
0.00 0.25 0.50 0.75 1.00

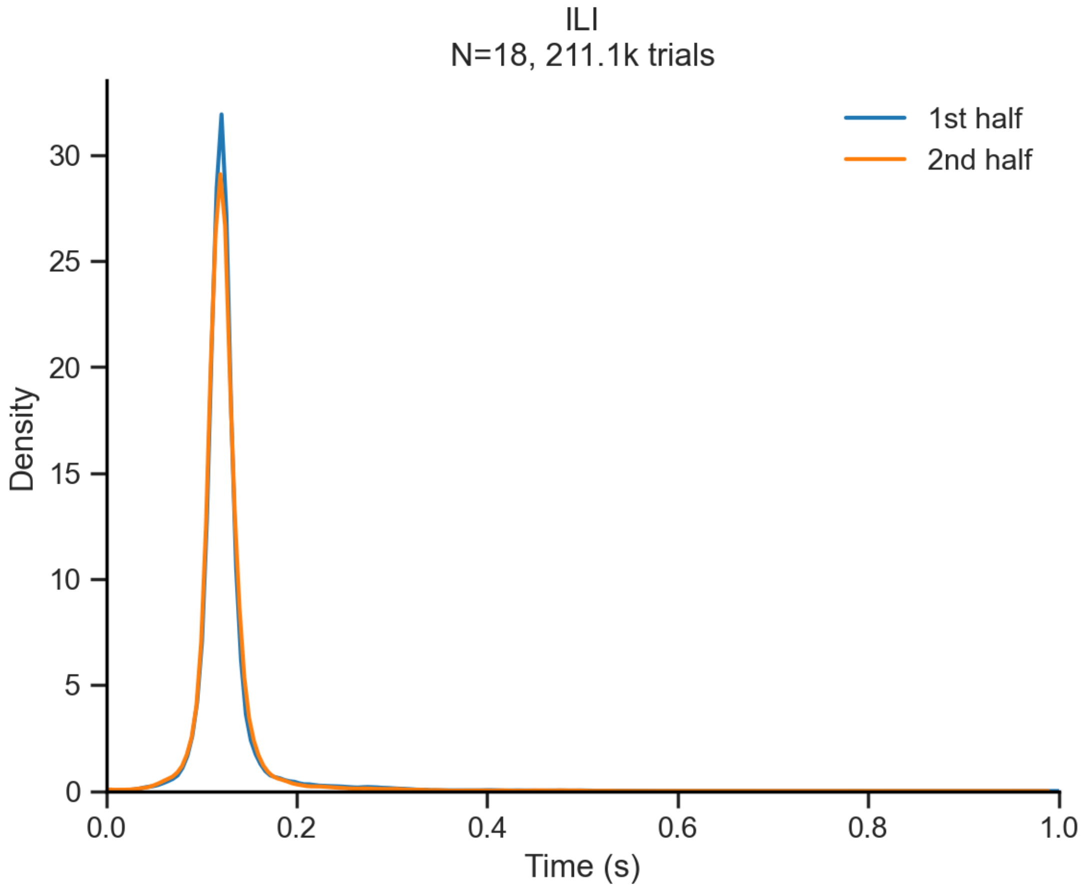
Stimulus

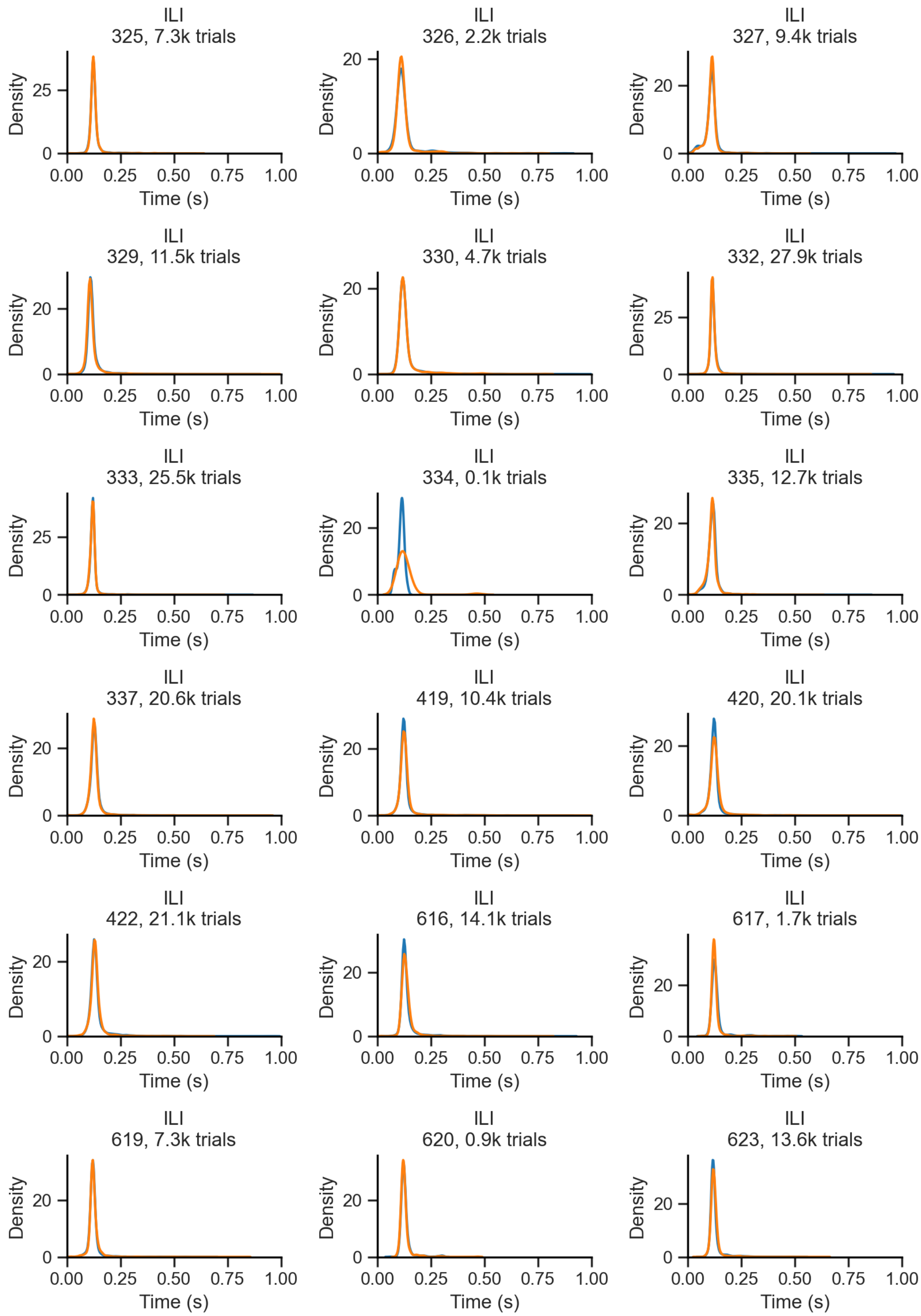


ILI
N=18, 211.1k trials
Left
Right
Density
Time (s)
0
5
10
15
20
25
30
0.0
0.2
0.4
0.6
0.8
1.0


ILI
325, 7.3k trials
ILI
326, 2.2k trials
ILI
327, 9.4k trials
ILI
329, 11.5k trials
ILI
330, 4.7k trials
ILI
332, 27.9k trials
ILI
333, 25.5k trials
ILI
334, 0.1k trials
ILI
335, 12.7k trials
ILI
337, 20.6k trials
ILI
419, 10.4k trials
ILI
420, 20.1k trials
ILI
422, 21.1k trials
ILI
616, 14.1k trials
ILI
617, 1.7k trials
ILI
619, 7.3k trials
ILI
620, 0.9k trials
ILI
623, 13.6k trials
Density
Time (s)
0.00
0.25
0.50
0.75
1.00

1st vs 2nd session half




ILI
325, 7.3k trials
ILI
326, 2.2k trials
ILI
327, 9.4k trials
ILI
329, 11.5k trials
ILI
330, 4.7k trials
ILI
332, 27.9k trials
ILI
333, 25.5k trials
ILI
334, 0.1k trials
ILI
335, 12.7k trials
ILI
337, 20.6k trials
ILI
419, 10.4k trials
ILI
420, 20.1k trials
ILI
422, 21.1k trials
ILI
616, 14.1k trials
ILI
617, 1.7k trials
ILI
619, 7.3k trials
ILI
620, 0.9k trials
ILI
623, 13.6k trials
Density
Time (s)
0.00 0.25 0.50 0.75 1.00
0
20
25

8 Save all notebook's figures

```
[NbConvertApp] Converting notebook notebook.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 63 image(s).
[NbConvertApp] Writing 13270696 bytes to notebook.html
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Saved 63 images to notebook_files/
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[NbConvertApp] Converting notebook notebook.ipynb to pdf
[NbConvertApp] Support files will be in notebook_files\
[NbConvertApp] Making directory .\notebook_files
[NbConvertApp] Writing 50275 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | b had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 8676892 bytes to notebook.pdf
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